The Nature for Cooling Rapid Assessment Tool

A transparent, evidence-grounded screening methodology for prioritising nature-based solutions for urban cooling

Methodology Report

Version 2 of the methodology, released for expert review

Criterra

https://criterra.eu

Methodology version	2026.08.05
Document date	1 August 2026
Status	Released for expert and public review
Licence	Apache License 2.0

Openness statement. The Nature for Cooling Rapid Assessment Tool, its complete methodology configuration, its source code, and this document are public. The tool is released under the Apache License, Version 2.0. Every quantitative value the tool applies is derived in this document, recorded in machine-readable configuration, and traced to a cited source whose verification status is disclosed in Appendix E. Readers are able to inspect, reproduce, contest, and modify every element of the methodology described here.

How to cite this document. Criterra (2026). *The Nature for Cooling Rapid Assessment Tool: a transparent, evidence-grounded screening methodology for prioritising nature-based solutions for urban cooling.* Methodology Report, version 2026.08.05.
https://github.com/Dimitrios-Kafetzis/CriterraNatureCoolingTool

Corresponding methodology artefacts. The normative configuration files are `config/*.yaml`; the per-archetype derivations are `docs/methodology/EVIDENCE-TABLES.md`; the source register with verification status is `docs/methodology/BIBLIOGRAPHY.md`. All carry the version stamp 2026.08.05 and move as a single unit.

Invitation to review, and how to reach us. This document is written to be contested. Reviewers are specifically invited to challenge the adopted cooling envelopes (Section 6), the aggregation weights (Section 7.12), the derivation of energy savings (Section 7.9), and the treatment of uncertainty (Section 8). Questions, corrections, and methodology challenges are handled openly as issues at
https://github.com/Dimitrios-Kafetzis/CriterraNatureCoolingTool/issues,
so that critique and its resolution remain part of the public record; Section 13 describes what a well-formed challenge looks like.

Abstract

Cities facing intensifying heat must decide where to intervene and with which nature-based solution, typically across many more candidate sites than can be studied in detail. The available options are unstructured judgment, which is fast but neither comparable nor defensible, and microclimate simulation, which is rigorous but too slow and costly for triage. This paper specifies a screening-level, multi-criteria methodology that occupies the gap between them. The methodology converts a structured description of an urban site and one or more proposed nature-based interventions into a Heat Priority Index, a Cooling Potential Score, an indicative daytime pedestrian-level air-temperature reduction range, derived cooling-energy and greenhouse-gas estimates, equity and co-benefit scores, and a single NbS Cooling Opportunity Score on a 0–100 scale, each accompanied by an explicit confidence rating. Its intervention library has two levels: 110 curated catalogue entries, each inheriting one of 18 cited *cooling archetypes* that carry every performance value and its citations, because solution-specific cooling literature does not exist for 110 intervention types and inventing 110 envelopes would manufacture precision the source catalogue does not contain. Every result names the evidence class it inherited from. The archetype envelopes are derived from published evidence, principally an umbrella review of sixty-four systematic reviews (Keravec-Balbot et al. 2026), an empirical meta-analysis of park cooling (Bowler et al. 2010), and a scale-dependent canopy study (Ziter et al. 2019); the composite-indicator construction follows established practice (Nardo et al. 2008) and the prioritisation logic follows Norton et al. (2015). Three calibration choices are emphasised: a downward correction for the modelling bias present in the synthesised literature, a refusal to claim super-additive cooling for multi-intervention packages, whose reported temperature is capped at the best-evidenced component and never summed, and the clipping of all reported temperature ranges to the literature-supported envelope so that favourable site conditions can never generate a claim exceeding published evidence. The methodology's limitations are stated rather than minimised: all cooling values are daytime only; the quantity estimated is air temperature, not thermal comfort; the evidence base is geographically skewed towards temperate and subtropical Northern Hemisphere cities; no default costs are shipped; and the aggregation weights are declared expert judgment, including a disclosed equity-forward emphasis that gives vulnerability an effective weight of 0.25 against heat exposure at 0.15. A published sensitivity analysis, regenerated for this version across 24 golden scenarios, finds priority ordering substantially stable under ±25 % single-weight perturbation: worst-case pairwise rank stability 0.9529, pooled mean score displacement 0.47 points, and 8 band crossings in 288 scenario-perturbation combinations, four of them attributable to a single scenario sitting 0.11 points below a band boundary. The tool, its configuration, and this document are public under the Apache License 2.0.

Keywords: urban heat island; nature-based solutions; urban cooling; composite indicator; screening assessment; heat vulnerability; decision support; evidence traceability.

Contents

1 Introduction

1.1 Urban heat as a decision problem

Urban populations are exposed to temperatures systematically higher than those of their rural surroundings, and the excess is not uniform across a city. The distinction between the surface, canopy-layer, and boundary-layer urban heat island, established by Oke (1976), remains the necessary starting point for any serious treatment of the subject, because the three are different physical quantities measured in different ways and they do not agree in magnitude. Within a single city, daytime near-surface air temperature can vary by several degrees over short distances: Ziter et al. (2019) measured a mean daytime intra-urban air temperature range of 3.5 °C (spanning 1.1–5.7 °C) and a mean nighttime range of 2.1 °C (1.2–3.0 °C) across a single urban area. Variation of this magnitude, over distances a person walks in minutes, is what makes urban heat a spatial planning problem rather than only a meteorological one.

The consequences of that variation are distributed unevenly across the population. Reid et al. (2009) reduced ten candidate vulnerability variables — six demographic, two describing household air-conditioning, one describing vegetation cover, and one describing diabetes prevalence — to four factors explaining more than 75 % of total variance: social and environmental vulnerability, social isolation, air-conditioning prevalence, and the proportion of elderly and diabetic residents. Two people experiencing the same air temperature do not face the same risk. Any instrument that ranks sites for intervention on physical heat alone is answering a narrower question than the one cities are actually asking.

1.2 Nature-based solutions as a cooling response

Vegetation and water cool urban environments through shade, evapotranspiration, and evaporation, and the empirical evidence for the effect is well established at the scale of individual green sites. Bowler et al. (2010), restricting their meta-analysis to empirical studies, found that an urban park was on average 0.94 °C cooler during the day than its non-green surroundings, with larger parks and parks containing trees cooler than smaller or open ones. Ziter et al. (2019) showed that the relationship between canopy cover and daytime air temperature is nonlinear, with the greatest cooling occurring above approximately 40 % canopy cover, and that the effect is scale-dependent, peaking at radii of 60–90 m — roughly the scale of a city block. The same study found that impervious cover raises temperature approximately linearly and that its warming effect persists overnight, where canopy cooling largely does not.

Aggregating across measures and climates, Keravec-Balbot et al. (2026) conducted an umbrella review of 64 systematic reviews with a multilevel meta-regression of pedestrian-level daytime air temperature reductions. That synthesis is the single most useful cross-typology anchor currently available, and it carries a finding with direct consequences for tool design: the type of intervention alone explains only 12 % of the variance in observed cooling effectiveness, rising to 30 % when climate zone is added and to 78 % once the Köppen–Geiger climate subzone is included. An instrument that reports one global cooling figure per intervention type is therefore discarding the majority of the explanatory signal in the evidence base.

1.3 The decision gap

A city preparing a heat adaptation programme must choose among many more candidate sites and interventions than it can study in detail. In practice it has two options, and neither serves the triage step.

The first is unstructured judgment, often mediated by a spreadsheet. It is fast and cheap, and it draws on genuine local expertise. Its weakness is not that it is wrong but that it is not *comparable*: two officers assessing two sites apply different implicit weights, cite no evidence that a third party can check, and produce numbers that cannot be defended to a funder or a public inquiry. Nothing about the process records why one site outranked another.

The second is detailed simulation — microclimate models of the ENVI-met class, computational fluid dynamics, or building energy simulation. These are the appropriate instruments for design, and they are rigorous. The work of Zölch et al. (2016) illustrates both their value and their cost: a micro-scale ENVI-met evaluation of heat mitigation measures which established, among other things, that trees achieved a mean physiologically equivalent temperature (PET) reduction of 13 % at pedestrian level, that green façades achieved 5–10 %, that green roof effects at pedestrian level were negligible, and — a finding of general importance — that increasing green cover did not correspond linearly to PET reduction, because placement matters more than quantity. Producing such results requires geometry, material properties, meteorological forcing, and specialist time. A city cannot run them across two hundred candidate street segments before deciding which twenty to investigate.

The gap between the two is the screening step: a structured, repeatable, evidence-referenced first pass that narrows a long list to a short one and records why. That is the step this methodology addresses.

1.4 Contribution and structure of this paper

This paper specifies, in full, the methodology of the Nature for Cooling Rapid Assessment Tool at methodology version 2026.08.05. It is written to be read independently of the software, by reviewers who wish to evaluate the scientific basis of the tool's outputs before deciding whether to trust, adopt, or contest them. Its contributions are four.

1. **A complete, auditable specification.** Every formula the engine applies is stated and typeset here, every weight is declared with its justification or with an explicit statement that it is a value choice, and every performance value is traced to a cited source whose verification status is disclosed (Appendix E).
2. **An evidence-derived library with its evidence classes made explicit.** 110 curated catalogue entries each inherit one of 18 *cooling archetypes*, and the archetype — not the entry — carries the envelope, the base score, the evidence rating, and the citations. The reason is stated plainly rather than finessed: solution-specific cooling literature does not exist for 110 intervention types, so an entry inherits a *cited* envelope and every result names the evidence class it inherited from (Section 6.1). Calibration decisions that materially lower the tool's headline numbers — a downward correction for modelling bias, the refusal of super-additivity for multi-intervention packages, the deliberate non-adoption of newer and larger published figures, and conservative

treatment where sources conflict — are presented for scrutiny in Section 6.6.

3. **A discipline of honest uncertainty.** All quantitative outputs are ranges; every output block carries its own confidence rating; missing data reduces confidence rather than being silently imputed; and the temperature-envelope clipping rule prevents favourable site conditions from generating claims that exceed published evidence (Section 7.6).
4. **Reproducibility as an engineering property.** The methodology lives in versioned configuration files rather than in code, and its evidence rules are enforced by the build: an uncited performance value or a citation pointing at an unrecorded source fails continuous integration (Section 10). This is what allows the claims in this paper to be checked rather than merely believed.

Section 2 states what the tool is and is not, and the design commitments that constrain it. Section 3 positions it against published work and argues for the standards by which a screening instrument should be judged. Section 4 presents the three-layer conceptual structure. Sections 5 to 7 specify the inputs, the typology library, and every scoring formula. Sections 8 and 9 cover uncertainty treatment and the executed sensitivity analysis. Section 10 describes the implementation and the machine-enforced evidence rules. Section 11 works one illustrative assessment through by hand. Sections 12 and 13 state the limitations and the governance process. Appendices reproduce the archetype library and the catalogue entries that inherit from it, the complete weight tables, the input field reference, a glossary, and the source verification register.

2 Scope and intended use

2.1 What the tool is

The Nature for Cooling Rapid Assessment Tool is a screening-level, multi-criteria decision-support instrument. It converts a structured description of an urban site and one or more proposed nature-based interventions into comparable scores and indicative quantitative estimates, so that early planning and investment decisions can be made on a transparent, evidence-referenced basis.

A user describes a site through six input groups, selects one or more of the 110 catalogue entries the tool offers for that site (Section 6), and receives a Heat Priority Index, a Cooling Potential Score with an indicative daytime pedestrian-level air temperature reduction range, derived cooling-energy and greenhouse-gas estimates, cost outputs where cost data has been supplied, equity and co-benefit scores, a single NbS Cooling Opportunity Score for comparing options, per-block confidence ratings, and an itemised list of every default the engine applied. Where several interventions are selected, each is scored and reported individually and the package's headline outputs are combined under the rules of Section 7.7, which never sum a temperature. Interventions and packages for the same site can be assessed and compared side by side.

2.2 What the tool is not

The following exclusions are definitional, not caveats to be read past.

- **It is not a microclimate simulation.** It does not solve an energy balance, does not represent site geometry, and does not predict the temperature at a specific location at a specific time. Its outputs are not comparable to ENVI-met-class or CFD results and must never be presented as such.
- **It is not a building energy model.** Cooling-energy savings are derived from an estimated air temperature reduction through a published temperature–electricity-demand sensitivity (Section 7.9). No building is modelled.
- **It is not a planting design or species selection tool.** The evidence that species and leaf area index materially change delivered cooling (Rahman et al. 2020) is used only to justify that canopy condition modulates performance. The tool gives no species guidance.
- **It is not a regulatory, engineering, or approval instrument.** Outputs are indicative ranges for comparison and prioritisation. They are not design specifications and they are not contractual performance guarantees.
- **It does not replace site investigation or local expertise.** It structures a judgment; it does not substitute for one.

2.3 Intended users and decisions

Three user groups are supported: municipal planners and public agencies; developers and consultants; and researchers and non-governmental organisations. The tool is designed to inform three questions:

1. Does this site merit attention at all, given how hot it is and who is exposed?
2. Which of several candidate interventions best suits this particular site?
3. How does a proposed intervention perform simultaneously across cooling, energy, climate, cost, equity, and co-benefit dimensions?

Each of these is a comparative question. The tool is built to rank and to compare, and its outputs should be interpreted comparatively. An Opportunity Score of 72 is meaningful principally in relation to the score another option or another site receives under the same methodology version; it is not a physical measurement.

2.4 The five design commitments

Five commitments constrain the methodology. They are listed here because several of them cost the tool apparent capability, and a reader should understand that those costs were chosen.

Table 1: Design commitments and their consequences for the methodology.

Commitment	**Consequence**
Transparency	Every score has exactly one documented formula, reproduced in this paper. No machine learning, no opaque calibration, no undisclosed adjustment. A user who disagrees with a number can locate the rule that produced it.
Determinism	Identical inputs under an identical methodology version always yield identical outputs. There are no stochastic components anywhere in the scoring or reporting path, including the recommendation text, which is composed from templates rather than generated.
Evidence traceability	Every performance value cites a source. Values that could not be grounded in retrieved literature are either omitted — as unit costs are (Section 7.11) — or shipped with an explicit low-confidence flag.
Honest uncertainty	All quantitative outputs are ranges. Every result carries a confidence rating, computed separately per output block. Missing data reduces confidence rather than being silently imputed, and every applied default is itemised in the result.
Auditability	The methodology lives in versioned configuration files, not in code. Every result records the methodology version that produced it, and the evidence rules are enforced by continuous integration rather than by review convention (Section 10).

The commitment to honest uncertainty is the one that most visibly reduces the tool's apparent authority. A screening instrument that reported single numbers would look more decisive and would be easier to put in a slide. The narrowest adopted cooling envelope spans a factor of three; the widest with a non-zero floor, the large open water body, spans a factor of thirty because remote sensing and field measurement disagree by an order of magnitude (Section 6.7.1); and four archetypes carry a genuine zero floor, for which no ratio is defined at all. The climate-differentiated estimates in Keravec-Balbot et al. (2026) carry uncertainty intervals that in several cases approach or exceed ± 1 °C, and in the case of street trees in arid climates reach ± 1.4 °C on a central estimate of 1.7 °C; a point estimate would imply a precision the evidence does not support. The ranges are the honest representation of what is known.

3 Related work and positioning

3.1 The closest published antecedent

Norton et al. (2015) is the closest published antecedent to this methodology. It sets out a framework for prioritising and selecting urban green infrastructure to mitigate high temperatures, built on a review of the relationships between urban geometry, green infrastructure, and temperature. Its logic — assess the thermal condition of a place, consider who is exposed, then determine which intervention is suited to that place — is the logic this tool implements, and the three-layer structure of Section 4 follows it directly.

The verification status of this source should be stated plainly, because the paper leans on it for framing. The bibliographic record was confirmed from the Monash University research portal, and the framework description was confirmed from an authoritative secondary description rather than from the full text. The tool takes no numeric value from Norton et al. (2015); it takes a structural argument. Appendix E records this alongside every other source.

Two differences from that antecedent are worth stating. First, this tool is an operational instrument with a fixed, versioned, machine-readable parameterisation rather than a conceptual framework, which means every judgment it embodies must be written down as a number and can therefore be disputed as a number. Second, it treats climate as a first-order term rather than as context, for the empirical reason given in Section 3.4.

3.2 Composite-indicator practice

The tool is, structurally, a composite indicator: a set of normalised sub-indicators combined by weighted aggregation into a headline score. The construction sequence adopted here — theoretical framework, variable selection, treatment of missing data, normalisation, weighting and aggregation, and presentation — follows Nardo et al. (2008), the standard reference for this class of instrument.

Two of that handbook's positions are load-bearing for this paper and are adopted without qualification.

The first is that **weighting is a value choice, not a technical derivation**. No literature can supply the correct relative importance of cooling performance against equity against cost, because that ratio expresses a policy preference rather than a fact about the world. The aggregation weights in Section 7.12 are therefore declared as expert judgment, versioned, adjustable in configuration, and are to be defended empirically through the sensitivity analysis specified in Section 9 rather than by appeal to authority. Presenting them as derived would be the more persuasive move and the dishonest one.

The second is that **uncertainty and sensitivity analysis are integral to a credible composite indicator, not optional additions**. Because the aggregation weights are the tool's most consequential unsourced choice, their influence on the output must be quantified and published rather than asserted to be small. Section 9 specifies that analysis in full and Section 9.4 reports its results at version 2026.08.05.

3.3 The risk framing

The Baseline layer follows the standard climate-risk decomposition, in which risk arises from the interaction of a *hazard* with *exposure* and *vulnerability*. This is why a physically hot site with no exposed population scores differently from an equally hot site surrounded by vulnerable residents, and it is the reason the tool computes a Heat Priority Index rather than reporting heat alone.

The decomposition is applied concretely: the Heat Exposure Score (Section 7.2) represents hazard and physical exposure, drawing on surface-energy-balance drivers for which Ziter et al. (2019) supplies direct empirical support; the Vulnerability Score (Section 7.3) represents population susceptibility, with its indicator selection grounded in Reid et al. (2009); and the two are combined into a single priority index (Section 7.4).

3.4 Climate as a first-order term

The tool's most consequential departure from common screening practice is its refusal to publish a single global cooling number per intervention type. The justification is empirical. Keravec-Balbot et al. (2026) reports that the type of solution alone explains 12 % of the variance in observed cooling effectiveness; adding climate zone raises this to 30 %; and adding the Köppen–Geiger climate subzone raises it to 78 %.

Figure 1 shows what that variance decomposition looks like at the level of individual measures. For green walls, the same synthesis that reports a 3.0 °C global central estimate reports subzone-specific estimates ranging from (5.7 ± 1.0) °C in Cwa and (5.5 ± 1.0) °C in BWh down to 1.3 °C in Cfa/Cfb and (1.0 ± 0.8) °C in Af — an approximately fivefold spread within a single measure. For street trees, the same review reports (4.0 ± 1.6) °C in tropical (A) climates against (1.7 ± 1.4) °C in arid (B) climates.

A tool reporting one number per measure would discard this. Climate suitability is consequently a first-order adjustment in the tool's Layer 2 (Section 7.5), not a refinement applied afterwards.

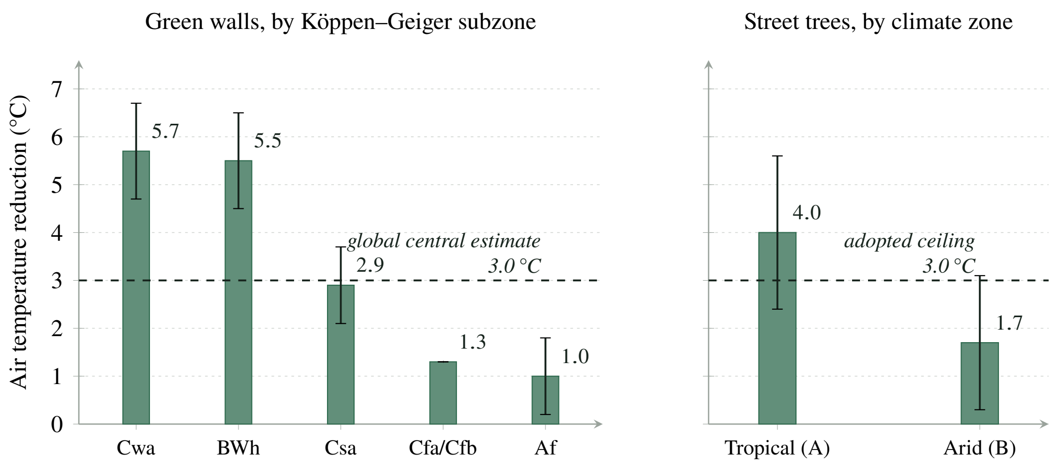

Figure 1: Climate dependence of reported cooling effectiveness, from the umbrella review of Keravec-Balbot et al. (2026). All values are daytime pedestrian-level *air* temperature reductions. Error bars show the uncertainty the source states; no bar is drawn for Cfa/Cfb, which this methodology's evidence tables record as a single combined central estimate without a stated interval. The dashed line marks 3.0 °C: the global central estimate for green walls in the left panel, and the ceiling this methodology adopts for any single site-level intervention in the right panel. The spread within each measure is the empirical reason climate suitability is treated as a first-order adjustment rather than as context.

3.5 By what standard should a screening tool be judged?

A recurring failure in the reception of screening instruments is that they are evaluated against simulation standards, which they cannot meet and were not built to meet. If the question is "does this reproduce ENVI-met to within 0.2 °C?", the answer is no, and it would be no for any instrument that asks forty-five questions and runs in seconds. That is not a finding about this tool; it is a restatement of what a screening instrument is.

The productive questions are different, and this paper invites reviewers to apply them.

1. **Is every value traceable?** Can a reader locate the source of every number the tool applies, and see what was actually verified about that source? Appendices A and E exist to make this checkable.
2. **Are the uncertainties honestly represented?** Does the tool report ranges that reflect the spread in the underlying evidence, and does it lower its stated confidence when it knows less? Section 8 specifies the mechanism.
3. **Is the ranking stable?** Because prioritisation is the tool's actual function, the relevant question about the weights is not whether they are correct — they are a value choice — but whether the ordering they produce survives reasonable perturbation. Section 9 specifies how this is measured, and Section 9.4 reports the answer.
4. **Are the value judgments disclosed?** Where the methodology takes a position that a reasonable person could reject — most notably the equity-forward weighting of Section 7.13 — is that position stated openly and quantified, or is it buried in an aggregation?
5. **Does it refuse to answer when it should?** An instrument that always produces a number is less trustworthy than one that reports *not estimated* when its inputs do not support an estimate. The cost policy of Section 7.11 and the missing-data rules of Section 8.4 are deliberate refusals.

A screening tool that satisfies these five is doing its job. One that fails them is not rescued by agreeing with a simulation on a test case.

4 Conceptual framework

4.1 Three layers, three questions

The methodology is organised into three layers, each answering a distinct question that a planner asks in sequence. The structure is shown in Figure 2.

Layer 1 — Baseline assessment: ***does this site deserve attention?***
A Heat Exposure Score and a Vulnerability Score are computed from the site description and combined into a Heat Priority Index. This layer is independent of the proposed intervention: it characterises the place, not the project. Two options assessed for the same site share an identical Layer 1 result, which is what makes side-by-side comparison meaningful.

Layer 2 — NbS performance: ***what can this intervention deliver here?***
A base value drawn from the typology library is modulated by a site adjustment factor derived from conditions the user has already described. This layer produces the Cooling Potential Score and the indicative temperature reduction range. It is where the tool's evidence base does its work, and where the clipping rule of Section 7.6 constrains what may be claimed.

Layer 3 — Impact and feasibility: ***what does it cost, and what else does it do?***
Cooling-energy savings are derived from the Layer 2 temperature estimate and converted to avoided greenhouse-gas emissions; capital cost, where supplied, yields payback and a derived cost-feasibility score; and co-benefit and equity scores capture what a purely thermal assessment would miss.

The three layers are then aggregated into a single NbS Cooling Opportunity Score, accompanied by per-block confidence ratings, any suitability or evidence flags, and a deterministically composed recommendation.

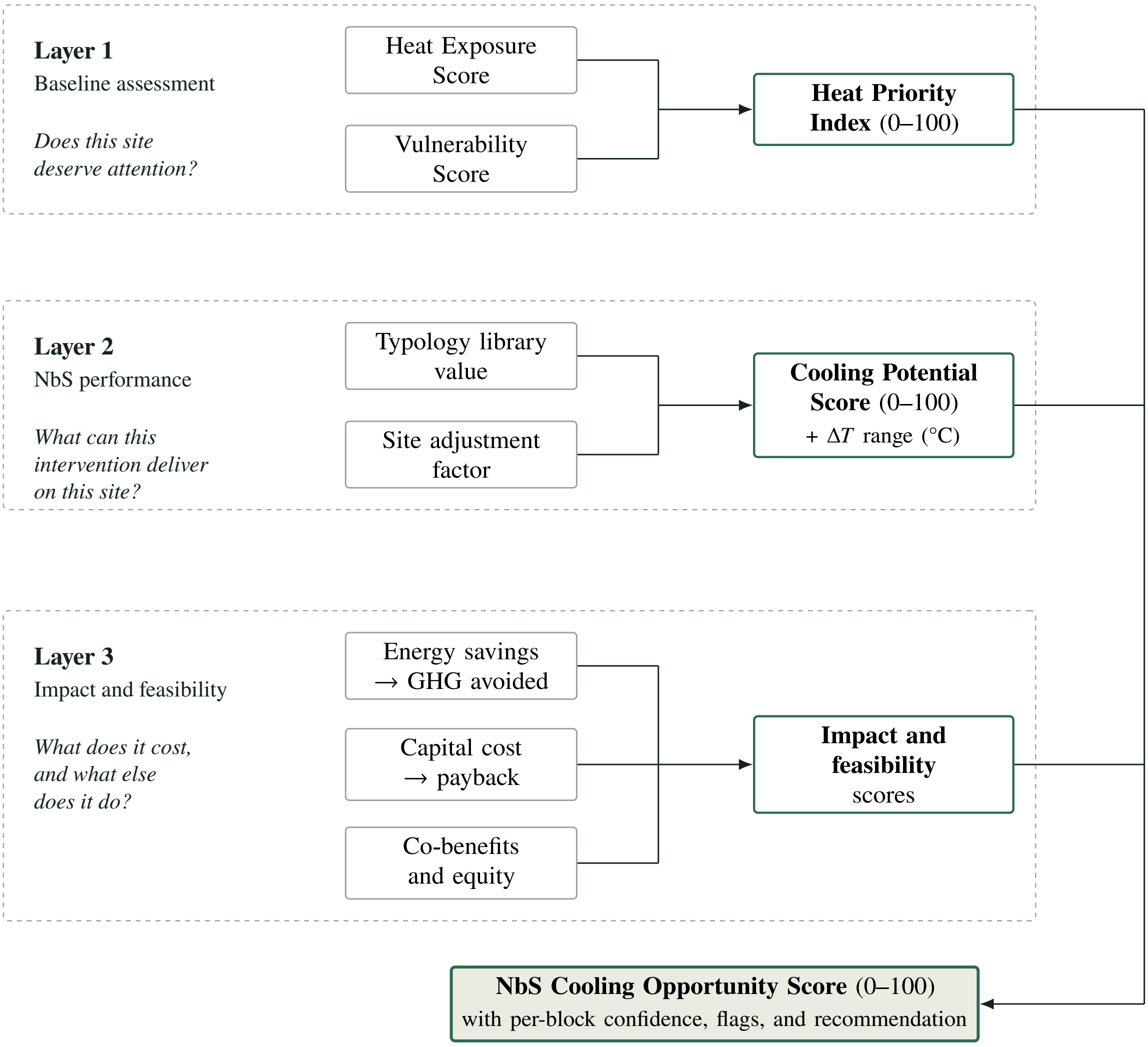

Figure 2: The three-layer conceptual framework. Layer 1 characterises the site independently of any intervention; Layer 2 evaluates one proposed intervention against that site; Layer 3 adds economic, climate, and social dimensions. The layers are aggregated by the weighted sum of Section 7.12. Because Layer 1 does not depend on the intervention, two options assessed for the same site differ only from Layer 2 onward.

4.2 Why this decomposition

The separation is not merely presentational; it has three consequences that matter for how outputs should be read.

It separates the place from the project. A high Heat Priority Index with a low Cooling Potential Score identifies a site that badly needs intervention but for which the proposed intervention is poorly matched — a result that a single blended score would conceal. This pattern is common and decision-relevant: it argues for a different intervention, not for abandoning the site.

It keeps the evidence base localised. Layer 2 is where published cooling evidence enters, and it enters in one place: the typology library of Section 6. Revising the evidence base means revising that library and its adjustment rules, not tracing consequences through the whole methodology. This is what makes the version-controlled configuration described in Section 10 tractable.

It makes the confidence structure natural. A site typically has good physical data and poor economic data, or the reverse. Because the layers are distinct, confidence can be computed and reported per block (Section 8.3) rather than collapsed into a single figure that would be dominated by whichever block happened to be worst.

4.3 From inputs to outputs

Figure 3 shows the same structure as an implementation data flow. Three boundaries in that diagram carry the design commitments of Section 2.4 into the software.

First, methodology values reach the engine only as configuration. The scoring code contains no typology value, no weight, and no threshold; all of these live in `config/*.yaml` and are validated on load. Changing the methodology never requires changing code, and the methodology version travels with the result.

Second, the engine is a pure function of its inputs and that configuration. It performs no input or output, makes no network call, holds no global state, and contains no stochastic component. Determinism is therefore a structural property rather than a testing aspiration.

Third, every output is a range or a categorical judgment accompanied by its confidence, and every default the engine applied is itemised in the result. A reader of a report can see exactly which numbers came from the user and which from the tool.

5 Inputs and normalisation

5.1 Input structure

Inputs are collected in six groups: project information; site characteristics; climate and heat exposure; vulnerability and equity; the proposed NbS intervention; and cost and energy parameters. The full field reference, with required or optional status and the normalisation applied to each field, is reproduced in Appendix C.

Two structural changes arrive at version 2026.08.05. The intervention field `nbs_type` is now a *list*: an assessment may propose a package of simultaneously available entries, scored under Section 7.7. And the site-characteristics group gains four questions that decide which interventions are offered and feed no score at all (Section 5.6).

Every input is either *required*, in which case the assessment cannot proceed without it, or *optional*, in which case its absence reduces confidence but never blocks the assessment. The interaction design anticipates roughly forty-five inputs, of which it expects about twenty to be mandatory; the implemented input schema requires exactly the four fields that a documented formula cannot be evaluated without, and Appendix C records the status and normalisation of every field at version 2026.08.05. Reducing the total count would not reduce uncertainty; it would move uncertainty from

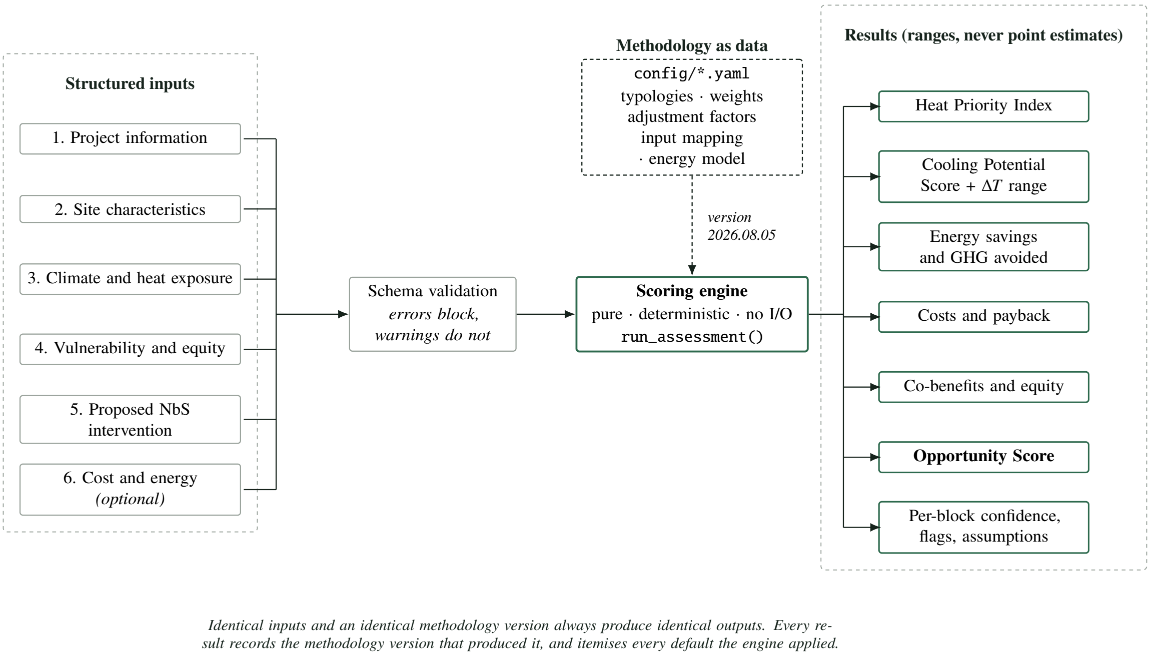

Figure 3: Data flow from the six input groups through validation and the scoring engine to the result. Methodology values enter only as versioned configuration (dashed path); the engine is a pure, deterministic function of inputs and configuration; and every output is reported as a range or a categorical judgment with its own confidence rating. Validation distinguishes errors, which block an assessment, from warnings, which do not.

visible — a stated confidence level — to invisible — a silent default. The interface consequently asks everything, makes skipping free, and shows the user what skipping costs.

5.2 Qualitative normalisation

Most inputs accept a qualitative level, because most users at screening stage do not have measured data. Levels map to a 0–100 scale as shown in Table 2.

Table 2: Normalisation of qualitative input levels. The standard scale applies where a higher level indicates higher risk or higher quantity; the inverted scale applies where a higher level indicates *lower* risk.

Level	Standard scale	Inverted scale
Low	25	100
Medium	50	50
High	75	25
Very high	100	—
Unknown	50	50

Three fields use scales of their own, because their natural vocabularies are not the four standard levels. Soil availability maps *none* → 25, *limited* → 50, *moderate* → 75, *high* → 100. Irrigation availability maps *none* → 25, *occasional* → 50, *reliable* → 100. Current shade level maps *very* *low* → 25, *low* → 50, *medium* → 75, *high* → 100. In each case *unknown* maps to 50.

5.3 The inverted mapping for cooling access

The principal application of the inverted scale is access to cooling refuge, measured by two fields — access to cooled *indoor* space and access to a cool *outdoor* refuge — where in each case *high* access maps to 25, *medium* to 50, and *low* to 100. Both enter the Vulnerability Score (Section 7.3) through a single *cooling access deficit* term: low access indicates higher vulnerability, so it must produce a higher score.

The justification is Reid et al. (2009). In that factor analysis of ten heat vulnerability variables, air-conditioning prevalence emerged as one of four factors explaining more than 75 % of total variance — and, importantly, as an *independent* dimension. That independence is what justifies giving cooling access its own term in the vulnerability formula instead of allowing it to be absorbed into a general deprivation indicator: a factor that emerges separately alongside the social and environmental vulnerability factor is not a restatement of it. That last step is this methodology's inference from what a factor analysis means, not a claim the source itself makes.

Two qualifications should be read alongside this. The study population was North American, and access to mechanical cooling has a different distribution and a different meaning in cities where air conditioning is rare across all income groups. And the tool asks about access to cooled indoor space generally rather than about air-conditioning ownership specifically, which is a deliberate broadening — a shaded, ventilated, publicly accessible cool refuge counts — but a broadening that the underlying study does not itself validate.

The second field, added at version `2026.08.03`, addresses part of the first qualification directly. Where mechanical cooling is rare, refuge from heat is found outdoors, and the evidence for that is its own: Burkart et al. (2016) and Sera et al. (2019) measure reachable vegetation and water as modifiers of heat mortality rather than inferring them. The combination rule for the two indicators, and the reason an unanswered indicator is excluded rather than averaged in at the neutral 50, are given with the formula in Section 7.3.

5.4 The treatment of *unknown*

Assigning *unknown* a neutral score of 50 is a deliberate and disclosed choice. Three alternatives were available and each carries a bias.

Treating an unknown as the best case flatters sites with poor data; treating it as the worst case penalises them; and imputing a value from other fields introduces a correlation the user never asserted and cannot see. The neutral midpoint avoids both optimistic and pessimistic bias in the score, at the cost of pulling every score toward the middle in proportion to how much is unknown.

That cost is not concealed. The confidence mechanism of Section 8.3 records, per output block, how much of the relevant input set was actually supplied, so a score dominated by neutral defaults is reported at low confidence. The two mechanisms are complementary: the score says what the evidence supports, and the confidence says how much evidence there was. Neither alone is sufficient, and quoting a score without its confidence rating is identified in Section 12.11 as a misuse of the tool.

5.5 Land surface temperature as an optional proxy

Where a user can supply a land-surface-temperature (LST) anomaly — degrees Celsius above the city mean, typically derived from satellite imagery — the tool converts it linearly to a 0–100 score, with 0 °C mapping to 0 and 10 °C mapping to 100, clamped at both ends:

$$S_{\mathrm{LST}} = \mathrm{clamp}(10 \times a_{\mathrm{LST}})\,, \qquad (1)$$

where a_{LST} is the supplied anomaly in degrees Celsius and clamp bounds its argument to $[0, 100]$. The 10 °C ceiling is consistent with observed intra-urban surface anomaly ranges while keeping the normalisation interpretable.

This proxy is used under a stated restriction, and the restriction is the most important sentence in this subsection. Land surface temperature is not air temperature, and the two diverge by a large and systematic factor.

Venter et al. (2021) contrasted crowdsourced air temperature measurements against satellite estimates and found the satellite-derived *surface* urban heat island to substantially exceed the air-temperature (*canopy-layer*) urban heat island: a reported mean surface UHI of 1.45 °C against a canopy-layer UHI of 0.26 °C, an approximately sixfold difference. This reflects the physical distinction between surface, canopy-layer, and boundary-layer heat islands established by Oke (1976).

The tool therefore treats LST **only as a relative indicator of which parts of a city are hotter than others**. It is never interpreted as an air-temperature value, never converted into a °C outcome, and never compared against the tool's own air-temperature reduction estimates. It influences only the ranking of sites within the Heat Exposure Score.

The verification status of Venter et al. (2021) is relevant here: the bibliographic record was confirmed from the ADS record and an author-hosted PDF, and the specific surface-versus-canopy comparison was confirmed from an indexed secondary description, publisher full text being blocked. The claim the methodology rests on — that surface UHI substantially exceeds canopy-layer UHI — is corroborated conceptually by Oke (1976), and the tool's use of the finding is qualitative (treat LST as relative only) rather than numerical, so no tool output depends on the precise sixfold ratio.

5.6 The four availability questions: inputs that gate but never score

Version 2026.08.05 adds four site-characteristic questions. They are unlike every other input in this section, and the difference is the point of this subsection.

Table 3: The four availability questions added at version 2026.08.05. Each decides which of the 110 catalogue entries the tool offers for this site. None of them appears in any formula in Section 7, in any confidence block of Section 8.3, or in any aggregation.

Field	Answers	Gates
`includes_railway`	yes / no	Exactly one entry, the railway green corridor
`existing_woodland`	yes / no	Woodland *restoration* entries only
`waterfront_type`	lake / river / dry river / covered river / sea	Twelve water-landscape and ecological-network entries, by category
`productive_governance`	multi-select over community / individual / institutional / commercial	The seven productive entries, by who delivers them

Gating is not scoring. These four fields decide which interventions the tool *offers*, and they feed no score.

Making a gating answer a scoring input would need its own justification and would move every existing result. The engine's test suite asserts the separation as a property — the same site scored with all four answered and with none answered produces byte-identical output — so the distinction cannot erode quietly through a later edit. The fields are recorded in the input schema and disclosed as availability-only in Table 23.

What an unanswered question means differs by kind, deliberately. Three of the four describe a *physical fact* about the site, and they gate on positive confirmation: an entry that acts on an existing river is not offered until the user says there is a river, because offering it would be offering an intervention that cannot be built there. The fourth describes *who could deliver* the intervention, and it subtracts nothing until it is answered.

That asymmetry is deliberate. The question originally specified for productive landscapes was a yes/no — is there interest in creating productive landscape from individuals or communities — which on a "no" would have suppressed the urban farm and the agroforestry system: the highest-canopy, highest-cooling entries in the group, and the ones a municipality is most likely to deliver. A multi-select over four delivery models gates correctly at no extra cost in user effort, and an *empty* selection is not evidence that no delivery model exists. The tool does not assert a negative from absent information, here as in the soil–water pair of Section 7.5 and the suitability rules of Table 9.

Two consequences follow from the same principle and are worth stating because both run against intuition. **Woodland *creation* entries are ungated**: urban woodland, microforest, afforestation, and the woodland buffer are offered whatever the answer, because creating woodland does not require woodland to be there already — only restoration does. And **the four constructed water features carry no waterfront condition at all**: a constructed wetland, retention pond, detention basin, or water square needs no existing water body, so gating them on one would withhold exactly the interventions available to a site that has no water.

Land use maps to interventions rather than naming one. The same mechanism carries the retirement of Section 6.4. 72 of the 110 entries name *school* in their land-use list, 93 name *campus*, 76 *healthcare*, and 41 *memorial*; the assessment scale then narrows the menu further, so a school *site* is offered 67. All four land-use totals and the nine published situation counts are asserted by the test suite, so a land use silently losing its interventions fails the build. Memorial is deliberately the sparsest: tree groves, meadows, woodland, and quiet permeable planting are in; plazas, playgrounds, tram corridors, living walls, and productive landscapes are out. All four counts are computed with the governance question unanswered, which is the governance-agnostic reading; answering it reduces them, and both readings are correct because they answer different questions.

5.7 Where the tool has no LST

Most screening-stage users will not have an LST anomaly. The methodology therefore provides a second path through the Heat Exposure Score (Section 7.2) that requires only a qualitative heat exposure level. The two paths are not equivalent in information content, and the confidence mechanism reflects that. The data-poor path is not a degraded mode to be avoided; it is the expected mode for most assessments, and the tool is designed to give a defensible answer in it.

5.8 Inputs a map may derive, and the many it may not

Version 2026.08.05 adds an optional map step at the head of the questionnaire. It is skippable in one action and the questionnaire is fully usable without ever opening it; nothing in this subsection applies to an assessment that did not use it.

Exactly three inputs can be derived from a point or a polygon on a map without inventing anything. `site_area_m2` is pure geometry: the geodesic area of the ring the user drew, evaluated on the authalic sphere, carrying no uncertainty beyond the drawing itself. `country` is a point-in-polygon test against public-domain boundaries (Natural Earth 2022), deterministic given the data, and feeds only the emission-factor and currency defaults. And `climate_zone` is a cited lookup plus a documented mapping: the classification is published (Beck et al. 2023), and the mapping of its thirty classes onto this tool's six zones is a methodology value in its own right, derived below.

Everything else the questionnaire asks about the site is *not* derived from the map, and the reason is the whole point of this subsection. Canopy cover, green cover, imperviousness, LST anomaly, land use, shade level, population density and vulnerable-population presence all require satellite or census data. The temptation is worth naming because it is strong: a map that filled in canopy cover from a quick NDVI lookup would demonstrate well. It would also produce this tool's *most decision-relevant site inputs* from an unvalidated pipeline, with no evidence table behind it and no confidence treatment—which is precisely the defect the methodology refused when it declined to ship default cost values (Section 7.11), appearing in a new place. The restriction is enforced at the service boundary rather than left to the interface, so it holds as the questionnaire grows.

Mapping Köppen–Geiger onto six zones

The classification is not ours; the mapping is. Thirty classes onto six zones is a judgement that selects a row of the climate adjustment matrix (Section 7.5) and therefore changes results, so it is declared as a methodology value with its own citation, derivation and version rather than buried in the picker's code. It follows four principles, and every one of the thirty assignments records which principle produced it, because a thirty-row table of individual opinions could not be reviewed.

Tropical Group A splits on its dry season, which is the same split the tool's two tropical zones name: `Af` and `Am` to `tropical_wet`, `Aw` to `tropical_dry`.

Arid Group B splits on desert (BW) versus steppe (BS) and the tool on arid versus semi-arid; these are the same axis. The hot/cold third letter is deliberately ignored, because the tool's arid downgrade encodes water limitation constraining evapotranspiration, not temperature—a cold desert limits evapotranspiration as a hot one does.

Warm season Groups C and D whose third letter is `a` or `b` map to `temperate`. This is where the mid-latitude cooling literature was measured. Group D is *not* sent to `other` wholesale: Chicago is `Dfa`, Seoul `Dwa` and Moscow `Dfb`, all three have serious urban heat problems, and all three sit inside the evidence base behind the temperate row.

No warm season Groups C and D whose third letter is `c` or `d`, and all of group E, map to `other`—fewer than four months above 10 °C, or none. There is no warm season for the cooling literature to describe.

Mapping the polar and boreal classes to `other` rather than to `temperate` matters more than it may appear. `other` is already the tool's neutral climate condition—every family resolves to condition *unknown*, factor 1.0—so an unclassifiable location receives neither a boost nor a penalty. Forcing them into `temperate` would assert that a tundra site cools like a temperate one, which no source in this bibliography supports, and *an uncited claim introduced through a lookup table is still an uncited claim*.

The bundled classification layer is 0.1°, roughly 11 km. That resolution is stated in the interface and in the report wherever a value derived from it appears: the classification is reliable for the city a site sits in, not for the site itself, which is the resolution at which climate zone enters this methodology in any case, since it selects a row of an adjustment matrix rather than a temperature.

An autofilled value counts as supplied, and says so

The confidence mechanism (Section 8.3) counts what the tool was *told*, not how hard the user worked to tell it. A Köppen–Geiger lookup at a location the user chose is a cited classification, not a guess, so it counts as an answer. Not counting it would mean that clicking the map and typing the same climate zone by hand produced *different confidence readings for identical inputs*, teaching users that the confidence meter measures effort rather than information.

The value therefore counts and the provenance is disclosed instead: beside the field in the interface, in the stored input, and itemised in the report's Inputs sheet, which distinguishes user-supplied, autofilled and defaulted values. Autofill never overwrites an answer the user has already given, and overriding a value removes its marking. In practice none of the three fields appears in any confidence block, so the question is one of principle rather than arithmetic—no confidence figure moves either way.

6 The NbS typology library

6.1 Two levels: cooling archetypes and catalogue entries

From version 2026.08.05 the library has **two levels**, and the separation between them is the central design decision of this release.

A **cooling archetype** is a cited evidence class. It carries every performance value — base cooling score, temperature reduction envelope, evidence confidence rating, suitability conditions, default co-benefit levels — together with the citations behind them. There are 18.

A **typology** is one of the 110 curated entries of the UNEP *Nature for Cooling* catalogue. It carries identity, family, availability, the primary cooling mechanism the catalogue ascribes to it, and the one archetype it inherits. It carries **no performance value of its own**.

The reason is worth stating without euphemism: solution-specific cooling literature does not exist for 110 typologies. Nobody has separately measured a felt-pocket living wall, a suspended-pavement tree pit, and a green parking lane.

The alternative to archetypes was inventing 110 envelopes. That would have manufactured precision the source catalogue does not contain, and it would have broken the evidence rule the tool exists to honour — that no performance value ships without a citation (Section 10.4).

Under the archetype model an entry inherits a *cited* envelope, and every result **names the evidence class it inherited from**. A Miyawaki forest reports its cooling on the dense-canopy evidence, and says so, rather than implying a measurement of Miyawaki forests that does not exist.

The catalogue's 243 entries were curated to 110 — 88 merged, 45 dropped — on the source document's own classification notes. Nearly every drop is an entry the document itself disqualifies as not an intervention: a land-use context, a management objective, an umbrella category, a planning model. Nothing is lost by dropping them. A land-use context is already carried by the `land_use` input, and a strategy is a *package* of kept entries, which Section 7.7 makes expressible for the first time.

The 110 entries fall into fourteen catalogue families — tree-based, street, park, public space, woodland and forest, ground level, water landscape, ecological network, district, green roof, vertical greening, building, vegetated shelter, and productive landscape — and are classified by the source document as 53 *elements* (a discrete thing that is built) and 57 *composites* (a coherent spatial system made of elements). Neither classification enters a formula; both are reported, because a reader comparing the tool's menu with the catalogue needs to see which is which.

6.2 The metric, stated once and enforced throughout

All adopted temperature values in this library are **daytime, pedestrian-level *air* temperature reductions**, relative to an equivalent unimproved site.

They are **not** land surface temperatures, and they are **not** thermal comfort indices such as the physiologically equivalent temperature (PET) or the universal thermal climate index (UTCI). These are different physical quantities, measured differently, with different magnitudes and different policy meanings.

Where a source reports surface temperature or a comfort index, that is stated explicitly, and the value is used only for the *relative ranking* of intervention families or to support a caveat. It is never converted into a °C air-temperature claim.

This discipline is not pedantry. Conflating the three metrics is the single most common way screening instruments mislead their users, and the divergence is large: Venter et al. (2021) report an approximately sixfold difference between surface and canopy-layer heat island magnitudes, and Zölch et al. (2016) report comfort improvements from trees (13 % PET reduction) that have no direct air-temperature equivalent. A decision-maker reading "3 °C of cooling" assumes air temperature. The methodology gives them air temperature, and where it cannot, it says so.

Two of the archetypes retrieved for this version exist *because* a source measured both quantities on the same plots and the two diverged by an order of magnitude (Sections 6.7.3 and 6.7.4). The discipline is therefore not only a presentational rule; it changed the library.

A direct consequence is that the tool systematically *understates* the benefit of shade-dominant interventions, because radiant load reduction — the mechanism through which shade most improves human thermal comfort — does not appear in an air-temperature figure. This is recorded as a limitation in Section 12.3 rather than corrected by an unsourced adjustment.

6.3 The eighteen archetypes

Table 4 gives the adopted values for all eighteen cooling archetypes and the number of catalogue entries that inherit each. Derivations are summarised in Section 6.5 and reproduced in full, with citations, in Appendix A.

Twelve of the eighteen archetypes are the previous version's typologies, unchanged — same envelopes, same base scores, same evidence ratings, same citations. They are the only classes in the library with individually retrieved literature behind them, and this release keeps that evidence rather than trading it for catalogue names.

No previously published cooling envelope moved in this release, with one exception: the newly retrieved large water body ships at a floor of 0.1 °C rather than the 0.5 °C proposed at design time, because source verification showed the field evidence did not support the higher floor. Section 6.7.1 tells that story in full, because it is a story about how the methodology is meant to work rather than an erratum to be buried.

How the base cooling score of a new archetype was placed. The base score is a relative 0–100 strength index, not a physical quantity, and the twelve inherited scores are not a function of the

Table 4: The eighteen cooling archetypes with their adopted values at methodology version 2026.08.05. All temperature envelopes are daytime, pedestrian-level *air* temperature reductions in degrees Celsius. The base cooling score is a 0–100 index expressing relative expected cooling performance before site adjustment; it is not a temperature. Evidence confidence is the rating defined in Section 8.2. *Provenance* records where the archetype's values come from: *v1.1* means the value is the previously published typology value, unchanged; *new* means the evidence was retrieved for this version; *derived* means no class-specific evidence exists and the envelope is bounded from an adjacent class with the bounding argument stated. *Entries* is the number of the 110 catalogue entries inheriting that archetype.

Archetype	**Provenance**	**Base score**	**Envelope (°C)**	**Confidence**	**Entries**
Green					
Street tree canopy	v1.1	75	0.5 – 3.0	High	15
Continuous shaded corridor	v1.1	80	0.5 – 3.0	Medium	2
Small green area with trees	v1.1	65	0.3 – 1.5	High	6
Dense tree canopy	v1.1	90	1.0 – 3.0	High	10
Established park	v1.1	70	0.5 – 2.0	High	10
Non-canopy vegetation	new	50	0.0 – 1.5	Medium	7
Productive tree canopy	derived	60	0.5 – 2.0	Low	3
Productive ground planting	derived	45	0.0 – 1.5	Low	6
Building					
Extensive green roof	v1.1	45	0.1 – 1.0	Medium	7
Green façade / living wall	v1.1	50	0.3 – 2.0	Low	11
Blue-green					
Bioretention	v1.1	55	0.1 – 0.8	Low	8
Blue-green corridor	v1.1	85	1.0 – 3.0	Medium	5
Riparian restoration	v1.1	85	1.0 – 3.0	Medium	6
Large open water body	new	75	0.1 – 3.0	Medium	3
Small constructed water	new	40	0.0 – 1.0	Medium	6
Hybrid					
Permeable shaded hardscape	v1.1	70	0.5 – 2.5	Medium	2
Enclosed courtyard greening	v1.1	60	0.3 – 1.5	Low	1
Vegetated shade structure	new	55	0.0 – 2.5	Low	2
Total entries					**110**

envelope midpoint — bioretention sits at 55 with a 0.45 °C midpoint while the green roof sits at 45 with a 0.55 °C midpoint. There is therefore no formula to apply, and inventing one would have moved twelve calibrated values in order to make six new ones look derived. Each new score is instead placed against a *named comparator already in the library*, with the argument stated where the archetype is derived below and in Appendix A.

6.4 Two retirements

Two typologies published at earlier versions no longer name a selectable intervention. Both are methodology changes rather than renames: each carried cited values and each appeared in a golden scenario.

Schoolyard greening is retired. The source document is explicit that a schoolyard is a *land-use context*, not an intervention. Keeping it would have meant asking the user "what is the land use here?" and then offering, as an intervention, the land use they had just described. A school site is instead offered the real interventions that suit it — 72 of the 110 entries name a school among their land uses — through the availability matrix of Section 5.6. The evidence that supported the retired typology is the same evidence supporting the small green area with trees, which the catalogue's green playground entry inherits; nothing was discarded except the duplicated question.

The mixed NbS package is retired in favour of real packages (Section 7.7). The finding it encoded — that no retrieved source quantifies super-additive cooling — has not changed and is not re-argued. What changed is that the tool no longer needs an opaque averaged card to express it: a user now selects the actual components, each is scored and shown individually, and the combination rules do the work the averaged card was standing in for.

6.5 How envelopes were derived

Figure 4 shows the principal cross-typology anchor: the per-measure daytime central estimates reported by Keravec-Balbot et al. (2026), with two contrasting estimates overlaid.

Three principles governed the derivation of an envelope from these estimates.

An envelope is not a prediction. The lower bound represents a poorly performing instance of the archetype; the upper bound represents a well-executed instance under favourable conditions. Site adjustment moves an assessment *within* this envelope (Section 7.6); it does not extend it.

Central estimates are ceilings, not midpoints. Because published central estimates already describe interventions that were built, studied, and in many cases modelled under favourable assumptions, adopted upper bounds sit at or below the synthesis central estimate rather than above it.

Where sources disagree, the conservative value is adopted and the disagreement is documented rather than averaged away. Three archetypes — the green roof, the green façade, and the large open water body — rest on genuinely conflicting evidence, and all three are treated below.

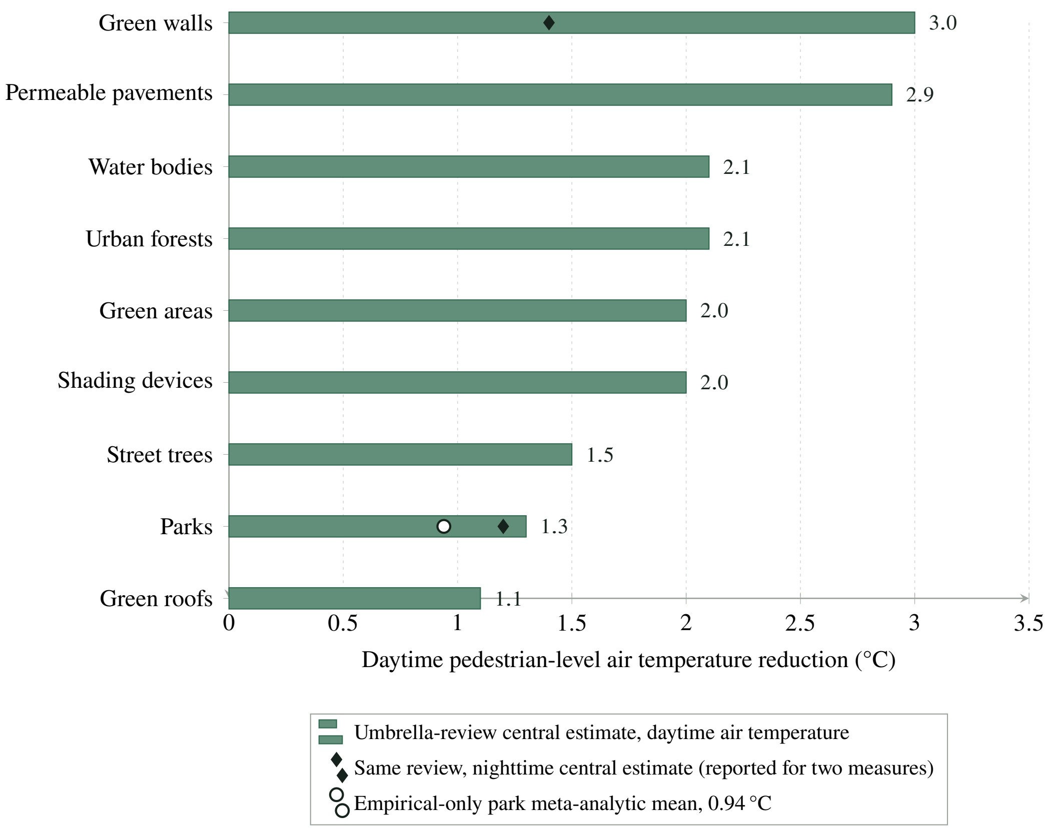

Figure 4: Central estimates of daytime pedestrian-level *air* temperature reduction by measure, from the umbrella review of 64 systematic reviews by Keravec-Balbot et al. (2026). Overlaid markers show, where an alternative estimate exists for the same measure, the nighttime central estimate from the same review and the empirical-only park meta-analytic mean of 0.94 °C reported by Bowler et al. (2010). The gap between the park bar (1.3 °C, modelling-inclusive) and the open marker (0.94 °C, empirical only) is the modelling bias discussed in Section 6.6.1; the diamonds show how much smaller nighttime effects are (Section 12.2).

6.6 Calibration decisions that warrant reviewer attention

Four decisions materially lowered the tool's headline numbers relative to a naive reading of the literature. Each is stated here because each is contestable.

6.6.1 Modelling bias is corrected downward

Keravec-Balbot et al. (2026) synthesises 64 systematic reviews, many of them dominated by simulation studies, and simulation systematically reports larger cooling effects than field measurement. The magnitude of this bias is visible within the tool's own source set. For parks, the modelling-inclusive synthesis gives a central estimate of 1.3 °C, while Bowler et al. (2010), whose meta-analysis is restricted to empirical studies, gives 0.94 °C — roughly 72 % of the modelling-inclusive figure, for the one measure where both are available.

Adopted upper bounds therefore sit at or below synthesis central estimates rather than above them. The clearest case is the street tree canopy: the synthesis reports (4.0 ± 1.6) °C for street trees in tropical climates, but the adopted ceiling is 3.0 °C. The tropical figure carries a ±1.6 °C interval and describes favourable tropical conditions; adopting it as a general ceiling would overstate typical performance.

This correction is a judgment, not a calculation. A single ratio observed for a single measure is not a bias correction factor, and the methodology does not present it as one — it is used to justify the *direction* of the adjustment, not its size. A reviewer who considers the correction too aggressive, or not aggressive enough, is disputing exactly the right thing.

6.6.2 Packages do not claim super-additivity

An earlier draft of the methodology proposed a 3.5 °C ceiling for a combined intervention package, on the intuition that combining measures should outperform any single measure. That value was rejected, and the rejection survives the move to real packages unchanged.

No retrieved source quantifies super-additive cooling from combining interventions. Gunawardena et al. (2017) describes green and blue features as offering synergistic benefits, but qualitatively and without quantification. Keravec-Balbot et al. (2026) reports no synthesis estimate for combined packages at all; its best-performing individual measures cluster at 2.0–3.0 °C.

A package's reported temperature is therefore **capped at its best-evidenced component and never summed** (Section 7.7). Claiming that a package outperforms its strongest component would be an unsourced assumption placed at exactly the point where a user is most likely to select it. The package's genuine advantage is expressed where evidence supports it — breadth of co-benefits, which are unioned across components — and not as additional degrees.

6.6.3 Green roofs and green façades are the weakest inherited points, and are labelled as such

Green roof. Three sources point in different directions. Keravec-Balbot et al. (2026) places green roofs at 1.1 °C, the lowest-performing green measure in the synthesis. Zölch et al. (2016) finds green roof effects at pedestrian level *negligible*. Santamouris (2014) reports a 0.3–3 K reduction in average urban ambient temperature — but the abstract states this explicitly for green roofs *applied on a city scale* in *simulation studies*.

That last qualification decides the matter. A site-level assessment evaluates one roof on one building, not mass deployment across a city, and the physics differ: a green roof's thermal benefit is concentrated at roof level and inside the building beneath, and it does not propagate to the street canyon in the way city-scale albedo and evapotranspiration changes would. The 0.3–3 K figure is therefore **not** used as a site-level value. The adopted envelope of 0.1–1.0 °C sits near the bottom of the published spread, and the tool states in its output that green roof benefits are principally building-level rather than street-level.

Green façade. Here the sources conflict outright. Keravec-Balbot et al. (2026) ranks green walls at the top of its air-temperature table with a 3.0 °C central estimate — but with an approximately fivefold spread across climate subzones (Figure 1). Zölch et al. (2016) ranks green façades *below* trees on thermal comfort, at 5–10 % PET reduction against 13 % for trees.

The two are not directly comparable, since one reports air temperature and the other a comfort index; but the ranking reversal is real and needs an explanation. The most plausible one is measurement position: near-wall measurements capture a strong local effect that does not extend across a street canyon. This is a hypothesis, not a finding from either source, and it is labelled as such.

Given an unresolved conflict between sources and a fivefold climate spread, the archetype receives a conservative envelope of 0.3–2.0 °C, an evidence confidence rating of *low*, and an explicit caveat in the tool's output. Its confidence rating also caps the cooling-block confidence at medium regardless of how complete the user's inputs are (Section 8.3.1). Eleven catalogue entries inherit it, which makes it the second most widely inherited class in the library and the one whose weakness reaches the most results.

And one further weak point. Bioretention deserves the same flag. No retrieved source quantifies air-temperature cooling for rain gardens or bioswales specifically. Keravec-Balbot et al. (2026) offers no bioswale-specific estimate, and its neighbouring categories — green areas at 2.0 °C, water bodies at 2.1 °C — are not transferable to a small drainage feature. Gunawardena et al. (2017) identifies tree-dominated greenspace as delivering heat-stress relief and does not identify small blue features as significant coolers. The adopted envelope of 0.1–0.8 °C is a reasoned lower bound from adjacent evidence, flagged low confidence, and the tool reports this archetype's primary benefits as stormwater management and biodiversity rather than cooling.

6.6.4 Newer, broader evidence is deliberately not adopted wholesale

Kumar et al. (2024) is a systematic review screening 27 486 papers to 202 analysed, covering 51 green–blue infrastructure types. It post-dates the calibration of the twelve inherited archetypes, which makes it the sharpest available external test of them, and it plays two distinct roles in this methodology.

As a validation, it reports street trees at up to 2.8 °C by in-situ monitoring — inside the tool's existing 0.5–3.0 °C envelope. Newer and broader evidence therefore supports the conservative calibration rather than overturning it, which is the outcome a conservative calibration is supposed to produce and is worth recording when it happens.

As a non-adoption, the same review reports means well above the tool's envelopes for other types: green walls at (4.1 ± 4.2) °C, vegetated balconies at (3.8 ± 2.7) °C, and botanical gardens at (5.0 ± 3.5) °C. These are **not** adopted. Taking them selectively would break internal consistency — vegetated balconies would outscore street trees — and adopting them wholesale is a full recalibration of the library, which would require its own sensitivity analysis and its own version. It is recorded here and in the decision log as a candidate for a future methodology review, not smuggled into this one. Note also that the green-wall figure's standard deviation exceeds its mean.

A reviewer who believes the tool should adopt these figures is disputing something specific and answerable, which is why the non-adoption is stated rather than left implicit in a set of numbers that simply happen to be lower.

6.7 The four archetypes retrieved for this version

Each of the four exists because the catalogue contains entries that would otherwise have inherited an envelope belonging to a different body of study. In two cases that inheritance would have overstated the entry roughly threefold.

6.7.1 The large open water body, and one envelope that moved at implementation

The large open water body archetype — inherited by urban lake restoration, living shorelines, and coastal ecological corridors — ships at 0.1–3.0 °C, not the 0.5–3.0 °C proposed when this release was designed.

A design may be approved; a citation is only settled once it has been read.

The approved design attributed a cooling magnitude near 2 °C to Yao, Sailor, Yang, et al. (2023). Source verification, carried out before the value was allowed into configuration, established that the paper reports no such figure: it measures a 0.1–0.6 °C daytime cool island at urban lakes in two humid subtropical cities — an order of magnitude below the attributed value — alongside a 1.2–1.3 °C *heat* island at night.

Holding the floor at 0.5 °C would have made the tool claim more than its most direct field evidence supports, which is precisely the inflation the clipping rule of Section 7.6 exists to prevent, and it would have shipped a citation that did not say what the configuration claimed it said. The floor was therefore lowered to the measured minimum.

Four further citation corrections were made in the same verification pass, none of which moved a value: an author list, a nocturnal-warming figure conflated between two papers by the same first author, a study of two parks under drought that had been described as a study of allotment gardens, and a mean-radiant-temperature range quoted more roundly than the source states. The verification policy of Appendix E is what caught all five, and it is reported here rather than silently repaired because a methodology that only publishes its successful checks is not publishing a check at all.

The resulting envelope spans a factor of thirty, and the width is the finding rather than a defect. A remote-sensing-inclusive meta-analysis of 27 studies gives a pooled 2.5 K (1.9–3.2 K at 95 %) (Völker et al. 2013), while direct field measurement of actual urban lakes gives 0.1–0.6 °C (Yao, Sailor, Yang, et al. 2023); and Ampatzidis and Kershaw (2020) identifies remote sensing as the overestimating measurement mode. A narrower envelope would have to pick a side that the evidence does not support picking, so the tool reports the disagreement and its output caveat says so in as many words rather than leaving a reader to wonder why the range is so wide.

The ceiling is unchanged at 3.0 °C, on Keravec-Balbot et al. (2026)'s 2.1 °C for water bodies and the upper bound of the Völker et al. (2013) confidence interval. The base score of 75 is placed below the blue-green corridor at 85, because a water body alone lacks the corridor's canopy component — the corridor's envelope rests on the canopy evidence as much as on the water evidence — and because its field-measured daytime performance is weak.

6.7.2 Small constructed water, and the interpolation the engine refuses

Constructed wetlands, retention ponds, detention basins, water squares, floating wetlands, and roof wetlands would naturally have inherited the blue-green corridor at 1.0–3.0 °C on family resemblance — they are water, and water cools. Three independent lines say otherwise for *small constructed* water.

Jacobs et al. (2020) simulates 16 representative urban water bodies and reports afternoon air temperature in surrounding spaces reduced by typically 0.2 °C, maximum 0.6 °C; synthesising 20 further papers, 14 of them (70 %) report an effect of 1 °C or less, and the paper concludes that the local thermal effect of small water bodies "can be considered negligible in design practice". Yao, Sailor, Zhang, et al. (2023) measures an urban pond at 0.6 °C of daytime cooling against a reference site, with 1.8 °C of nocturnal *warming*. Ampatzidis and Kershaw (2020) adds that sensible cooling can be offset by increased water-vapour content. The adopted envelope is 0.0–1.0 °C, with a genuine zero floor on Jacobs et al. (2020)'s own conclusion, and a base score of 40 — below the extensive green roof at 45, because the two share a ceiling near 1.0 °C but this class has a true zero floor and the roof does not. Without the split, six catalogue entries would have been overstated roughly threefold.

The two water archetypes are never interpolated between by area. They are separate bodies of study, not two points on a published dose–response curve: the size-threshold literature for water bodies is land-surface-temperature only, and no published curve connects the two on air temperature.

An area-based interpolation would look eminently reasonable and would be unsourced. The engine therefore does not perform one, and the test suite asserts its absence by scoring the same constructed wetland on two sites differing in area by a factor of 2 500 and requiring an identical ceiling, so that a future edit cannot introduce it quietly.

6.7.3 Non-canopy vegetation, and the surface/air conflation

Planting beds, groundcover, pocket meadows, roof meadows, and green medians — seven entries — inherit a class of their own rather than the tree-based classes, on the strength of one study that measures both metrics on the same plots. Armson et al. (2012) found that grass reduced maximum *surface* temperature by up to 24 °C while having little effect upon *globe* temperature, whereas shading reduced globe temperature by 5–7 °C, and concludes that grass "has little effect upon local air or globe temperatures . . . whereas tree shade can provide effective local cooling".

That contrast is decisive precisely because it is measured on the same plots: a 24 °C surface reduction alongside almost no globe-temperature change is not a small effect measured badly, it is a large effect on the wrong quantity. It is also the mechanism by which most impressive published figures for grass cooling are generated, and the tool's output caveat tells the user exactly that.

The floor is a genuine zero, on Gill et al. (2013), which models grass cooling as reduced or lost in summer droughts when soils dry out, and on Kraemer and Kabisch (2022), whose field measurement under the 2018–2019 Central European drought found the grass-dominated park to be the hottest area during the daytime while the tree-dominated park stayed consistently cooler. The adopted envelope is 0.0–1.5 °C with a base score of 50, below the small green area with trees at 65, which shares the 1.5 °C ceiling but adds the canopy shade that Armson et al. (2012) shows is doing the work.

6.7.4 Vegetated shade structures, and a figure that does not transfer

Green pergolas and green waiting shelters take 0.0–2.5 °C at *low* confidence, from Chàfer et al. (2020) — the only measured comparison of a vine-covered pergola against an identical bare support, giving a maximum daytime difference of 2.5 °C for west-oriented pergolas in a Mediterranean climate. That paper's abstract headlines "up to 5 °C", but that figure compares greenery against *unshaded* conditions rather than against the bare pergola; the like-for-like contrast is the one adopted, and the discrepancy is recorded in Appendix E so that a reviewer checking the abstract does not read an error into the ceiling.

> **The "shading devices 2.0 °C" figure from Keravec-Balbot et al. (2026) is deliberately not used here.** That category aggregates area-scale shading, where the cooled air volume persists against advection; a pergola shades roughly 10 m^2. Applying the figure would rate a pergola above measured tree canopy, which Ouyang et al. (2024) puts at 1.42 °C.
>
> The same figure *is* used by the permeable shaded hardscape archetype, where the scale matches. The distinction is scale, not scepticism about the source, and the test suite asserts both halves of it.

Ouyang et al. (2024) also separates the two axes that matter for shade structures, measuring an air temperature reduction of 1.42 °C for natural shade against 1.31 °C for artificial, alongside mean radiant temperature reductions of 15.93 °C and 13.71 °C respectively — an order of magnitude apart. Users will feel substantially cooler under a pergola than the air-temperature figure suggests, and the output caveat says so: under-selling an intervention is as misleading as over-selling it, only in the other direction. Colter et al. (2019) supplies the ordering that keeps the base score at 55, well below the permeable shaded hardscape at 70: sparse and open canopies group with solid constructed shade rather than with dense canopy.

6.8 The two derived archetypes, and a citable absence

Retrieval established that no class-specific cooling evidence exists for productive landscapes. Both productive archetypes are therefore *derived* — bounded from an adjacent class with the bounding argument stated — and both are rated *low* confidence for that reason.

What makes this different from the bioretention case, where no source was found at all, is that here **the absence itself is citable**. Kumar et al. (2024) places allotments and city farms among the least-studied of its 51 infrastructure types, with insufficient data to quantify their cooling. The methodology can therefore state the gap as a finding rather than as a search failure, which is a materially stronger position.

Productive tree canopy — urban orchards, food forests, agroforestry systems — takes 0.5–2.0 °C at a base score of 60, bounded *below* dense tree canopy on two grounds: orchard and agroforestry canopies are deliberately kept open for access and yield, so they sit below the 40 % canopy-cover inflection identified by Ziter et al. (2019), and Colter et al. (2019) establishes that sparse canopies do not perform like closed ones. The envelope matches the established park at a lower evidence rating: the same range, honestly labelled as inherited rather than measured.

Productive ground-level planting — community gardens, allotments, school gardens, urban farms, productive roofs, rooftop farms — inherits the non-canopy vegetation envelope of 0.0–1.5 °C at a base score of 45, below non-canopy vegetation at 50, because cultivated ground-level cover is periodically broken between crops.

The best allotment-garden measurement available cannot be used for a value. Rost et al. (2020) measured 13 Berlin allotment complexes at an average 2.7 K cooler than built-up areas — *at night*. The restriction is in the study's title, abstract, and design, and this tool reports daytime air temperature. The study is cited as evidence of what the productive-landscape literature does and does not measure, and for no numeric value.

Irrigation is not credited as extra degrees. Cheung et al. (2022) documents irrigation as *sustaining* performance through drought rather than raising it, measuring agricultural cooling at 0.09–0.43 °C against modelled potentials of 2.1–2.5 °C, and predicting an increase in the daily *minimum* temperature as stored heat is released overnight. One nuance is carried carefully: that measured-versus-modelled gap is specific to *agricultural field* studies, and the same paper measures −2–−1 °C in irrigated urban parks, much closer to the modelled values. The methodology therefore does *not* claim that models overstate irrigation cooling in general.

6.9 Suitability conditions, inherited with twenty-one overrides

Each archetype declares the conditions under which an intervention is unsuitable: a minimum viable site area, a soil requirement, an irrigation requirement, and any unsuitable climate zones. Catalogue entries **inherit** these, with per-entry overrides only where the catalogue's own description of the entry makes an inherited value plainly wrong.

Inventing a minimum area and a soil requirement for each of 110 entries would have manufactured exactly the precision the archetype model exists to avoid. There are 21 overrides in total, each listed with its reason in Appendix A, and **no new number enters the methodology through an override**: every replacement value — 20 m^2, 50 m^2, 200 m^2, 500 m^2, 1 000 m^2 and 2 000 m^2 — is one already present in the cited library, borrowed from the archetype that established it. A test fails if the override list grows past 25 entries, because a long list would mean the inheritance ruling had quietly been abandoned.

The overrides fall into three groups, and the second is the one worth reading. Area overrides correct a scale mismatch: the dense tree canopy's 2 000 m^2 describes the woodland its *cooling evidence* comes from, not the footprint of a microforest on a constrained site. Soil overrides in the roof group set the requirement to *none*, because a roof's growing medium is built rather than ground, and requiring ground soil on a roof would flag every roof entry whose archetype happens to be a ground-level class. And the four ground-rooted façade entries override soil in the *stricter* direction, to *limited*, because the catalogue defines them as rooted in natural ground where the containerised living-wall archetype requires nothing — an override that removes an option rather than adding one.

Land-use context is not a second list. The urban-context sub-indicator of Table 9 compares the site's land use against the entry's own land-use list, which is the availability matrix's list (Section 5.6). An entry is therefore "in context" exactly where it is offered — one list, one meaning, and no opportunity for two lists to drift apart.

When a disqualifying condition is met, the tool computes the assessment transparently but attaches a prominent *"not suitable for this site"* flag naming the reason, which is carried into the recommendation text and the exported report. Silently returning a merely lower score for a physically implausible intervention — riparian restoration on a site with no water feature, for instance — would misrepresent the finding as a matter of degree when it is a matter of kind.

Unsuitable entries nonetheless remain fully selectable, including entries the availability matrix did not offer for this site. A user deliberately testing a hypothesis should not be overridden by the tool, and the flag follows the result through to the report regardless.

One suitability rule is worth singling out because it encodes a caveat that would otherwise be lost in prose. Gunawardena et al. (2017) notes that poorly designed bluespace may *exacerbate* heat stress under oppressive humid conditions. Rather than leaving this as a written warning, the methodology encodes it as a climate-suitability penalty: blue-green archetypes are downgraded to the *moderate* condition level in tropical-wet zones (Table 7). A caveat that changes a number is a caveat that gets read.

7 Scoring formulas

7.1 Notation

All scores are on a 0–100 scale unless stated otherwise. Two bounding operators are used throughout:

$$\mathrm{clamp}(x) = \min\bigl(100,\ \max(0,\ x)\bigr), \tag{2}$$

$$\mathrm{clip}(x,\ l,\ u) = \min\bigl(u,\ \max(l,\ x)\bigr). \tag{3}$$

clamp bounds a score to the reporting scale; clip bounds a physical estimate to a stated envelope. The distinction matters in Section 7.6, where a score is clamped to [0, 100] while a temperature is clipped to a literature-derived interval.

Every weight appearing below is reproduced with its rationale in Appendix B. Within each weighted sum, the weights sum to unity.

7.2 Heat Exposure Score

There are two paths, selected by data availability.

Data-rich path. Where a land-surface-temperature anomaly is supplied and converted to S_{LST} by Equation (1):

$$S_{\mathrm{heat}} = \mathrm{clamp}\bigl(0.40\,S_{\mathrm{LST}} + 0.25\,S_{\mathrm{imp}} + 0.20\,S_{\mathrm{solar}} + 0.15\,S_{\mathrm{vegdef}}\bigr), \tag{4}$$

where S_{imp} is the impervious surface percentage of the site used directly as a 0–100 score, S_{solar} is the qualitative solar exposure level normalised by Table 2, and the vegetation deficit is

$$S_{\mathrm{vegdef}} = \mathrm{clamp}(100 - g), \tag{5}$$

with g the existing green cover percentage.

Data-poor path. Where no LST anomaly is supplied, the Heat Exposure Score is the normalised qualitative heat exposure level directly: $S_{\mathrm{heat}} = S_{\mathrm{heatlevel}}$.

Basis. The component selection reflects established surface-energy-balance drivers of urban heat: surface temperature, sealed surfaces, solar loading, and vegetation deficit. Ziter et al. (2019) provides direct empirical support for two of the four — impervious cover raises air temperature approximately linearly, and canopy cover reduces it nonlinearly — and additionally establishes that the warming effect of impervious surfaces persists at night while canopy cooling largely does not.

What is expert judgment here. The *relative* weights are not derived from literature. No retrieved source ranks the marginal contribution of LST anomaly against imperviousness against solar loading against vegetation deficit for the purpose of a screening index. The weights express a judgment that a directly measured thermal signal should dominate where it is available, that sealed surface is the strongest single proxy where it is not, and that vegetation deficit — already partly captured by the other three — should carry the least. This judgment is declared, versioned, and subject to the sensitivity analysis of Section 9.

7.3 Vulnerability Score

$$S_{\text{vuln}} = \text{clamp}\big(0.40\,S_{\text{dens}} + 0.40\,S_{\text{groups}} + 0.20\,S_{\text{coolaccess}}\big), \tag{6}$$

where S_{dens} is normalised population density, S_{groups} is normalised vulnerable-group presence, and $S_{\text{coolaccess}}$ is the cooling access *deficit* obtained through the inverted mapping of Section 5.3.

Basis. Indicator selection follows Reid et al. (2009), whose factor analysis of ten heat-vulnerability variables identified four factors explaining more than 75 % of total variance: social and environmental vulnerability, social isolation, air-conditioning prevalence, and the proportion of elderly and diabetic residents.

Population density and vulnerable-group presence are weighted *equally* because that analysis provides no basis for ranking demographic vulnerability above exposure density; asserting a difference would be inventing a finding. Sera et al. (2019) independently supports density as a modifier of the heat–mortality relationship across 340 cities in 22 countries. Cooling access carries the smaller weight because it is the field most frequently unknown at screening stage, so a larger weight would amplify the neutral default of Section 5.4.

One dimension, two indicators. From version `2026.08.03` the cooling access deficit is measured by two indicators rather than one, at the same 0.20 weight: access to actively cooled *indoor* space, and access to a cool *outdoor* refuge — shaded green or blue space the site's users can reach. The split is evidenced, not assumed. Reid et al. (2009) grounds the indoor indicator. Burkart et al. (2016) grounds the outdoor one directly: in Lisbon, above the 99th temperature percentile, mortality among residents over 65 rose 14.7 % per °C in the least-vegetated areas against 3.0 % in the most vegetated, and 7.1 % per °C beyond 4 km from water against 2.1 % within it.

The dimension is the *mean of the indicators actually supplied*:

$$S_{\text{coolaccess}} = \begin{cases} \dfrac{1}{|\mathcal{A}|}\displaystyle\sum_{i\in\mathcal{A}} S_i & \text{if } \mathcal{A} \neq \emptyset, \\ 50 & \text{otherwise,} \end{cases} \tag{7}$$

where $\mathcal{A}$ is the set of supplied indicators (an explicit *unknown* is not supplied). Averaging an *unsupplied* indicator in at the neutral 50 was considered and rejected: it would pull a signal the user did give toward the middle purely because a second question was skipped, letting an unknown alter the meaning of an answer — the defect removed from the soil–water pair in Section 7.5. Under the adopted rule a single answer speaks for the dimension, answering the second moves it in whichever direction that answer warrants, and skipping costs nothing. The two indicators correspondingly share one confidence slot (Section 8.3).

A disclosed overlap. The outdoor indicator describes cool refuges in the *surroundings* that the site's users can reach; the vegetation deficit of Section 7.2 describes greenness *of the site itself*. The two are distinct quantities but conceptually adjacent, and a bare site in a green district registers a low outdoor-refuge deficit alongside a high vegetation deficit. We state this rather than leave it for a reader to discover, as with the vulnerability double contribution of Section 7.12.

7.4 Heat Priority Index

$$\mathrm{HPI} = \mathrm{clamp}\big(0.60\, S_{\mathrm{heat}} + 0.40\, S_{\mathrm{vuln}}\big). \tag{8}$$

The index is interpreted by the bands in Table 5. Band boundaries drive recommendation text, which is why their stability under weight perturbation is one of the four quantities the sensitivity analysis reports (Section 9.4).

Table 5: Interpretation bands for the Heat Priority Index and the NbS Cooling Opportunity Score.

Range	Heat Priority Index	Opportunity Score
0 – 30	Low	Low priority
31 – 60	Medium	Moderate opportunity
61 – 80	High	Strong opportunity
81 – 100	Critical	High-priority project

7.5 Derived site adjustment factor

Four site conditions modulate archetype performance. Each condition resolves to a level, and each level maps to a multiplicative factor:

Condition level	Poor	Moderate	Good	Excellent	Unknown
Factor	0.5	0.8	1.0	1.2	1.0

The composite adjustment factor is

$$A = 0.40\, f_{\mathrm{canopy}} + 0.25\, f_{\mathrm{soilwater}} + 0.20\, f_{\mathrm{scale}} + 0.15\, f_{\mathrm{climate}}, \tag{9}$$

which ranges over $[0.5, 1.2]$, attaining its bounds only when all four conditions are poor or all four are excellent.

All four conditions are derived, not asked. No additional questions are put to the user; each condition is computed from inputs already supplied. Table 6 gives the rules.

Why canopy carries the largest weight. It is the condition with the strongest direct empirical support. Ziter et al. (2019) found that cooling increases nonlinearly with canopy cover and is greatest above approximately 40 % cover; the *excellent* boundary is placed at that empirically identified inflection rather than chosen for convenience. The same study locates maximum daytime cooling at 60–90 m radii, which is the basis for the scale condition: a district-scale or city-scale intervention is more likely to reach the spatial scale at which the effect is strongest than a single-building one.

Table 6: Derivation of the four site adjustment conditions from inputs the user has already supplied.

Condition	Derivation rule
Canopy	$c = \frac{\text{existing tree canopy + new canopy at maturity}}{\text{site area}} \times 100$, then: poor if $c < 10$; moderate if $10 \le c < 25$; good if $25 \le c < 40$; excellent if $c \ge 40$.
Soil–water	Soil availability and irrigation availability are each mapped to a condition level, and the *more limiting* of the two is taken. The mapping is *none* → poor; *limited*, *occasional* → moderate; *moderate* → good; *reliable*, *high* → excellent. When exactly one of the two is known, the condition is capped at *good*: a pair containing an unknown may never exceed the neutral factor, so declaring reliable irrigation is never worse — and with high soil strictly better — than skipping the question (D-026, version 2026.08.02).
Scale	From the declared assessment scale: *city*, *district* → excellent; *neighbourhood* → good; *site*, *building* → moderate.
Climate	A lookup of archetype family against climate zone (Table 7).

Why climate is present at all. Keravec-Balbot et al. (2026) finds that adding Köppen–Geiger subzone raises explained variance in cooling effectiveness from 12 % to 78 %. Omitting climate would discard the largest single explanatory term in the available evidence.

Table 7: Climate suitability matrix: the condition level assigned to each archetype family in each supported climate zone, resolved to a factor through the condition-level table above. Vegetation-based archetypes are downgraded in arid and semi-arid zones where water limitation constrains evapotranspiration; blue-green archetypes are downgraded in tropical-wet zones following the humidity caveat of Gunawardena et al. (2017), and upgraded in tropical-dry zones. An unrecognised zone yields *unknown*, whose factor is the neutral 1.0.

Climate zone	Green	Building	Blue-green	Hybrid
Tropical wet	Good	Good	Moderate	Good
Tropical dry	Good	Good	Excellent	Good
Arid	Moderate	Moderate	Moderate	Good
Semi-arid	Moderate	Good	Good	Good
Temperate	Good	Good	Good	Good
Other	Unknown	Unknown	Unknown	Unknown

7.6 Cooling Potential Score and temperature reduction

The Cooling Potential Score scales the base cooling score of the archetype the selected entry inherits by the site adjustment factor:

$$S_{\text{cool}} = \text{clamp}(B \times A)\,, \tag{10}$$

where B is the base cooling score from Table 4. Every quantity in this subsection is a property of the *archetype*, not of the catalogue entry, and the result names the archetype it was computed against (Section 6.1).

The reported temperature reduction range is computed differently, and the difference is the point of this subsection:

$$\Delta T_{\min} = \mathrm{clip}(\tau_{\min} \times A,\ \tau_{\min},\ \tau_{\max})\,, \tag{11}$$
$$\Delta T_{\max} = \mathrm{clip}(\tau_{\max} \times A,\ \tau_{\min},\ \tau_{\max})\,, \tag{12}$$

where $[\tau_{\min}, \tau_{\max}]$ is the archetype's literature envelope from Table 4.

The clipping rule is a deliberate correction to the draft methodology. Without it, an adjustment factor above 1.0 applied to a typology's upper bound produces temperature claims exceeding anything in the retrieved literature. An urban forest on an excellent site ($A = 1.2$, $\tau_{\max} = 3.0\,°\mathrm{C}$) would report up to 3.6 °C — a figure no retrieved source supports for a site-level intervention.

Because the published envelopes already represent well-executed interventions under favourable conditions, favourable site conditions must not be permitted to compound that favourability. The 0–100 Cooling Potential Score of Equation (10) remains free to rise above the base value, so the tool can still express that a site is exceptionally well suited — but it expresses this as a higher relative score, not as a larger physical claim.

Figure 5 shows the behaviour across the full range of the adjustment factor. The rule is asymmetric in effect, and it is worth being precise about what it does. For $A < 1$ the upper bound falls — a poor site is reported as delivering less — while the lower bound is held at $\tau_{\min}$, so the reported range narrows from the top. For $A > 1$ the upper bound is held at $\tau_{\max}$ while the lower bound rises, so the range narrows from the bottom. In both directions the reported interval remains a subinterval of the literature envelope: site conditions shift an estimate *within* the published evidence, and never outside it.

Derived reporting quantities. Two further quantities are reported directly. Shade potential is the fraction of the site brought under new canopy at maturity:

$$S_{\text{shade}} \;=\; \mathrm{clamp}\!\left(\frac{a_{\text{canopy}}}{a_{\text{site}}} \times 100\right), \tag{13}$$

with a_{canopy} the new canopy area at maturity and a_{site} the site area, both in square metres. And the midpoint of the adjusted temperature range, $\overline{\Delta T} = (\Delta T_{\min} + \Delta T_{\max})/2$, is assigned a heat-index improvement bucket: *low* below 0.5 °C, *medium* from 0.5 °C to 1.5 °C, and *high* from 1.5 °C upward.

Time to benefit. No maturity discount is applied to cooling values. Instead the expected maturity period is reported as a time-to-benefit class, so that a slow-maturing intervention is never presented as delivering its cooling immediately: under one year *immediate*, one to three years *short-term*, three to ten years *medium-term*, ten years and beyond *long-term* (half-open brackets). Absent a maturity period the output is *not estimated*.

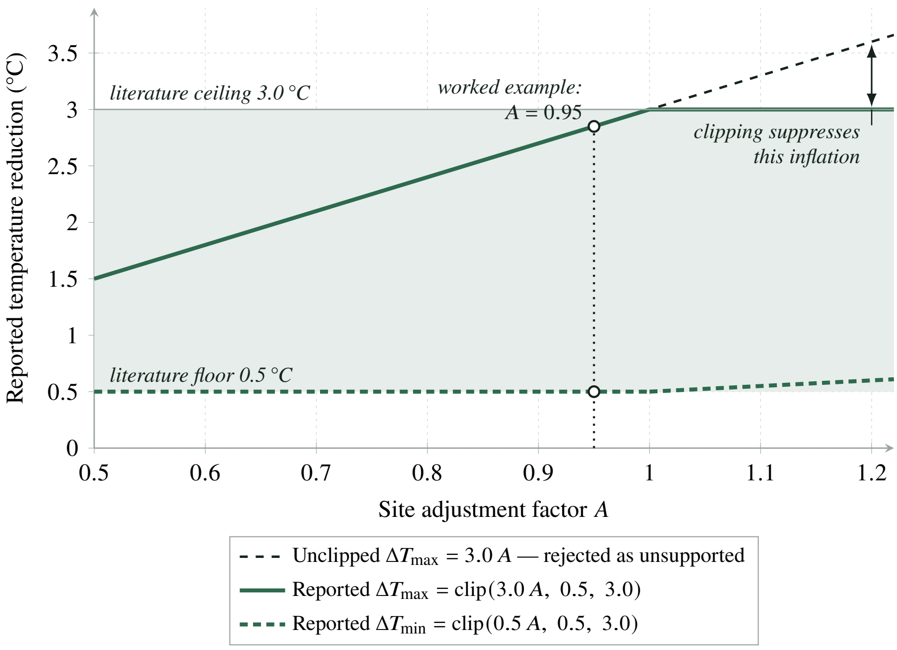

Figure 5: The temperature-envelope clipping rule of Equations (11) and (12), illustrated for street tree planting (envelope 0.5–3.0 °C) across the full 0.5–1.2 range of the site adjustment factor A. The dashed line is the unclipped product $3.0\,A$, which the methodology rejects because for $A > 1$ it exceeds the literature ceiling. The reported bounds remain within the shaded envelope throughout. The marked point locates the worked example of Section 11.

7.7 Multi-intervention packages

An assessment may propose several simultaneously available catalogue entries. Each component is **scored individually by the formulas above and reported individually**; only the package's headline outputs are combined, and they are combined under rules that add nothing the evidence does not support. No rule below computes a new methodology value: each either selects among the component results or takes an extremum of them.

Table 8: How a package's headline outputs are combined from its components. Every rule is a selection or an extremum; none introduces a value that is not already a component result.

Output	Rule, and why
Temperature	**The best-evidenced component's adjusted range; never summed.** No retrieved source quantifies super-additive cooling from combining measures (Section 6.6.2). Note that $\min\left(\max_i \Delta T_{\mathrm{max},i},\ \Delta T_{\mathrm{max,rep}}\right)$ and $\Delta T_{\mathrm{max,rep}}$ are the same number, so the cap and the selection are one rule, not two.
Co-benefits	**Union: the maximum per dimension**, then re-weighted by Equation (15). A package delivers each dimension at least as well as its best component does, and this is where a package's real advantage lies.
Suitability	**The minimum across components.** A package is no more deliverable on a site than its least suitable member; averaging would let a well-fitting component conceal one that cannot be built here, which is exactly what the flags of Section 6.9 exist to prevent.
Energy	Derived from the best-evidenced **building-energy-applicable** component. Energy is a function of one cooling estimate (Equation (17)); restricting rather than summing keeps the derivation traceable to exactly one component. If no component is building-energy applicable, the energy status is `typology_not_applicable`.
Capital cost	**Entered once for the whole package.** The tool ships no unit costs (Section 7.11), so there is nothing to apportion; Table 23 states that the figure covers every selected component.
Flags	Per component, all surfaced, each naming its entry. The user must be able to see *which* part of the package is the problem.

Which component speaks for the package. The component carrying the temperature is chosen by **evidence rating first, adjusted envelope width second**, with ties falling to the user's own selection order so that the choice is deterministic. A low-evidence component with a higher ceiling therefore does not speak for the package — otherwise the least certain estimate in a selection would set its headline number.

Package size is unbounded, but warned above five components.

Under the capped-never-summed rule a large package's headline temperature is carried by its strongest member while the remainder add co-benefit breadth and capital cost, so a hard cap would forbid something harmless. Above five components the tool states plainly that adding a further component will not raise the temperature estimate. That is the information a refusal would have conveyed only by refusing.

A worked instance from the regression suite. Golden scenario `s23` selects five components on one temperate neighbourhood site: a tree avenue (0.5–3.0 °C, high evidence), a green median (0.0–1.5 °C, medium), a blue-green corridor (1.0–3.0 °C, medium), a pocket park (0.5–2.0 °C, high), and courtyard greening (0.3–1.5 °C, low). The reported range is the tree avenue's alone — 0.5–3.0 °C at an adjustment factor of 1.0 — because it is the best-evidenced component and, among the two high-evidence components, the one with the wider envelope. The sum of the five ceilings would be 11.0 °C, and the tool does not claim it. The co-benefit score, unioned, is 71.25; the suitability score is the weakest component's 86.25; and the Opportunity Score is 76.10. The scenario exists to make a regression in any one of these rules fail the build.

7.8 NbS Suitability, co-benefit, and equity scores

Three further composite sub-scores enter the result. Their aggregation weights are declared in configuration and reproduced in Appendix B:

$$S_{\text{suit}} = \text{clamp}\big(0.30\,S_{\text{space}} + 0.25\,S_{\text{soil}} + 0.20\,S_{\text{water}} + 0.15\,S_{\text{maint}} + 0.10\,S_{\text{context}}\big), \tag{14}$$

$$S_{\text{cobenefit}} = \text{clamp}\big(0.25\,S_{\text{biodiv}} + 0.25\,S_{\text{storm}} + 0.20\,S_{\text{health}} + 0.15\,S_{\text{social}} + 0.15\,S_{\text{urbanq}}\big), \tag{15}$$

$$S_{\text{equity}} = \text{clamp}\big(0.35\,S_{\text{vulnben}} + 0.25\,S_{\text{access}} + 0.20\,S_{\text{safety}} + 0.20\,S_{\text{particip}}\big). \tag{16}$$

The co-benefit sub-indicators take per-archetype default levels, each cited in the library (Appendix A), which the user may override. In a package they are unioned across components before weighting (Table 8).

The sub-indicator rules, fixed at version 2026.08.01. An earlier version of this document disclosed that the rules mapping user inputs to the individual sub-indicator scores were unspecified and would be fixed alongside the engine implementation. They are now fixed, live in `config/derived_scores.yaml`, and — like the aggregation weights — are declared expert judgment, versioned and adjustable rather than derived from literature. All suitability sub-indicators derive from inputs the user has already supplied; no new questions are asked.

Table 9: Derivation of the NbS Suitability sub-indicators (decision D-022). A score of 25 on space, soil, or water coincides with the hard "not suitable" flag of Section 6.9; an unknown availability never raises a flag, because a disqualification is never asserted from absent information.

Sub-indicator	Rule
Space	r = site area/minimum viable area: $r < 1 \rightarrow 25$ *and the flag*; $1 \le r < 2 \rightarrow 50$; $2 \le r < 5 \rightarrow 75$; $r \ge 5 \rightarrow 100$.
Soil	Availability against the entry's requirement on the ordinal scale *none* < *limited* < *moderate* < *high*: requirement *none* → 100; availability unknown → 50; below requirement → 25 *and the flag*; meets exactly → 75; exceeds → 100.
Water	The same rule on *none* < *occasional* < *reliable* against the irrigation requirement.
Maintenance	Declared maintenance intensity through the inverted scale: low → 100, medium → 50, high → 25, unknown → 50.
Urban context	Site land use inside the entry's own land-use list → 100; outside them → 25 (a caution, not a disqualification); unknown → 50.

The four equity sub-indicators read how much equity benefit the intervention can deliver at this site, so a *deficit* raises the score, paralleling the risk framing of the Vulnerability Score: vulnerable-user benefit reuses the vulnerable-population input already collected; public accessibility and community participation are the two new optional inputs introduced at this version; and safety concern deliberately uses the standard (non-inverted) scale, because a site with greater safety and comfort problems stands to gain more from a well-designed intervention.

Two disclosures carry over unchanged from the previous version. The co-benefit weights include a *social inclusion* term for which the library supplies no default; absent a user override it falls to the neutral default of Section 5.4 and is itemised as an applied assumption. And the Equity Score of Equation (16) is computed and reported as an output block with its own confidence rating but does *not* enter the final aggregation of Section 7.12: equity influences the headline score through the Vulnerability Score instead. Whether that is the right design remains a legitimate question for reviewers.

7.9 Energy savings — derived, not assumed

An earlier draft of the methodology stored per-typology "energy reduction factors" in the range 2–15 %. No source could be found for any of them. They were removed and replaced by a derivation from the tool's own estimated cooling effect:

$$E = D \times \Delta T \times s, \qquad s \in [0.02,\ 0.04] \text{ per °C}, \tag{17}$$

where D is the user-supplied annual cooling energy demand in kW h and s is the temperature–electricity-demand sensitivity. Because both ΔT and s are intervals, the result is an interval:

$$E_{\min} = D \times \Delta T_{\min} \times 0.02, \qquad E_{\max} = D \times \Delta T_{\max} \times 0.04. \tag{18}$$

Basis. Akbari et al. (2001) reports that urban electricity demand increases by 2–4 % for each 1 °C of temperature increase, and estimates that 5–10 % of urban electricity demand serves to compensate for the 0.5–3.0 °C urban temperature elevation. Applying that sensitivity in reverse to an estimated temperature reduction yields an energy saving that is sourced, internally consistent with the tool's own cooling estimate, and automatically responsive to site conditions — a poorly suited site yields a smaller ΔT and therefore a smaller energy saving, with no separate parameter to maintain.

Preconditions. Energy savings are calculated only when all three of the following hold: the user confirms that nearby building cooling demand is relevant to the intervention; an annual cooling energy demand is supplied; and the archetype the selected entry inherits is flagged as capable of affecting adjacent building loads. In a package the derivation follows one building-energy-applicable component (Table 8). Otherwise the result carries an explicit status, checked in a fixed order — `typology_not_applicable` (the intervention cannot affect building loads, whatever the user supplied), `not_applicable` (the user states demand is not relevant), `relevance_not_confirmed` (the relevance question was skipped or answered *unknown*; a supplied demand figure cannot substitute for the unanswered confirmation), or `missing_energy_demand` — and **never a zero**, which would understate the result while appearing to be a finding.

An acknowledged weakness. The sensitivity is drawn from predominantly North American evidence, and its transferability to other building stocks, construction practices, and climates is untested. Building stock differs in envelope performance, in the prevalence and type of mechanical cooling, and in occupant behaviour, all of which affect how a change in outdoor air temperature translates into a change in electricity demand. This is an area where reviewer input is specifically sought. It is also worth noting that Akbari et al. (2001) is recorded in Appendix E as *metadata + finding (secondary)*: the bibliographic record and the 2–4 % figure were confirmed from multiple indexing records and an institutional listing, not from a retrieved full text.

7.10 Greenhouse gas emissions avoided

$$G = E \times \phi, \tag{19}$$

where ϕ is the grid emission factor in kg h/kW of carbon dioxide equivalent, applied to both ends of the energy interval.

Grid emission factors are sourced per country from Ember (2026), which publishes electricity carbon intensity for 215 countries under a CC-BY-4.0 licence permitting redistribution with attribution inside an Apache-2.0 tool. Where no verified factor is available for a country, the tool reports greenhouse gas savings as *not calculated* rather than substituting a global average.

> **At version 2026.08.05 no country emission factors ship with the tool.** The country defaults file declares the source, licence, unit, and per-entry schema, but its country table is empty pending per-country verification. In consequence, a deployment using the shipped configuration unmodified will report greenhouse gas savings as *not calculated* for every country. This is the intended failure mode of the verification policy — no unverified value ships — but it means Equation (19) is at present a specified formula awaiting its data, and any assessment reporting a greenhouse gas figure is necessarily using a locally supplied factor.

7.11 Costs — why the tool ships no default values

World Bank (2021) states that nature-based solution project costs vary significantly and are highly site- and project-specific, that unit costs differ sharply between developed and developing contexts — illustrated by dredging at 2 US\$/m^3 in Bangladesh against 59 US\$/m^3 in the United Kingdom, roughly a thirtyfold difference for the same operation — and that urban implementation costs typically exceed rural ones. No source was found offering globally applicable unit costs for the interventions in this library.

The tool therefore ships **no default cost values** in version 1. Where a user supplies a capital cost, the tool computes cost outputs; where they do not, capital cost, payback, and cost feasibility are all reported as *not estimated*.

The reasoning is worth stating explicitly, because shipping a plausible default would have been easy and would have made the tool look more complete. Fabricating a default per-square-metre cost would generate the most decision-relevant number in the entire assessment — simple payback — from an invented input. No confidence rating adequately qualifies a payback figure whose numerator was guessed: the reader sees a number of years, and the number of years is what enters the decision. Locally calibrated cost tables remain a defined extension point; a deployment may supply its own cost configuration, which the tool records as a documented assumption in the result.

Where costs are supplied:

$$C_{\text{annual}} = E \times p, \tag{20}$$

$$T_{\text{payback}} = \frac{C_{\text{capital}}}{C_{\text{annual}}}, \qquad \text{undefined if } C_{\text{annual}} \le 0, \tag{21}$$

with p the energy price per kW h. Because E is an interval, so is the payback, with the bounds inverted: the largest energy saving gives the shortest payback.

Cost feasibility is *derived* rather than user-asserted:

$$S_{\text{cost}} \;=\; f\big(\text{payback bracket},\ \text{implementation complexity},\ \text{maintenance intensity}\big), \tag{22}$$

with short payback, low complexity, and low maintenance yielding high feasibility. The bracket boundaries and combination rule, fixed at version 2026.08.01 (decision D-023), are as follows. Because the energy saving is an interval spanning an order of magnitude, the payback bracket is applied to the payback computed from the *central* energy estimate — the midpoint of the savings range — while the reported payback interval still carries both ends. The brackets are half-open and follow common public-investment screening horizons: under 5 years scores 100 (*short*); 5–10 years 75 (*medium*); 10–20 years 50 (*long*); 20 years and beyond 25 (*very long*). The combination is

$$S_{\text{cost}} = 0.50\, S_{\text{bracket}} + 0.25\, S_{\text{complexity}} + 0.25\, S_{\text{maintenance}}, \tag{23}$$

with complexity and maintenance normalised through the inverted scale so that low complexity and low maintenance raise feasibility, and unknowns taking the neutral 50, itemised. Payback carries half the weight as the only quantitative economic signal. An *investment readiness* level is reported alongside: the bracket sets the base (*short* → high, *medium* → medium, otherwise low), downgraded one level for high implementation complexity and one level for low energy-block confidence, with a floor at low; it is *not estimated* whenever cost feasibility is.

Exclusion rather than substitution. When payback cannot be computed, cost feasibility is reported as unavailable and is *excluded* from the final aggregation, with the remaining weights redistributed proportionally (Equation (25)). The alternative — substituting a neutral 50 — would silently reward projects that have provided no economic evidence at all, placing them ahead of projects that supplied a genuinely poor payback figure. Exclusion preserves the ordering among options that did supply data, and the missing block is reported rather than hidden.

7.12 Aggregation and the weighting question

The headline score is a weighted sum of six components:

$$S_{\text{opp}} = 0.25\,\text{HPI} + 0.25\,S_{\text{cool}} + 0.15\,S_{\text{suit}} + 0.15\,S_{\text{vuln}} + 0.10\,S_{\text{cobenefit}} + 0.10\,S_{\text{cost}}. \tag{24}$$

Where a component is unavailable — in practice, cost feasibility — the sum is taken over the available set $\mathcal{A}$ and renormalised:

$$S_{\text{opp}} = \frac{\sum_{i\in\mathcal{A}} w_i\,S_i}{\sum_{i\in\mathcal{A}} w_i}. \tag{25}$$

The result is interpreted through the bands of Table 5.

These weights are expert judgment, and the tool says so. Nardo et al. (2008) is explicit that weighting in composite indicators is a value choice rather than a technical derivation, and no literature can supply the relative importance of cooling performance against equity against cost. The weights are declared here, versioned in configuration, adjustable by any deployment, and are to be defended empirically through the sensitivity analysis specified in Section 9 rather than by appeal to authority.

7.13 The equity-forward weighting, stated openly

Vulnerability enters the final score *twice*. It contributes 0.40 of the Heat Priority Index, which itself carries 0.25 in Equation (24), and it appears again directly at 0.15. Decomposing the Heat Priority Index in Equation (24) gives

$$\begin{aligned} S_{\text{opp}} &= 0.25\big(0.60\,S_{\text{heat}} + 0.40\,S_{\text{vuln}}\big) + 0.25\,S_{\text{cool}} + 0.15\,S_{\text{suit}} + 0.15\,S_{\text{vuln}} + 0.10\,S_{\text{cobenefit}} + 0.10\,S_{\text{cost}} \\ &= 0.15\,S_{\text{heat}} + 0.25\,S_{\text{vuln}} + 0.25\,S_{\text{cool}} + 0.15\,S_{\text{suit}} + 0.10\,S_{\text{cobenefit}} + 0.10\,S_{\text{cost}}. \end{aligned} \tag{26}$$

The effective weights are set out in Table 10.

Table 10: Effective weights in the NbS Cooling Opportunity Score, after decomposing the Heat Priority Index. The nominal weights are those of Equation (24); the effective weights are those of Equation (26). Both columns sum to unity, as the total row records. The two rows below that total are not additional weights: they restate, for emphasis, the totals that vulnerability and heat exposure reach once the Heat Priority Index is decomposed.

Component	Nominal weight	Effective weight
Heat Priority Index	0.25	—
of which heat exposure		0.15
of which vulnerability		0.10
Cooling Potential	0.25	0.25
NbS Suitability	0.15	0.15
Vulnerability (direct)	0.15	0.15
Co-benefits	0.10	0.10
Cost Feasibility	0.10	0.10
Total	**1.00**	**1.00**
Restated from the rows above, not added to them:		
Vulnerability, total		**0.25**
Heat exposure, total		**0.15**

The tool therefore weights *who is affected* (0.25) above *how hot the site is physically* (0.15). Between two equally hot sites, the one serving a more vulnerable population ranks higher.

This is a value position, not a technical necessity. There is no finding in the literature that requires it, and a reasonable reviewer may reject it. A city whose mandate is to reduce peak thermal load across its territory, rather than to direct adaptation investment toward disadvantaged communities, would be right to rebalance it.

It is disclosed here, and quantified, for a specific reason: an undisclosed double contribution would be a defect, indistinguishable from an arithmetic error, and a reviewer discovering it unaided would be entitled to distrust everything else in the methodology. Disclosed and quantified, it is a defensible position that reviewers may legitimately dispute. Any deployment may rebalance it in `config/weights.yaml` without touching code, and every result is to record the methodology version under which it was produced.

The double contribution is also not an accident of construction that happened to be noticed afterwards. It follows from a deliberate decision, recorded in the project's decision log as D-007, that the tool should prioritise who is affected slightly above how hot a place is. The alternative designs — dropping the direct vulnerability term, or removing vulnerability from the Heat Priority Index — were both available; each would have produced a defensible instrument with a different value stance.

8 Uncertainty, confidence, and missing data

8.1 Ranges, not point estimates

All quantitative outputs — temperature reduction, energy savings, greenhouse gases avoided, cost savings, payback — are reported as ranges. This is not a presentational preference; it is what the evidence supports.

The adopted cooling envelopes of Table 4 are wide, and they are wider at version 2026.08.05 than before. The narrowest are threefold (dense tree canopy, blue-green corridor, and riparian restoration, each 1.0–3.0 °C); the extensive green roof at 0.1–1.0 °C is tenfold; the large open water body at 0.1–3.0 °C is thirtyfold, because remote sensing and field measurement disagree by an order of magnitude and the tool declines to pick a side (Section 6.7.1); and four archetypes carry a genuine zero floor, for which no ratio is defined at all. The climate-differentiated estimates in Keravec-Balbot et al. (2026) carry their own uncertainty on top of that: street trees in arid climates are reported at (1.7 ± 1.4) °C, an interval nearly as wide as the estimate itself. A single figure drawn from within a spread of that magnitude would imply a precision that does not exist, and would be quoted, in practice, without its provenance.

The ranges compound. An energy estimate combines a temperature interval with a sensitivity interval (Equation (18)), so its width is the product of the two — the worked example in Section 11 produces an energy range spanning roughly an order of magnitude. That width is uncomfortable to present and it is the honest answer. A user who needs a tighter energy figure needs a building energy model, which is precisely the boundary Section 2.2 draws.

8.2 Confidence levels

Two distinct things are called confidence in this methodology, and separating them prevents a common confusion.

Evidence confidence is a property of an *archetype*, recorded in the library and inherited by every catalogue entry that resolves to it, expressing how well grounded its adopted values are:

High multiple converging sources, including a synthesis.

Medium a synthesis estimate with limited class-specific corroboration.

Low sparse, indirect, or conflicting evidence. The tool flags these results explicitly in its output.

Block confidence is a property of an *assessment*, expressing how completely the user filled in the inputs that a given output block depends on. It is computed as described below.

An archetype may have high evidence confidence in an assessment with low block confidence, or the reverse. Reporting a single merged figure would lose the distinction between “we know this intervention well but you told us little about your site” and “you described your site thoroughly but the evidence for this intervention is thin”. Those call for different responses.

8.3 Branched confidence

A single confidence rating would be dominated by whichever input block happened to be least complete, and would obscure the common case of a site with good physical data and no economic data. Confidence is therefore computed separately for each output block — cooling, energy, economic, and equity — from the completeness of the inputs feeding that block:

Table 11: The block confidence model. Completeness is the proportion of the fields relevant to that output block which the user supplied.

Completeness of relevant fields	Confidence	How the result should be read
Below 40 %	Low	A screening signal for further investigation
40–70 %	Medium	Indicative; the main drivers are known
Above 70 %	High	As reliable as a screening instrument gets

The fields counted for each block are enumerated in `config/derived_scores.yaml` (fixed at version 2026.08.01, decision D-024): ten slots for cooling — with the LST anomaly and the qualitative heat exposure level sharing a single either-of slot, since they are alternative signals for the same quantity — three for energy, four for economic, and six for equity. A field answered *unknown* counts as **not supplied**: an explicit unknown carries no more information than a skipped question. The thresholds apply with exact boundaries (below 40 % low, 40–70 % inclusive medium, above 70 % high), and the overall rating attached to the headline score is the *lower median* of the four block ratings — exact, slightly conservative, and never higher than three of the four blocks.

Two additional constraints apply.

8.3.1 Evidence confidence propagates

An archetype rated *low* evidence confidence in the library — at version 2026.08.05 the green façade / living wall, bioretention, the enclosed courtyard, the vegetated shade structure, and both productive classes, six of the eighteen — caps the cooling-block confidence at *medium* regardless of how complete the user's inputs are. The cap follows the inheritance, so every catalogue entry resolving to one of those six carries it.

The reason is straightforward: complete inputs cannot compensate for thin underlying evidence. A user who answers every question about a site where a green façade is proposed has told the tool a great deal about the site and nothing at all about whether green façades reliably reduce street-level air temperature, which is the question the disagreement between Keravec-Balbot et al. (2026) and Zölch et al. (2016) leaves open. The interface states the reason for the cap explicitly rather than showing an unexplained ceiling.

8.3.2 A high score with low confidence is a flag, not a verdict

The combination of a high Opportunity Score and low block confidence is presented as an invitation to investigate further, never as a conclusion. In practice this is the most dangerous output the tool can produce, because the headline number is exactly what gets extracted and quoted. The interface and the exported report attach the confidence rating to the score wherever the score appears, and Section 12.11 identifies quoting a score without its confidence as a misuse.

8.4 Missing data

Required inputs must be present; the assessment does not proceed without them. Optional inputs, when absent, follow explicit rules:

- **Qualitative inputs** default to *unknown*, which normalises to the neutral 50 of Section 5.4.
- **Quantitative inputs** propagate as *not calculated*. They are **never** treated as zero. A zero would understate the result while appearing to be a finding — an energy saving reported as 0 kW h reads as "this intervention saves no energy", which is a claim, whereas the truth is "we were not told the demand", which is an absence.
- **Derived outputs** whose inputs are missing carry an explicit status naming the reason, as the energy statuses of Section 7.9 illustrate.

Every default the engine applies is itemised in the result and reproduced in the exported report, so that a reader can see exactly which numbers came from the user and which from the tool. This list is not a diagnostic aid; it is part of the output, and a report circulated without it has removed the information needed to judge the numbers it contains.

9 Sensitivity analysis

This section specifies the design and reports its results.

The design below was published at methodology version 2026.07.30, before the calculation engine existed, precisely so that it could be criticised before execution and so that the results could not be shaped after the fact to suit the outcome. It has not been altered since. Section 9.4 reports the results at version 2026.08.05, produced by `tools/sensitivity_analysis.py` and regenerated whenever any aggregation weight changes.

The results are **re-confirmed unchanged at version 2026.08.05**. That version adds one methodology value—the Köppen–Geiger mapping of Section 5.8—and moves no aggregation weight, no typology value, no scoring formula and no golden scenario. The analysis was *re-executed rather than assumed*, and every figure below is identical to the one published at version 2026.08.04; the only difference in the generated file is its version stamp. A release that changes nothing the analysis measures should be able to demonstrate that rather than assert it.

The figures themselves were regenerated at version 2026.08.04, when the library moved from fourteen typologies to 110 entries over 18 archetypes and the scenario set grew from twenty to 24. The aggregation weights have been **unchanged** throughout: what moved at that version was the population of scenarios they are applied to.

9.1 Why sensitivity analysis is obligatory here

Nardo et al. (2008) treats uncertainty and sensitivity analysis as integral to a credible composite indicator rather than as an optional addition. The argument applies with particular force to this methodology.

The aggregation weights of Equation (24) are the tool's most consequential unsourced choice. Every other quantitative element — the typology envelopes, the canopy threshold, the energy sensitivity, the LST normalisation — traces to a published source that a reviewer can consult and dispute on its own terms. The weights trace to a judgment. That judgment determines the headline number, and therefore determines which projects a city funds.

Two responses to this are available. One is to argue that the weights are reasonable and hope the reader agrees. The other is to measure how much the output actually depends on them and publish the answer, including if the answer is inconvenient. Only the second is checkable, and it is the one adopted.

There is a further reason specific to this tool. If rank ordering turns out to be robust to weight perturbation, then the disclosed equity-forward stance of Section 7.13 — while remaining a genuine value choice — changes fewer decisions than its prominence suggests, and deployments can adopt the tool without first having to resolve a philosophical dispute. If ordering turns out to be fragile, users need to know that a different but equally defensible weight set would have produced a different priority list, and the tool should say so at the point of use. Both outcomes are informative; only running the analysis distinguishes them.

9.2 Design of the analysis

Each aggregation weight in Equation (24) is varied by ± 25 % of its nominal value, with the remaining weights renormalised to preserve a unit sum, and the full golden-scenario set is re-scored under each perturbation. Formally, for the perturbation of weight w_k by a factor $\lambda \in \{0.75,\ 1.25\}$:

$$w'_k = \frac{\lambda w_k}{\lambda w_k + \sum_{i \neq k} w_i}, \qquad w'_j = \frac{w_j}{\lambda w_k + \sum_{i \neq k} w_i} \quad \text{for } j \neq k. \tag{27}$$

The golden-scenario set is the same collection of hand-verified input/output cases that serves as the engine's regression suite (Section 10.5), so the scenarios exercised by the sensitivity analysis are the same ones whose correctness is independently asserted.

9.3 What is reported

Four quantities are reported for each perturbation, chosen because they correspond to four distinct ways the tool could mislead a user.

1. **Rank stability** — the proportion of scenario pairs whose relative ordering is preserved under the perturbation. This is the primary measure. Prioritisation is the tool's actual function, so ordering matters more than absolute values: a perturbation that shifts every score by five points but reorders nothing has not changed a single decision.

2. **Score displacement** — the distribution of the absolute change in the Opportunity Score across the scenario set, reported as a distribution rather than a mean, because a small average displacement concealing a few large ones is a materially different result from a uniformly small one.
3. **Category migration** — how often a scenario crosses a band boundary in Table 5, for instance from *Strong opportunity* to *Moderate opportunity*. Categories drive the recommendation text, so a category change is a change in what the tool *says*, not merely in what it scores. A score displacement of two points matters greatly at 80.5 and not at all at 45.
4. **Influence ranking** — which weights the output is most sensitive to, ordered. This tells a deployment considering recalibration which weight to argue about first, and it identifies which of the tool's value judgments are actually load-bearing.

9.4 Results at version 2026.08.05

The full scenario set (24 scenarios, 276 scenario pairs) was re-scored under each of the twelve perturbations — six weights, each at $\lambda \in \{0.75, 1.25\}$ of Equation (27).

Table 12: Sensitivity of the NbS Cooling Opportunity Score to ±25 % perturbation of each aggregation weight, across the 24 golden scenarios at version 2026.08.05. Rank stability is the proportion of the 276 scenario pairs whose ordering is preserved; displacement is the absolute change in the Opportunity Score (0–100 scale); migrations counts scenarios whose interpretation band changed.

Weight	Shift	Rank stab.	Mean disp.	Max disp.	Migrations
Heat Priority Index	−25%	0.9529	0.65	2.00	1
	+25%	0.9891	0.57	1.77	1
Cooling Potential	−25%	0.9819	0.79	2.05	2
	+25%	0.9601	0.69	1.79	1
NbS Suitability	−25%	0.9891	0.54	1.49	0
	+25%	0.9891	0.49	1.38	1
Vulnerability	−25%	0.9891	0.50	1.75	0
	+25%	0.9928	0.45	1.62	1
Co-benefits	−25%	0.9964	0.34	1.01	1
	+25%	0.9891	0.32	0.96	0
Cost Feasibility	−25%	0.9928	0.14	1.04	0
	+25%	0.9928	0.13	1.00	0

The four committed quantities, in the order of Section 9.3:

1. **Rank stability.** Worst case 0.9529, under the −25% Heat Priority Index perturbation. Ten of the twelve perturbations preserve at least 98 % of pair orderings. Reordering still occurs only between scenarios whose baseline scores differ by around two points or less — pairs the methodology itself would describe as materially equivalent.

2. **Score displacement.** Pooled over all 288 scenario-perturbation combinations: mean 0.47 points, median 0.34, 75th percentile 0.67, maximum 2.05 on the 0–100 scale.
3. **Category migration.** 8 of 288 combinations cross a band boundary. Six of the eight involve two scenarios sitting almost exactly on the Moderate–Strong line, at 59.58 and 59.89 — within half a point of the boundary. The remaining two are a Strong–High-priority crossing from 80.92 and a further crossing of the same near-boundary scenario. No scenario moves by more than one band, and no scenario far from a boundary migrates.
4. **Influence ranking.** By mean absolute displacement: Cooling Potential (0.74), Heat Priority Index (0.61), NbS Suitability (0.51), Vulnerability (0.48), Co-benefits (0.33), Cost Feasibility (0.14). Cost feasibility ranks last partly by construction: it is excluded and its weight redistributed in the nineteen of 24 scenarios that supply no cost data, so the weight binds only where economic evidence exists.

9.5 What last moved these figures, and why

The figures have not moved at version 2026.08.05. They last moved at version 2026.08.04, when every headline figure became slightly worse than at `2026.08.03`—and the honest reading of that change is that it is a property of the scenario set rather than a degradation of the methodology.

Headline figure	2026.08.03	2026.08.04	2026.08.05
Worst-case rank stability	0.9737	0.9529	0.9529
Pooled mean displacement	0.42	0.47	0.47
Category migrations	3 of 240	8 of 288	8 of 288
as a proportion	1.25 %	2.8 %	2.8 %

The cause is visible in the migration table. **Four of the eight migrations are a single new scenario**, the productive-landscape evidence gap, whose baseline score of 59.89 sits 0.11 points below a band boundary. A scenario that close to a boundary will cross it under almost any perturbation.

The four scenarios added at that version were chosen to exercise its *evidence* edge cases — the small-water envelope, the pergola-scale ceiling, the package cap, and the productive evidence gap — and not to sit comfortably inside bands. One of them landing on a boundary is a coincidence of that choice. Reporting it as though the tool had become less stable would be the wrong reading; suppressing it would be worse.

Interpretation. Under ±25 % perturbation of any single aggregation weight, the tool's priority ordering remains substantially stable: at least 95.29 % of pairwise orderings survive every perturbation, scores move by well under half a point on average, and category changes occur only for sites already sitting on a band boundary. The disclosed equity-forward stance of Section 7.13 therefore changes fewer decisions than its prominence suggests: a deployment that rebalances the weights within this range should expect a materially similar priority list unless two options were near-tied to begin with — and near-ties are exactly the cases the methodology already tells users not to resolve on the headline score alone.

A third qualification is now visible in the numbers, and it is the one this version adds. **The migration count is a weak statistic on a set of 24 scenarios**, because it is dominated by where those scenarios happen to sit relative to band boundaries: one near-boundary scenario contributes half of the migrations. Rank stability and displacement are the figures to read. The migration count is best read as "which scenarios sit on a boundary", which is what the paragraph above states directly.

Two further qualifications are owed and are unchanged. The analysis perturbs one weight at a time, so these figures are not bounds for simultaneous re-weightings. And the golden scenarios are designed to span the methodology's behaviour — climates, archetypes, data completeness, cost availability — not to be a probability sample of real assessments; the stability figures are properties of this set. Both qualifications argue for reading near-boundary results as ranges rather than verdicts, which the confidence mechanism already enforces at the point of use.

9.6 Publication and maintenance

The results above are also committed to the repository in machine-generated form (`docs/methodology/SENSITIVITY-ANALYSIS.md`) and in the Methodology Report, and must be regenerated whenever any weight changes. Because a weight change requires a methodology version bump (Section 13), and because the version stamp is enforced across all configuration files by continuous integration, a weight change that does not carry a regenerated sensitivity analysis is visible in review rather than silent.

The threshold at which a result would prompt a methodology change is stated in advance to avoid its being negotiated afterwards: if rank stability under any single ± 25 % weight perturbation falls below a level at which the tool's prioritisation could be considered reliable, the appropriate response is to report that limitation prominently at the point of use — telling the user that their priority ordering is weight-sensitive — rather than to adjust the weights until the number improves.

10 Implementation and reproducibility

This section describes how the methodology is realised in software. It is not an appendix to the science: it is the part of the paper that makes the preceding claims *checkable* rather than merely asserted. A methodology whose values live in prose can drift from the code that applies them, and a reader has no way to detect the drift. The arrangement described here is designed so that such drift fails the build.

10.1 Configuration as data

Every methodology value lives in schema-validated, version-stamped YAML under `config/`:

Table 13: The methodology configuration files.

File	Content
`nbs_typologies.yaml`	Both levels of the library (Section 6.1): the 18 cooling archetypes with their base cooling scores, temperature envelopes, evidence confidence, suitability conditions, co-benefit defaults, output caveats, and a `sources` array carrying for each citation the source key and the specific finding it supports; and the 110 catalogue entries with their identity, family, kind, primary cooling mechanism, availability, the archetype each inherits, and the 21 per-entry suitability overrides.
`availability.yaml`	The gating policy of Section 5.6: what each of the four site conditions means, what an *unanswered* condition means, what each assessment scale offers, the package size warning threshold, and the land-use mapping counts. It carries no performance value and feeds no score.
`weights.yaml`	Every weighted sum in Section 7, with its rationale, plus the sensitivity-analysis perturbation parameter.
`adjustment_factors.yaml`	The condition-level to factor table, the four derivation rules of Table 6, and the climate suitability matrix of Table 7.
`input_mapping.yaml`	Qualitative to numeric normalisation, the inverted scale, the LST normalisation, and the heat-index improvement buckets.
`energy_model.yaml`	The cooling-energy derivation of Section 7.9: the formula, the sensitivity interval of 0.02–0.04 per °C with its citation, the three preconditions, the reported statuses, and the recorded limitation on transferability.
`country_defaults.yaml`	Grid emission factor source, licence, and per-country schema; energy price policy. Deliberately contains no cost values.
`recommendation_templates.yaml`	The deterministic text fragments from which recommendations are composed, and the conditions under which climate-dependent caveats fire.
`derived_scores.yaml`	The derivation rules fixed at version 2026.08.01: suitability and equity sub-indicator rules, cost-feasibility payback brackets, investment-readiness rules, time-to-benefit buckets, the confidence field-to-block mapping, and the score interpretation bands.

The scoring code contains no typology value, no weight, and no threshold. Changing the methodology never requires changing code, which has three consequences. A deployment can rebalance the weights it disagrees with — including the equity-forward stance of Section 7.13 — without forking the software. A reviewer can read the complete parameterisation of the tool in nine configuration files rather than by tracing constants through source code. And the diff of a methodology change is legible: it shows values moving, not logic moving.

10.2 What is implemented at version 2026.08.05

This section distinguishes throughout between what the repository *contains* at methodology version 2026.08.05 and what it *specifies* for the next development phase. The distinction is stated because a reviewer can and should check it by cloning the repository, and because a paper that described planned software in the present tense would forfeit precisely the credibility this section exists to establish.

Implemented and running in continuous integration: the configuration layer — schema-validated loading of all nine methodology files, version lock-step enforcement, and the citation cross-check against the bibliography — the six evidence rules of Section 10.4; the calculation engine itself, comprising the input and result schemas, one scoring module per formula family, the availability matrix, the package combination rules of Section 7.7, branched confidence, the deterministic recommendation composer, and the orchestrating entry point, under a 100 % line-coverage gate with 24 hand-verified golden scenarios, a byte-identical determinism test, and the executed sensitivity analysis of Section 9.4; the report and workbook export; the HTTP API with its explicit stored-draft migration (Section 13); and the web interface.

The claims in this section are about what the repository contains, and a reviewer should check them by cloning it. Where this paper describes a behaviour the engine is *specified* to have rather than one it exhibits, it says so.

Every number in the worked example of Section 11 was first computed by hand from the specification and is now also asserted, digit for digit, by the engine's golden-scenario suite — the hand derivation is the expected value, never the engine's own output.

10.3 The engine contract

The calculation engine is specified as a pure function

$$\texttt{run\_assessment}(\textit{AssessmentInput},\ \textit{MethodologyConfig}) \longrightarrow \textit{AssessmentResult}$$

with no input or output, no network access, no global mutable state, and no stochastic component. Determinism is therefore a structural property of the design rather than a behaviour that testing hopes to confirm: there is no source of variation for a test to detect. A determinism test is nonetheless run, on the principle that a structural guarantee worth relying on is worth verifying.

Each result records both the engine version and the methodology version that produced it, and comparisons between results computed under differing methodology versions raise a warning rather than being silently rendered side by side.

The contract was published in advance of its implementation, which is the point at which it could still be criticised cheaply, and it has not been relaxed since. A reviewer who believes it is too strong, too weak, or unenforceable is invited to say so.

10.4 Machine-enforced evidence rules

The project's evidence policy is a build gate, not a review convention. Configuration loading fails — and continuous integration fails with it — if any of the following holds.

1. **A performance value carries no citation.** The schema requires a non-empty `sources` array on every archetype, and each entry in it must carry both a source key and the finding it supports. Because catalogue entries inherit their values, the check runs over the *resolved* library, so an entry cannot acquire an uncited value through inheritance either.
2. **A citation points at an unrecorded source.** Every `sources[].key` appearing anywhere in the configuration tree is collected recursively and checked against the set of keys declared in the bibliography. A citation naming a source the bibliography does not record is a hard error. This is the rule that makes the traceability claim of Section 2.4 enforceable: a value cannot cite something that does not exist.
3. **The configuration files fall out of version lock-step.** Every file carries a top-level date-stamped `version`, and all must agree. A methodology version moves as a single unit or not at all.
4. **A temperature envelope is inverted or implausible.** The maximum must not fall below the minimum, and no resolved entry may declare a maximum above 3.5 °C. The upper guard encodes a methodology judgment: no retrieved source supports a daytime pedestrian-level air temperature reduction above roughly 3 °C for a single site-level intervention, so a larger value in configuration indicates a calibration error rather than new evidence. Raising the guard requires changing the test, which makes the change visible in review.
5. **The low-confidence declarations are removed.** A test asserts that the green façade / living wall, bioretention, the enclosed courtyard, the vegetated shade structure, and both productive archetypes remain declared as low evidence confidence, guarding against a future edit quietly promoting a weakly evidenced class.
6. **Default costs appear.** A test asserts that the country defaults ship no cost or emission values pending verification, and that the typology library carries no per-square-metre unit cost. The cost policy of Section 7.11 holds in configuration, not only in prose.

The practical effect is that the integrity claims made in this paper degrade loudly rather than quietly. A contributor who adds a typology without a citation, or who bumps one configuration file without the others, does not produce a subtly wrong tool; they produce a red build.

10.5 Testing and continuous integration

The configuration layer. The test suite asserts that all 18 archetypes and all 110 entries load, resolve, and validate, that every entry names an archetype that exists, that every configuration file shares one version stamp, that every resolved entry carries at least one citation with a stated finding, that every source key resolves against the bibliography, that temperature envelopes are ordered and below the 3.5 °C guard, that the six low-confidence archetypes remain declared as such, and that no default cost or unverified emission value ships. Negative cases are tested explicitly: an invented citation key, a version mismatch, an uncited class, and an inverted envelope must each fail the load.

These are the six evidence rules of Section 10.4. Continuous integration runs linting, format checking, strict static type checking, and the full test suite — configuration, scoring, confidence, recommendation, scenarios, determinism — with the 100 % engine coverage gate on every push and pull request, across supported Python versions.

The engine. Unit tests cover every scoring module, with the 100 % engine line-coverage target enforced as a build gate; 24 golden scenarios, each a stored input with a hand-verified expected output and its derivation, serve as regression armour against unintended methodology drift and as the basis of the sensitivity analysis of Section 9.4; and a determinism test asserts byte-identical results across repeated runs and freshly loaded configurations.

Three methodology properties are asserted as tests rather than stated as prose, because each is a rule a plausible future edit would break. *Availability feeds no score:* one site is scored twice, with all four gating questions answered and with none of them answered, and the two results must be byte-identical (Section 5.6). *The two water archetypes are never interpolated by area:* the same constructed wetland scored on two sites differing in area by a factor of 2 500 must report an identical ceiling (Section 6.7.2). And *the area-scale shading figure does not transfer to pergola scale:* the vegetated shade structure must not cite it, while the permeable shaded hardscape must (Section 6.7.4). The nine published availability situation counts and the four land-use totals — including the 72 entries that name *school* in their land-use list — are likewise asserted, so a land use silently losing its interventions fails the build.

The ordering is deliberate rather than incidental. The evidence rules were implemented before the scoring code they protect, so that no scoring value can enter the repository uncited at any point in the project's history — not even temporarily, during development.

10.6 Methodology versioning stamped into every result

The methodology version — 2026.08.05 at the time of writing — stamps this document, the Methodology Report in the repository, and every configuration file, and is recorded, together with the engine version, in every assessment result the engine produces. Lock-step across the configuration files is enforced by the build.

This has a consequence that matters for how the tool behaves over time. Existing assessments are *never* to be silently recomputed. A result retains the version that produced it, and the interface indicates when a newer methodology version is available, leaving the decision to re-run with the user. A city whose priority list was produced under version 2026.08.05 can therefore explain, years later, exactly which rules generated it — because those rules are a tagged, public, immutable artefact rather than the current state of a codebase.

10.7 What this arrangement does and does not guarantee

It is worth being precise about the limits of these guarantees, since this section makes the strongest reproducibility claims in the paper.

The arrangement guarantees that every methodology value in the repository carries a citation to a recorded source, that no citation names a source the bibliography does not hold, that a change to any value is versioned and visible across every file at once, and — since the engine and its coverage gate exist — that the tool applies exactly the methodology described in this document. These are properties of internal consistency and traceability, not of correctness.

It does *not* guarantee that the methodology is correct, that the adopted envelopes are the right ones, or that the cited sources support the weight placed on them. Those are scientific questions, and no build gate can settle them. What the arrangement does is make them *answerable* by an external reviewer: the complete parameterisation is public, each value names its source, and Appendix E states what was actually verified about each source. The engineering makes the science auditable; it does not make it right.

11 Worked example

This example is illustrative. The site does not exist, and the result is not a finding about any real place.

The calculation was performed **by hand** from the formulas and configuration values specified in this paper, and every number below is also asserted, digit for digit, by the engine's golden-scenario suite — the hand derivation is the expected value, never the engine's own output. Its purpose is to make the specification concrete enough to check: a reader who disagrees with any number below can locate the equation that produced it and recompute it.

One quantity still rests on an assumption: the grid emission factor of Section 7.10, which no shipped configuration supplies. Section 11.7 states how much the conclusion depends on it.

11.1 The illustrative site and intervention

A neighbourhood-scale street corridor in a dense, largely sealed inner-city area in a temperate climate. The proposed intervention is a *tree avenue* — one of the 110 catalogue entries — which inherits the *street tree canopy* archetype (Table 4) and with it an envelope of 0.5–3.0 °C, a base cooling score of 75, and a *high* evidence rating. Every performance number below is the archetype's, and the result says so. This is a package of one: the combination rules of Section 7.7 apply but have nothing to combine. Table 14 gives the inputs.

11.2 Layer 1 — Baseline assessment

An LST anomaly is available, so the data-rich path of Equation (4) applies. Normalising by Equation (1), $S_{\mathrm{LST}} = \mathrm{clamp}(10 \times 4.2) = 42.0$. The impervious percentage enters directly as

Table 14: Inputs to the illustrative assessment. Fields left unsupplied are shown explicitly, because their absence changes both the result and its confidence.

Group	Field	Value
Project	Assessment scale	Neighbourhood
Site	Site area	6 000 m^2
	Existing tree canopy	8 %
	Existing green cover	12 %
	Impervious surface	85 %
	Soil availability	Limited
	Irrigation availability	Occasional
Climate	Climate zone	Temperate
	Land surface temperature anomaly	4.2 °C
	Solar exposure	High
Vulnerability	Population density	High
	Vulnerable population presence	High
	Access to cooled indoor space	Low
Intervention	Intervention selected	Tree avenue
	Archetype inherited	Street tree canopy
	New canopy area at maturity	1 200 m^2
Energy	Nearby building cooling demand relevant	Yes
	Annual cooling energy demand	450 000 kW h
	Energy price	*not supplied*
Cost	Capital cost	*not supplied*

$S_{\text{imp}} = 85$; solar exposure *high* normalises to $S_{\text{solar}} = 75$ by Table 2; and the vegetation deficit is $S_{\text{vegdef}} = \text{clamp}(100 - 12) = 88$ by Equation (5). Then

$$\begin{aligned} S_{\text{heat}} &= 0.40(42.0) + 0.25(85) + 0.20(75) + 0.15(88) \\ &= 16.80 + 21.25 + 15.00 + 13.20 \;=\; \mathbf{66.25}. \end{aligned}$$

For vulnerability, population density and vulnerable-group presence are both *high* and normalise to 75. Access to cooled indoor space is *low*, which under the inverted mapping of Section 5.3 yields a cooling access *deficit* of 100. By Equation (6),

$$S_{\text{vuln}} = 0.40(75) + 0.40(75) + 0.20(100) = 30.00 + 30.00 + 20.00 = \mathbf{80.00}.$$

The Heat Priority Index follows from Equation (8):

$$\text{HPI} = 0.60(66.25) + 0.40(80.00) = 39.75 + 32.00 = \mathbf{71.75},$$

which falls in the *High* band of Table 5. Note that vulnerability (80.0) exceeds heat exposure (66.25) here, so the equity-forward weighting of Section 7.13 raises this site's final score relative to a heat-only instrument. That is the disclosed value choice operating as intended, and a reviewer who rejects the choice should read this result as correspondingly inflated.

11.3 Layer 2 — NbS performance

Each of the four adjustment conditions of Table 6 is derived from inputs already supplied.

Canopy. Existing tree canopy is 8 % of 6 000 m^2, or 480 m^2. Adding 1 200 m^2 of new canopy at maturity gives 1 680 m^2, which is 28.0 % of the site — inside the *good* band ($25 \le c < 40$), so $f_{\text{canopy}} = 1.0$. The site falls short of the 40 % inflection identified by Ziter et al. (2019), which is why it does not reach *excellent*.

Soil–water. Soil availability *limited* maps to *moderate*; irrigation availability *occasional* also maps to *moderate*. The more limiting of the two is *moderate*, so $f_{\text{soilwater}} = 0.8$.

Scale. Assessment scale *neighbourhood* maps to *good*, so $f_{\text{scale}} = 1.0$.

Climate. The street tree canopy is a *green*-family archetype and the site is in a *temperate* zone, which Table 7 rates *good*, so $f_{\text{climate}} = 1.0$.

By Equation (9),

$$A = 0.40(1.0) + 0.25(0.8) + 0.20(1.0) + 0.15(1.0) = 0.400 + 0.200 + 0.200 + 0.150 = \mathbf{0.95}.$$

The inherited street tree canopy archetype has base cooling score $B = 75$ and envelope $[\tau_{\min}, \tau_{\max}] = [0.5,\ 3.0]$. By Equation (10),

$$S_{\text{cool}} = \text{clamp}(75 \times 0.95) = \mathbf{71.25}.$$

The temperature range follows from Equations (11) and (12), and the clipping rule binds on the lower bound:

$$\Delta T_{\text{min}} = \text{clip}(0.5 \times 0.95,\ 0.5,\ 3.0) = \text{clip}(0.475,\ 0.5,\ 3.0) = \mathbf{0.50}\ ^\circ\text{C},$$
$$\Delta T_{\text{max}} = \text{clip}(3.0 \times 0.95,\ 0.5,\ 3.0) = \text{clip}(2.85,\ 0.5,\ 3.0) = \mathbf{2.85}\ ^\circ\text{C}.$$

The raw product $0.5 \times 0.95 = 0.475$ falls below the literature floor and is clipped up to it; the upper bound is unaffected because $A < 1$. This is the point marked on Figure 5.

The midpoint is $\overline{\Delta T} = (0.50 + 2.85)/2 = 1.675\,^\circ\text{C}$, which exceeds $1.5\,^\circ\text{C}$ and therefore falls in the *high* heat-index improvement bucket. Shade potential, by Equation (13), is $\text{clamp}(1200/6000 \times 100) = 20.0\,\%$.

11.4 Layer 3 — Impact and feasibility

Energy. All three preconditions of Section 7.9 hold: nearby building cooling demand is confirmed relevant, an annual demand of 450 000 kW h is supplied, and the street tree canopy archetype is flagged as capable of affecting adjacent building loads. By Equation (18),

$$E_{\text{min}} = 450\,000 \times 0.50 \times 0.02 = 4\,500\,\text{kW h/yr},$$
$$E_{\text{max}} = 450\,000 \times 2.85 \times 0.04 = 51\,300\,\text{kW h/yr}.$$

The upper bound is roughly eleven times the lower. That width is not a defect of the arithmetic; it is the product of a sixfold temperature envelope and a twofold sensitivity range, and it is the honest consequence of what the evidence supports. A user who needs a tighter figure needs a building energy model.

Greenhouse gases. Version 2026.08.05 ships no country emission factors (Section 7.10). *For this illustration only*, assume the deploying authority has supplied a locally verified grid emission factor of 0.30 kg h/kW. This value is not a Criterra estimate, is not shipped with the tool, and carries no citation; it is a placeholder input chosen to demonstrate Equation (19). On that assumption,

$$G = 1\,350\text{–}15\,390\ \text{kg CO}_2\text{e per year, or about } 1.4\text{–}15.4\ \text{t CO}_2\text{e per year.}$$

Under the shipped configuration with no factor supplied, this output would read *not calculated*.

Costs. No capital cost and no energy price were supplied. Consistent with Section 7.11, capital cost, annual cost savings, simple payback, and cost feasibility are all reported as *not estimated*. Cost feasibility is therefore excluded from the final aggregation and its weight redistributed.

Co-benefits. The street tree canopy archetype carries library default levels of *medium* biodiversity, *medium* stormwater, *high* public health, and *high* urban quality. As noted in Section 7.8, the library supplies no default for social inclusion, so it falls to the neutral *unknown* value of 50 and is itemised as an applied assumption. By Equation (15),

$$\begin{aligned} S_{\text{cobenefit}} &= 0.25(50) + 0.25(50) + 0.20(75) + 0.15(50) + 0.15(75) \\ &= 12.50 + 12.50 + 15.00 + 7.50 + 11.25 = \mathbf{58.75}. \end{aligned}$$

NbS Suitability. Applying the rules of Table 9: the site offers sixty times the 100 m^2 minimum inherited from the archetype ($S_{\text{space}} = 100$); soil availability *limited* exactly meets the inherited *limited* requirement ($S_{\text{soil}} = 75$); irrigation *occasional* exactly meets the *occasional* requirement ($S_{\text{water}} = 75$); maintenance intensity and land use were not supplied, so both take the neutral 50, itemised. Then

$$S_{\text{suit}} = 0.30(100) + 0.25(75) + 0.20(75) + 0.15(50) + 0.10(50) = \mathbf{76.25},$$

close to — and slightly below — the 78 assumed in the previous version of this example, because the neutral defaults for maintenance and land use hold the score down where full inputs would not.

11.5 Aggregation

Cost feasibility is unavailable, so Equation (25) applies over the remaining five components, whose nominal weights sum to $\sum_{i \in \mathcal{A}} w_i = 0.90$.

Table 15: Aggregation of the illustrative assessment. Cost feasibility is excluded and its 0.10 weight redistributed proportionally across the remaining five components.

Component	**Score**	**Nominal w_i**	**Renormalised w'_i**	**Contribution**
Heat Priority Index	71.75	0.25	0.2778	19.93
Cooling Potential	71.25	0.25	0.2778	19.79
NbS Suitability	76.25	0.15	0.1667	12.71
Vulnerability	80.00	0.15	0.1667	13.33
Co-benefits	58.75	0.10	0.1111	6.53
Cost Feasibility	*not estimated — excluded*			—
NbS Cooling Opportunity Score				**72.29**

The NbS Cooling Opportunity Score is **72.3**, which falls in the *Strong opportunity* band (61–80) of Table 5.

11.6 Confidence and reported result

Applying the branched confidence model of Section 8.3 with the field-to-block mapping enumerated at version 2026.08.05: the cooling block has eight of its ten slots supplied (80 % → *high*; the street tree canopy archetype carries *high* evidence confidence, so no cap applies). The energy block has the relevance confirmation and the demand but no energy price (two of three, 66.7 % → *medium*). The economic block has none of its four fields (*low*). The equity block has three of its six fields (50 % → *medium*). The overall rating is the lower median of {high, medium, low, medium}: *medium*.

Table 16: The illustrative assessment as it would be reported, with confidence ratings and applied assumptions. All temperature values are daytime, pedestrian-level air temperature reductions.

Output	Value	Confidence
Heat Priority Index	71.8 — *High*	High
Cooling Potential Score	71.3	High
Temperature reduction	0.50–2.85 °C (daytime, pedestrian-level air; on the street tree canopy evidence)	High
Heat-index improvement	*High* (midpoint 1.675 °C)	High
Shade potential	20.0 % of site	High
Energy savings	4 500–51 300 kW h per year	Medium
GHG avoided	1.4–15.4 t CO_2e per year[b]	Medium
Capital cost	*Not estimated — no cost data supplied*	Low
Simple payback	*Not estimated*	Low
Cost feasibility	*Not estimated — excluded from aggregation*	Low
Co-benefit Score	58.8	Medium
Opportunity Score	**72.3 — *Strong opportunity***	Medium

[b] On an assumed locally supplied grid emission factor of 0.30 kg h/kW. The shipped configuration would report this as *not calculated*.

Assumptions the engine itemises for this run: maintenance intensity and land use defaulted to the neutral 50 (suitability); the grid emission factor supplied by the user rather than from configuration; energy price absent, so cost savings not computed; the four cited co-benefit library defaults applied; social inclusion defaulted to the neutral 50 (no library default); public accessibility, safety concern, and community participation defaulted to the neutral 50 (equity); cost feasibility not estimated, excluded from the aggregation with its weight redistributed.

11.7 How much the result depends on the assumed values

At the previous methodology version the NbS Suitability Score was assumed rather than derived, and this subsection quantified the assumption's reach: across the full 0–100 range of the sub-score the Opportunity Score spans 67.9–76.3 and never leaves the *Strong opportunity* band. The sub-score is now derived (Table 9), and its derived value of 76.25 lands within two points of the earlier assumption — the conclusion is unchanged. Category-migration behaviour across the whole scenario set is quantified in Section 9.4.

The one remaining assumed input, the grid emission factor, affects only the greenhouse-gas output, which scales linearly with it and enters no score.

11.8 What this example demonstrates

Five properties of the methodology are visible in the arithmetic above, and each corresponds to a design commitment from Section 2.4.

The evidence class is named, not implied. Every performance value used above — the base score, the envelope, the minimum area, the soil and irrigation requirements, the co-benefit defaults, the evidence rating, and the building-energy applicability — belongs to the street tree canopy archetype, not to the tree avenue entry, and the result says so. A reader can therefore see at what resolution the claim is made, which is the point of Section 6.1 and the limitation of Section 12.5.

The clipping rule binds in practice. The raw lower bound $0.5 \times 0.95 = 0.475$ was clipped up to the literature floor. On a better site the rule would bind at the ceiling instead, and the reported range would not exceed 3.0 °C regardless of how favourable the conditions were.

Refusal is a first-class output. Three of the twelve reported quantities are *not estimated*, and a fourth — annual cost savings — is not reported at all for want of an energy price. The tool did not substitute a plausible capital cost, and consequently did not produce a payback figure — the single number most likely to be extracted from a report and quoted without qualification.

Exclusion preserves ordering. Because cost feasibility was excluded rather than set to a neutral 50, this assessment remains comparable to another option for the same site that also lacks cost data, without either being rewarded for the absence.

The ranges are wide, and stay wide. The reported temperature range spans nearly a factor of six and the energy range a factor of eleven. Nothing in the methodology narrows them for presentational comfort.

12 Limitations and responsible use

The limitations below are not a disclaimer appended for legal comfort. Several of them — the daytime restriction, the air-temperature-only scope, the geographic skew of the evidence — materially constrain what the tool's outputs mean, and a user who does not know about them will misread the results. They are stated at length for that reason.

12.1 Screening level

Outputs are indicative estimates produced from simplified typology factors and from user-supplied or default assumptions. They are not microclimate simulation outputs, and they must not be presented as predicted temperatures at a location. The tool has no representation of site geometry, no surface energy balance, no wind field, and no time dimension beyond the daytime/nighttime distinction it declines to model.

The practical consequence is that small differences in an Opportunity Score between two options carry little meaning. The tool distinguishes a strong opportunity from a weak one; it does not rank two strong ones against each other with any reliability. Users comparing closely scored options should treat them as tied and decide on grounds the tool does not represent.

12.2 Daytime only

All cooling values in this methodology are daytime values. The tool produces no nighttime estimate.

This is a substantive restriction, not a scoping convenience, because the evidence indicates that nighttime behaviour differs both in magnitude and in mechanism. Keravec-Balbot et al. (2026) reports substantially smaller nighttime effects — parks at 1.2 °C against 1.3 °C by day, green walls at 1.4 °C against 3.0 °C. Ziter et al. (2019) finds canopy cooling minimal at night while the warming effect of impervious surfaces *persists* overnight, which means that at night the two mechanisms represented in the tool's Heat Exposure Score no longer offset one another.

The importance of this is that nighttime heat is strongly associated with heat-health outcomes: sustained overnight temperatures prevent physiological recovery, and it is that failure to recover which drives excess mortality during heat events. A favourable daytime result from this tool should **not** be read as evidence of overnight heat relief, and an intervention selected on the strength of a daytime score may deliver much less of the health benefit a city is actually seeking.

The evidence retrieved for version 2026.08.05 sharpens this limitation into a warning for two classes of intervention. They do not merely cool less at night; they warm.

Water. Yao, Sailor, Zhang, et al. (2023) measures 1.8 °C of nocturnal warming at an urban pond, Yao, Sailor, Yang, et al. (2023) measures a 1.2–1.3 °C nocturnal heat island at urban lakes, and Ampatzidis and Kershaw (2020) states that blue space may exacerbate the nighttime urban heat island outright, with sensible cooling partly offset by increased water-vapour content.

Irrigated productive landscapes. Cheung et al. (2022) predicts an increase in the daily *minimum* air temperature, as heat stored in wet soil through the day is released overnight.

In both cases the effect is **outside** the tool's daytime estimate rather than netted off against it, and both archetypes carry an output caveat saying so. A daytime-only tool that quietly omitted this would be reporting only the favourable half of what its own sources found.

A user whose primary concern is nighttime heat should treat this tool's output as addressing a related but different question.

12.3 Air temperature only — thermal comfort is not modelled

The tool estimates air temperature reduction. It does not quantify thermal comfort, which is what a person standing in the street actually experiences.

The two diverge most where shade is the dominant mechanism. Shade improves comfort substantially by reducing radiant load, even where the effect on air temperature is small, and comfort indices capture this while air temperature does not. Zölch et al. (2016) reports a 13 % PET reduction from trees at pedestrian level, which illustrates the magnitude of what is omitted; that study also found that placement matters more than quantity, a design insight this methodology cannot express through an air-temperature envelope.

The direction of the resulting error is known and is worth stating plainly: **the tool systematically understates the benefit of shade-dominant interventions.** Every entry inheriting the street tree canopy, the continuous shaded corridor, the permeable shaded hardscape, or the vegetated shade structure delivers more human benefit than its air-temperature figure conveys. The vegetated shade structure is the starkest case, and it is measured: Ouyang et al. (2024) reports an air temperature reduction of 1.42 °C alongside a mean radiant temperature reduction of 15.93 °C for the same natural shade — an order of magnitude apart. The tool reports the first number and says so.

This bias was not corrected, and the decision not to correct it deserves explanation. Applying an uplift to shade-dominant typologies would require a conversion factor between air temperature and comfort that no retrieved source supplies, and inventing one would trade a known, stated, one-directional bias for an unknown one concealed inside a number. A stated bias that users can reason about is preferable to a hidden correction they cannot.

12.4 The evidence base is geographically uneven

The underlying literature is dominated by temperate and subtropical Northern Hemisphere cities. Tropical and arid-climate estimates rest on fewer studies with wider intervals — visible in Figure 1, where the tropical street-tree estimate carries a ±1.6 °C interval and the arid estimate ±1.4 °C on a central value of 1.7 °C, an interval nearly as large as the estimate itself.

The energy sensitivity of Section 7.9 derives largely from North American evidence, and its transferability to other building stocks and climates is untested.

Applications in under-represented contexts should treat outputs as more uncertain than the confidence rating alone suggests. The confidence model of Section 8.3 measures input completeness; it does not measure how well the evidence base covers the user's part of the world. A complete assessment of a site in a tropical city may report high cooling confidence while resting on thinner evidence than the same assessment in a temperate one. This is a known gap in the confidence model and a candidate for a future methodology version.

12.5 An entry's cooling value is its archetype's, not its own

This is the limitation created by the archetype model of Section 6.1, and it is the price of the honesty that model buys.

110 catalogue entries are served by 18 cited evidence classes. Fifteen entries therefore report the street tree canopy's 0.5–3.0 °C and eleven report the green façade's 0.3–2.0 °C, whatever distinguishes one entry from another within the class. A suspended-pavement tree pit and a continuous tree trench are not the same intervention, and the tool does not pretend to distinguish their cooling.

The alternative was worse. Assigning 110 separate envelopes would have required inventing 92 of them, since solution-specific measurement does not exist at that resolution, and a fabricated envelope is indistinguishable in a report from a measured one. What the tool does instead is inherit a *cited* envelope and **name the evidence class in every result**, so that a reader can see the resolution of the claim rather than having to infer it. Two entries reporting the same range because they inherit the same class is a visible fact about the output, not a hidden assumption inside it.

The consequence for use is direct: **the tool does not discriminate between entries within one archetype on cooling**. Two such entries are separated, if at all, by their suitability conditions, their co-benefit defaults, and their availability — not by their temperature range. A user choosing between them on the strength of a cooling difference is reading a difference that is not there.

12.6 Newer and larger published figures are not adopted

Section 6.6.4 records that Kumar et al. (2024) reports figures substantially above this library's envelopes for several intervention types — green walls at (4.1 ± 4.2) °C, vegetated balconies at (3.8 ± 2.7) °C, botanical gardens at (5.0 ± 3.5) °C — and that they are deliberately not adopted, because selective adoption would break internal consistency and wholesale adoption is a full recalibration requiring its own sensitivity analysis.

A reader should understand what that means for the outputs: **if those figures are right, this tool understates the corresponding interventions.** The non-adoption is a defensible methodological choice and it is also a live disagreement with a recent review, held open rather than resolved. It is recorded as a candidate for a future methodology version, and a reviewer who believes the recalibration should happen now is making an argument the project would need to answer with evidence rather than with precedent.

12.7 Modelling bias is mitigated, not eliminated

Much of the synthesised evidence derives from simulation rather than field measurement, and simulation tends to report larger effects. Section 6.6.1 describes the downward correction applied through conservative bound-setting. That correction is a judgment about direction informed by a single observable ratio; it is not a calibrated adjustment, and it does not eliminate the bias. Adopted values may still be optimistic.

12.8 No default costs

Economic outputs require user-supplied costs (Section 7.11). Absent them, the tool reports no capital cost, no payback, and no cost feasibility, and excludes the latter from aggregation. A user seeking an investment case from the tool alone will not get one. This is deliberate, and the reasoning is given in Section 7.11; but it is a real functional limitation and should be planned for.

12.9 The weights are contestable

The aggregation weights, including the equity-forward emphasis quantified in Section 7.13, are value choices. Users who disagree should adjust them in `config/weights.yaml` and re-run, rather than reinterpreting outputs produced under weights they reject. Reinterpretation is untraceable; reconfiguration is versioned, and the resulting scores carry the version stamp of the configuration that produced them.

12.10 Specification gaps at version 2026.08.05

The three specification gaps disclosed at version 2026.07.30 — the input-to-sub-indicator rules for the NbS Suitability and Equity scores (Section 7.8), the payback bracket boundaries and combination rule for cost feasibility (Equation (22)), and the enumeration of which input fields feed which confidence block (Section 8.3) — were closed at version 2026.08.01 and remain closed. They are named here because a reader comparing versions should be able to see what happened to them, not because they are still open.

One gap remains, and it is a data gap rather than a specification gap: the country emission factor table ships **empty** (Section 7.10), so greenhouse-gas outputs are reported as *not calculated* unless a deployment supplies its own factor. This is the verification policy operating as designed — no unverified value ships — and it is nonetheless a functional limitation. Closing it requires per-country verification, a version bump, and a corresponding update to this document in the same change set.

12.11 Misuse cases

Three specific misuses are anticipated. They are named because each is likely, each is damaging, and each is easy to commit without intending to.

12.11.1 Treating outputs as predicted temperatures

The most likely misuse is quoting a temperature reduction range as a prediction: "this project will reduce temperatures by up to 2.85 °C". It will not. The range is an envelope of what comparable interventions have delivered in the published literature, narrowed by a coarse site adjustment. It carries no location, no time of day beyond "daytime", no meteorological condition, and no confidence interval in the statistical sense.

The correct formulation names the metric, the range, and the basis: "a screening assessment indicates an indicative daytime pedestrian-level air temperature reduction in the range 0.5–2.85 °C, based on published evidence for this intervention type adjusted for site conditions". This is longer, and it is what the tool actually supports.

The failure mode is particularly acute where a range is collapsed to its upper bound for a headline. The upper bound of an envelope describes a well-executed instance under favourable conditions; it is not the expected outcome.

12.11.2 Quoting scores without their confidence

An Opportunity Score of 72.6 at medium confidence and an Opportunity Score of 72.6 at low confidence are different results, and the difference is not cosmetic — the second may rest largely on the neutral defaults of Section 5.4 rather than on anything the user knew about the site.

Scores extracted into a slide, a table, or a funding proposal without their confidence rating and their methodology version have been stripped of the information needed to judge them. The tool attaches confidence wherever a score appears, and reports circulated onward should preserve that pairing. Where a score is quoted, the methodology version should be quoted with it, because the same site may score differently under a later version.

12.11.3 Using the tool to justify a decision already taken

The most consequential misuse is the least visible one. Because the tool is transparent, configurable, and produces a defensible-looking number, it can be run after a decision has been made in order to furnish that decision with evidence — by selecting the option already chosen, by adjusting weights until the preferred site ranks first, or by presenting a single assessment without the alternatives that were considered.

Nothing in the software can prevent this, and no methodology can. What the design does is leave traces: the methodology version is stamped on every result, the weights used are recorded in configuration, applied defaults are itemised, and comparison of options for the same site is a first-class feature rather than an afterthought. An assessment presented without its alternatives, or produced under non-default weights that are not disclosed, should be questioned on those grounds.

The tool is built to compare options and set priorities. Used to ratify a conclusion reached elsewhere, it lends the appearance of rigour to a process that had none, which is worse than having no tool at all.

12.12 Responsible use in summary

- Use the tool to compare options and set priorities, not to justify a decision already taken.
- Combine results with local knowledge and stakeholder input; the tool represents neither.
- Validate assumptions through site visits before committing resources.
- Re-run assessments as better data becomes available, and expect the result to change.
- Report the confidence rating and the methodology version alongside any score that is quoted.
- Name the metric whenever a temperature is reported: daytime, pedestrian-level, air temperature.
- Carry the evidence class with the number. A cooling range is the archetype's, not the entry's (Section 12.5), and the result states which archetype it came from for exactly this reason.

13 Governance and methodology evolution

13.1 Versioning

The methodology version — 2026.08.05 — stamps this document, the Methodology Report in the repository, and every configuration file, and is to be recorded in every assessment result the engine produces. The version is a date stamp rather than a semantic version, because a methodology does not have a meaningful notion of a backward-compatible change: any alteration to a value may alter a ranking.

A change to any methodology value requires a version bump and a corresponding update to this document in the same change set. Continuous integration enforces the mechanical half of this — version lock-step across configuration files, and citations for every performance value (Section 10.4) — while the substantive half, that the documentation actually describes the new value, is enforced in review.

Existing assessments are never to be silently recomputed: a result retains the version that produced it, the interface indicates when a newer methodology version is available, leaving the decision to re-run with the user, and comparisons between results produced under differing versions raise a warning.

Version 2026.08.05 is the first release to exercise this properly, because it is the first to break something. The intervention field became a list and the input schema gained four fields, so stored *drafts* cannot be read unchanged. They are migrated **explicitly** rather than silently: the single intervention is wrapped into a one-element list, the four new fields are left absent rather than defaulted, and the migration is itemised for the user. The migration notes distinguish two cases that would otherwise be confused — a slug that no longer names any selectable entry, whose draft must be re-chosen, and a slug that still names a catalogue entry but whose inherited evidence class changed, which needs no re-selection. Telling a user to re-select an intervention that is still there would be a false alarm about their own saved work. Stored *results* are not touched in either case.

13.2 The public change process

Methodology changes are proposed as public pull requests, accompanied by the evidence supporting them. A proposal that changes a quantitative value is expected to state which source supports the new value, what was verified about that source and how, and why the existing value should be displaced. The verification standard is the one described in Appendix E: bibliographic details checked against the publisher record, an institutional repository, or an indexing service, with the verification level recorded honestly rather than overstated.

> **A design may be approved; a citation is only settled once it has been read.** Version 2026.08.05 supplies the case that makes this concrete. An envelope floor was approved on the strength of a figure attributed to Yao, Sailor, Yang, et al. (2023); verification before the value entered configuration found that the paper reports no such figure and in fact measures an order of magnitude less by day, and the floor moved (Section 6.7.1). Four further citation corrections were made in the same pass without moving a value.
>
> The process rule that follows is stated here rather than left as folklore: **approval of a design is not approval of the citations inside it**, and verification runs after approval and before configuration, with the authority to change the approved value.

Because the methodology lives in configuration rather than in code, the diff of such a proposal is legible to a domain expert who does not read the implementation language. This is a deliberate property. A reviewer evaluating a change to the green façade envelope should be able to see a temperature range, a confidence rating, and a citation change, and nothing else.

13.3 How to challenge this methodology

Methodology critique is a first-class contribution to this project, not an inconvenience to be managed. The route is to open an issue in the repository at `https://github.com/Dimitrios-Kafetzis/CriterraNatureCoolingTool`, referencing the section number of this document and, where possible, the literature supporting the alternative position.

Six areas are in particular need of external scrutiny, and are listed in descending order of how much a change would affect the tool's outputs.

1. **The aggregation weights** (Section 7.12), and specifically the equity-forward double contribution of vulnerability (Section 7.13). This is the tool's most consequential unsourced choice and a value position rather than a finding. The sensitivity analysis of Section 9.4 quantifies its influence: rank orderings prove substantially stable under single-weight perturbation, which bounds — but does not settle — the dispute.
2. **The green façade and bioretention envelopes** (Section 6.6.3). For green façades the sources conflict outright and the tool adopts a conservative resolution based on a hypothesis about measurement position that neither source states; eleven catalogue entries inherit the result. For rain gardens and bioswales no source quantifies air-temperature cooling at all, and the adopted range is reasoned from adjacent evidence. Both are candidates for replacement by better evidence, and both would benefit from a reviewer who knows this literature better than the derivation in Appendix A demonstrates.
3. **The two derived productive archetypes, and the width of the large water body envelope** (Sections 6.7.1 and 6.8). The productive classes are bounded from adjacent evidence rather than measured, and the tool can cite the absence (Kumar et al. 2024) but cannot fill it. The water envelope spans a factor of thirty because remote sensing and field measurement disagree by an order of magnitude; a reviewer who can resolve that disagreement on air temperature would narrow the widest range in the library.
4. **The non-adoption of Kumar et al. (2024)'s higher figures** (Sections 6.6.4 and 12.6). Adopting them selectively would break internal consistency and adopting them wholesale is a full recalibration with its own sensitivity analysis. This is a live disagreement with a recent review, held open deliberately, and an argument for closing it either way is welcome.
5. **The transferability of the energy sensitivity** (Section 7.9). The 2–4 % per °C figure is applied globally from predominantly North American evidence. Reviewers with access to equivalent evidence for other building stocks and climates would materially improve this part of the methodology, and evidence that the figure does *not* transfer would be equally valuable.
6. **The downward correction for modelling bias** (Section 6.6.1). The correction rests on a single observable ratio for a single measure, used to justify a direction rather than a magnitude. A reviewer who considers the resulting envelopes too conservative — or not conservative enough — is disputing exactly the right thing.

13.4 What a good challenge looks like

The most useful critiques share a shape. They identify a specific value or rule, by section number; they state what is wrong with it, distinguishing between an arithmetic error, a misread source, a source that does not support the weight placed on it, and a value judgment the critic rejects; and where the critique is evidential rather than a difference of values, they name a source and what it actually reports, including which metric it reports it in.

The last point is the one most often missed. A critique of a cooling envelope that cites a figure without stating whether it is air temperature, land surface temperature, or a comfort index cannot be acted on, because the methodology cannot adopt a value whose metric is unknown (Section 6.2).

This is the same standard the methodology holds itself to.

Critiques that reject a value judgment — the equity weighting being the obvious case — are welcome and will be recorded, but they will generally be resolved by making the configuration easier to rebalance and the consequences easier to see, rather than by replacing one contested judgment with another.

14 Conclusion

This paper has specified, in full, the methodology of the Nature for Cooling Rapid Assessment Tool at methodology version 2026.08.05: every input and its normalisation, the two-level evidence library of 110 catalogue entries over 18 cited cooling archetypes and the derivations behind them, the rules by which a package of interventions is combined, every scoring formula and every weight, the treatment of uncertainty and missing data, the design and the executed results of the sensitivity analysis, and the engineering arrangement that keeps the software in correspondence with the specification.

The instrument occupies a deliberately narrow position. It is faster and more comparable than unstructured judgment, and far less capable than microclimate simulation. It exists for the triage step: narrowing a long list of candidate sites and interventions to a short one, on a transparent and evidence-referenced basis, and recording why. Judged as a simulation it fails; judged as a screening instrument, the questions that matter are whether its values are traceable, whether its uncertainties are honestly represented, whether its rankings are stable, whether its value judgments are disclosed, and whether it declines to answer when its inputs do not support an answer (Section 3.5).

Several of the methodology's choices lower its apparent authority, and they were the substance of the work rather than concessions to it. Adopted cooling envelopes sit at or below synthesis central estimates because much of the underlying evidence is modelling-derived and modelling reports larger effects than measurement. A multi-intervention package reports the temperature of its best-evidenced component and never the sum, because no retrieved source quantifies super-additive cooling. Reported temperature ranges are clipped to the literature envelope, so that a favourable site raises a relative score but never a physical claim. Per-typology energy factors that no source supported were deleted and replaced by a derivation from a published sensitivity. No default unit costs ship, so the tool declines to produce the payback figure that would otherwise be its most quotable output. Six of the eighteen archetypes — the green façade, bioretention, the enclosed courtyard, the vegetated shade structure, and both productive classes — are labelled low confidence, and their confidence caps propagate into results.

The library's structure carries the same discipline. 110 catalogue entries are served by 18 cited evidence classes rather than by 110 invented envelopes, because solution-specific cooling literature does not exist at that resolution; every result names the class it inherited from, so the resolution of a claim is visible in the output rather than inferred. Where a retrieval found that an entry would have been overstated by inheriting a neighbouring class, a separate archetype was retrieved rather than the resemblance accepted — which is why small constructed water sits at 0.0–1.0 °C and not at the corridor's 1.0–3.0 °C. And one approved envelope moved during implementation, downward, because verification found the citation behind it did not say what the design claimed it said.

The limitations are correspondingly explicit. All values are daytime, and nighttime heat — the more consequential quantity for health outcomes — is not estimated. The quantity modelled is air temperature, not thermal comfort, which means the tool systematically understates shade-dominant interventions; the bias is stated rather than corrected by an unsourced factor. The evidence base is skewed toward temperate and subtropical Northern Hemisphere cities, and the confidence model does not yet measure that skew. The aggregation weights are expert judgment, including a disclosed equity-forward stance that gives vulnerability an effective weight of 0.25 against heat exposure at 0.15. A further limitation arrives with the archetype model itself: an entry's cooling range is its class's, so the tool does not discriminate between two entries that inherit the same evidence.

The sensitivity analysis of Section 9.4, regenerated for this version rather than carried forward, finds the ranking substantially stable under ±25 % single-weight perturbation — worst-case pairwise stability 0.9529, pooled mean displacement 0.47 points, and 8 near-boundary category changes in 288 scenario-perturbation combinations — which bounds how much the contestable weights actually move decisions. Every one of those headline figures is slightly worse than at version 2026.08.04, and the reason is stated rather than smoothed: four of the eight migrations are one new scenario sitting 0.11 points below a band boundary, and the new scenarios were chosen to exercise this release's evidence edge cases rather than to sit comfortably inside bands.

The methodology was first offered for review with its gaps disclosed rather than closed, because the values most in need of scrutiny are settled most cheaply before an engine, a user base, and a set of published assessments have accumulated around them. Version 2026.08.01 closed the disclosed specification gaps — the sub-indicator rules, the cost-feasibility derivation, and the confidence field mapping — without altering any evidence-derived value, and version 2026.08.05 replaces the library without moving a single previously published envelope, save the one that verification required. The tool, its complete configuration, and this document are public under the Apache License 2.0. Every value it applies names its source; Appendix E states what was actually verified about each of those sources, including where publisher restrictions meant a finding was confirmed from an abstract or an indexing record rather than a full text. That disclosure is not a weakness in the evidence base. It is the property that makes the evidence base checkable, and it is what the authors would want to know before trusting an instrument of this kind.

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A The library: archetypes, entries, and their evidence

This appendix reproduces the full library at methodology version 2026.08.05, as it appears in `config/nbs_typologies.yaml`. All temperature values are daytime, pedestrian-level *air* temperature reductions in degrees Celsius, relative to an equivalent unimproved site.

The library has two levels (Section 6.1). Tables 17 to 20 give the 18 cooling archetypes, which carry every performance value and every citation. Table 21 lists the 110 catalogue entries and the archetype each one inherits; an entry carries no performance value of its own. Table 22 lists the 21 per-entry suitability overrides.

A.1 Archetype values and inherited suitability conditions

Table 17: Adopted values and suitability conditions for all 18 cooling archetypes. *Bldg. energy* indicates whether the archetype is flagged as capable of affecting adjacent building cooling loads, a precondition for the energy derivation of Section 7.9. The minimum area and the soil and irrigation requirements are inherited by every catalogue entry resolving to the archetype, subject to the overrides of Table 22.

Archetype	Base score	Envelope (°C)	Conf.	Bldg. energy	Min. area (m^2)	Soil / water required
Green						
Street tree canopy	75	0.5–3.0	High	Yes	100	Limited / occas.
Continuous shaded corridor	80	0.5–3.0	Medium	No	200	Limited / occas.
Small green area with trees	65	0.3–1.5	High	No	200	Limited / occas.
Dense tree canopy	90	1.0–3.0	High	Yes	2000	Moderate / occas.
Established park	70	0.5–2.0	High	No	500	Limited / occas.
Non-canopy vegetation	50	0.0–1.5	Medium	No	50	Limited / occas.
Productive tree canopy	60	0.5–2.0	Low	No	500	Moderate / reliable
Productive ground-level planting	45	0.0–1.5	Low	No	200	Moderate / reliable
Building						
Extensive green roof	45	0.1–1.0	Medium	Yes	50	None / occas.
Green facade or living wall	50	0.3–2.0	Low	Yes	20	None / reliable
Blue-green						
Bioretention and drainage planting	55	0.1–0.8	Low	No	50	Moderate / none
Blue-green corridor	85	1.0–3.0	Medium	No	1000	Moderate / none
Riparian restoration	85	1.0–3.0	Medium	No	1000	Moderate / none
Large open water body	75	0.1–3.0	Medium	No	1000	Moderate / none
Small constructed water feature	40	0.0–1.0	Medium	No	50	Moderate / none
Hybrid						
Permeable shaded hardscape	70	0.5–2.5	Medium	No	200	Limited / occas.
Enclosed courtyard greening	60	0.3–1.5	Low	Yes	50	Limited / occas.

continued on next page

Table 17 continued from previous page

Archetype	Base	Envelope	Conf.	Bldg.	Min. area	Soil / water
Vegetated shade structure	55	0.0–2.5	Low	No	20	None / reliable

No archetype declares an unsuitable climate zone at version 2026.08.05. Climate influences performance through the adjustment matrix of Table 7 rather than through outright disqualification.

A.2 Co-benefit defaults

The primary cooling mechanism is declared *per catalogue entry* rather than per archetype, because it is the one descriptive attribute the source catalogue supplies at entry resolution; it is reported in the result and enters no formula. The co-benefit defaults below are archetype properties, inherited by every entry, and are unioned across components in a package (Table 8).

Table 18: Default co-benefit levels per archetype. The library supplies no default for the *social inclusion* sub-indicator of Equation (15); absent a user override it falls to the neutral *unknown* value and is itemised as an applied assumption.

Archetype	Biodiv.	Storm.	Health	Urban q.
Street tree canopy	Med	Med	High	High
Continuous shaded corridor	Med	Low	High	High
Small green area with trees	Med	Med	High	High
Dense tree canopy	High	High	High	High
Established park	High	Med	High	High
Extensive green roof	Med	High	Low	Med
Green facade or living wall	Low	Low	Low	Med
Bioretention and drainage planting	Med	High	Low	Med
Blue-green corridor	High	High	High	High
Riparian restoration	High	High	Med	High
Permeable shaded hardscape	Low	High	Med	High
Enclosed courtyard greening	Med	Med	Med	High
Large open water body	High	High	Med	High
Small constructed water feature	Med	High	Low	Med
Non-canopy vegetation	Med	Med	Low	Med
Vegetated shade structure	Low	Low	Med	Med
Productive tree canopy	Med	Med	High	High
Productive ground-level planting	Med	Low	High	Med

A.3 Citations and the findings they support

Every performance value in the library carries at least one citation, and each citation records the specific finding it supports. Configuration loading fails if a resolved entry is uncited or if a citation names a source absent from the bibliography (Section 10.4). The findings below are reproduced from the configuration.

Table 19: Citations supporting each archetype's adopted values, with the finding each citation supports. Metrics are stated where a source reports something other than air temperature.

Archetype	Source	Finding relied upon
Street tree canopy	`keravec2026`	Street trees: 1.5 °C central estimate for pedestrian-level daytime air temperature; 4.0 ± 1.6 °C in tropical (A) and 1.7 ± 1.4 °C in arid (B) climates.
	`ziter2019`	Air temperature decreases nonlinearly with canopy cover, greatest cooling above ~40% cover; peak effect at 60–90 m scale.
	`bowler2010`	Empirical studies broadly support cooler air beneath tree canopy.
	`rahman2020`	Species and leaf area index materially change delivered cooling.
	`kumar2024`	Street trees measured at up to 2.8 °C by in-situ monitoring, inside this envelope: newer and broader evidence supports the conservative calibration rather than overturning it.
Continuous shaded corridor	`keravec2026`	Street trees 1.5 °C and shading devices 2.0 °C pedestrian-level daytime air temperature reduction.
	`zolch2016`	Shade placement in heat-exposed locations outperforms maximising total green cover; trees -13% PET.
Small green area with trees	`bowler2010`	Meta-analysis: park on average 0.94 °C cooler by day; the review reports that larger parks and parks containing trees tended to be cooler, stated qualitatively.
	`keravec2026`	Parks 1.3 °C pedestrian-level daytime air temperature reduction.

continued on next page

Table 19 continued

Archetype	Source	Finding relied upon
Dense tree canopy	`keravec2026`	Urban forests 2.1 °C pedestrian-level daytime air temperature reduction, among the highest-performing green measures.
	`ziter2019`	Greatest cooling where canopy cover exceeds 40%; effect grows with patch scale to 60–90 m.
	`bowler2010`	Larger, tree-rich green sites cooler during the day.
Established park	`bowler2010`	Park mean 0.94 °C daytime cooling; parks with trees reported cooler than open turf.
	`keravec2026`	Parks 1.3 °C, green areas 2.0 °C pedestrian-level daytime air temperature reduction.
	`kraemer2022`	Under 2018–2019 drought and heat, maximum spatially averaged park-to-built cooling was 1.1 °C in the morning, falling below 1 °C in the afternoon and confined to shaded areas.
Extensive green roof	`keravec2026`	Green roofs 1.1 °C pedestrian-level daytime air temperature reduction, lowest-performing green measure in the synthesis.
	`zolch2016`	Green roof effects at pedestrian level negligible.
	`santamouris2014`	0.3–3 K average urban ambient reduction, but only for city-scale deployment in simulation studies, not single-building installation.
Green facade or living wall	`keravec2026`	Green walls 3.0 °C central, with fivefold spread across climate subzones (Cwa 5.7, BWh 5.5, Csa 2.9, Cfa/Cfb 1.3, Af 1.0 °C).
	`zolch2016`	Green facades -5% to -10% PET, below trees at -13%.
Bioretention and drainage planting	`keravec2026`	No bioswale-specific estimate available; neighbouring categories (green areas 2.0 °C, water bodies 2.1 °C) are not transferable to small drainage features.

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Table 19 continued

Archetype	Source	Finding relied upon
	`gunawardena2017`	Tree-dominated greenspace delivers heat-stress relief; small blue features are not identified as significant coolers.
Blue-green corridor	`keravec2026`	Water bodies 2.1 °C and urban forests 2.1 °C pedestrian-level daytime air temperature reduction; water bodies and parks reported as largely climate-independent.
	`gunawardena2017`	Green and blue features together offer synergistic benefits; poorly designed bluespace may exacerbate heat stress in humid conditions.
Riparian restoration	`keravec2026`	Water bodies 2.1 °C, climate-robust; urban forests 2.1 °C pedestrian-level daytime air temperature reduction.
	`gunawardena2017`	Bluespace cooling mechanisms; humid-condition heat-stress caveat.
Permeable shaded hardscape	`keravec2026`	Permeable pavements 2.9 °C, cool pavements 2.0 °C, shading devices 2.0 °C pedestrian-level daytime air temperature reduction.
	`zolch2016`	Shade positioned in heat-exposed locations is the decisive design factor.
Enclosed courtyard greening	`keravec2026`	Green areas 2.0 °C and green walls 3.0 °C; enclosed-courtyard geometry not separately resolved.
	`ziter2019`	Cooling scales with patch size; courtyards fall below the 60–90 m optimum.
	`zolch2016`	Micro-scale greening in dense residential fabric reduces PET; placement dominates quantity.
Large open water body	`keravec2026`	Water bodies 2.1 °C pedestrian-level daytime air temperature reduction, reported as largely climate-independent.

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Table 19 continued

Archetype	Source	Finding relied upon
	`volker2013`	Meta-analysis of 27 studies: pooled cooling of 2.5 K (95% CI 1.9–3.2 K) in the warmest months, attributed to urban blue sites WHEN REMOTE-SENSING DATA ARE INCLUDED. A single pooled estimate across ponds, lakes and rivers together; it does not differentiate cooling by water-body size.
	`yao2023a`	Year-long field measurement of urban lakes in Nanjing and Guangzhou: weak daytime urban cool island intensity of 0.1–0.6 °C, against a 1.2–1.3 °C heat island intensity at night in the warm months.
	`ampatzidis2020`	A clear disparity exists between cooling potentials reported by remote sensing and those from field measurement or numerical simulation; at night blue spaces may exacerbate the urban heat island.
Small constructed water feature	`jacobs2020`	Sixteen simulated urban water bodies: afternoon air temperature in surrounding spaces reduced by typically 0.2 °C, maximum 0.6 °C. Of 20 papers synthesised for water bodies without fountains, 14 (70%) report an air-temperature effect of 1 °C or less; local thermal effects of small water bodies can be considered negligible in design practice.
	`yao2023b`	Field measurement of an urban pond: moderate daytime cooling of 0.6 °C against a reference soil site, and a pronounced nocturnal warming effect of 1.8 °C.
	`ampatzidis2020`	Blue-space cooling varies with time of day, season and location; at night blue spaces may exacerbate the urban heat island, and sensible cooling can be offset by increased water-vapour content.

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Table 19 continued

Archetype	Source	Finding relied upon
Non-canopy vegetation	`armson2012`	Measured on the same plots: grass reduced maximum SURFACE temperature by up to 24 °C but had little effect upon GLOBE temperature, while shading reduced globe temperature by 5–7 °C. Grass has little effect on local air or globe temperature; tree shade provides effective local cooling.
	`gill2013`	Modelled: grass cooling can be reduced or lost in summer droughts when soils dry out, an effect projected to become more pronounced under climate change.
	`kraemer2022`	Field measurement under drought: the grass-dominated park was the hottest area during the daytime, while the tree-dominated park was consistently cooler.
	`bowler2010`	Parks containing trees reported cooler than open turf, stated qualitatively.
	`kumar2024`	Typology-resolved review placing grass and herbaceous systems well below canopy systems for air-temperature cooling.
Vegetated shade structure	`chafer2020`	The only measured comparison of a vine-covered pergola against an identical bare support: maximum daytime air temperature difference 2.5 °C (45.3 °C over the bare ropes against 42.9 °C under the greenery), west-oriented pergolas, Mediterranean climate.
	`ouyang2024`	Field measurement of shading structures on both axes: air temperature reduced 1.42 °C (natural) and 1.31 °C (artificial), while mean radiant temperature was reduced 15.93 °C and 13.71 °C respectively — an order of magnitude apart.

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Table 19 continued

Archetype	Source	Finding relied upon
	`colter2019`	Sparse and open canopies group with solid constructed shade structures rather than with dense canopy: PET relief under Parkinsonia, Prosopis and ramadas was 2.9–4.3 °C worse than under dense-canopy species.
Productive tree canopy	`kumar2024`	Review of 51 green-blue infrastructure types across 202 publications: allotments and city farms are among the least-studied categories, with insufficient data to quantify their cooling.
	`ziter2019`	Cooling strengthens above approximately 40% canopy cover; orchard and agroforestry canopies are deliberately kept open for access and yield, so they sit below that inflection.
	`colter2019`	Sparse canopies perform no better than solid artificial shade, which bounds an open production canopy below a closed woodland canopy.
	`cheung2022`	Irrigation sustains rather than raises performance: measured agricultural cooling was 0.09–0.43 °C against modelled potentials of 2.1–2.5 °C, and irrigation increases the daily minimum air temperature as stored heat is released at night.
Productive ground-level planting	`kumar2024`	Allotments and city farms are among the least-studied of 51 green-blue infrastructure types, with insufficient data to quantify their cooling.
	`rost2020`	The best available allotment-garden measurement: 13 Berlin complexes averaged 2.7 K cooler than built-up areas AT NIGHT. Nocturnal only by design, so it cannot support a daytime value.

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Table 19 continued

Archetype	Source	Finding relied upon
	`armson2012`	Ground-level vegetation cuts surface temperature substantially while barely affecting globe temperature, which is the basis of the inherited envelope.
	`cheung2022`	Measured agricultural cooling 0.09–0.43 °C against modelled potentials of 2.1–2.5 °C; irrigation raises the daily minimum air temperature at night.

A.4 Calibration notes recorded in configuration

Table 20: Calibration notes carried in the library, explaining why an adopted value departs from a naive reading of its sources. The dense tree canopy, blue-green corridor, and riparian restoration carry no note; their values follow directly from converging sources.

Archetype	Note
Street tree canopy	Upper bound capped at 3.0 °C rather than the tropical central estimate of 4.0 °C, which carries a ±1.6 °C interval and reflects favourable conditions.
Continuous shaded corridor	Scored above street trees because shade is placed where exposure occurs. The narrow-street effect reported by `keravec2026` is an urban-form effect, not an NbS intervention, and is deliberately not inherited.
Small green area with trees	Explicitly the small end of the park distribution, bounded below the general park estimate by the size relationship reported in `bowler2010`. That relationship is qualitative and hedged in the source, and is worded here to match (corrected at 2026.08.04).
Established park	Represents added cooling over an already partly green baseline, so the achievable increment is bounded below new-forest values.
Extensive green roof	Benefits are concentrated at roof level and inside the building beneath. The tool states in its output that green-roof benefits are principally building-level rather than street-level.

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Archetype	Note
Green facade or living wall	Sources conflict outright: the synthesis ranks green walls near the top on air temperature while the micro-scale comparison ranks them below trees on comfort, plausibly explained by near-wall measurement position. Low confidence, conservative envelope, explicit caveat in tool output. `kumar2024` reports green walls at 4.1 ± 4.2 °C, which is deliberately not adopted (D-044): its own standard deviation exceeds its mean, and adopting it selectively would rate a wall above measured tree canopy.
Bioretention and drainage planting	No source quantifies air-temperature cooling for rain gardens or bioswales specifically. Value is a reasoned lower bound from adjacent evidence. Primary benefits are stormwater and biodiversity, not cooling.
Blue-green corridor	The corridor combines an open water body with continuous canopy, which is why it sits above the water-only archetype: its envelope rests on the canopy evidence as much as on the water evidence.
Riparian restoration	As with the blue-green corridor, the envelope rests on the combination of flowing water and riparian canopy, not on the water surface alone.
Permeable shaded hardscape	The permeable-pavement estimate rests on a narrower evidence base than the vegetation figures, so the adopted upper bound is set below it. The `keravec2026` shading-devices figure is admissible HERE, at plaza scale, because that category aggregates area-scale shading; it is deliberately not inherited by the pergola-scale archetype below.
Enclosed courtyard greening	Small, enclosed, often sheltered from ventilation. Bounded by scale limitation; no courtyard-specific synthesis exists.

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Archetype	Note
Large open water body	The envelope floor is 0.1 °C, not the 0.5 °C proposed in the v1.2 review pack. The pack attributed a cooling magnitude near 2 °C to `yao2023a`; retrieval at implementation established that the paper reports no such figure and instead measures 0.1–0.6 °C by day. Holding the floor at 0.5 would have made the tool claim more than its most direct field evidence supports, which is exactly the inflation D-008 exists to prevent. The ceiling stays at 3.0 °C on `keravec2026` and the `volker2013` confidence interval. The width of the envelope is real and is the finding: the remote-sensing-inclusive meta-analysis and the direct field measurements disagree by an order of magnitude, and `ampatzidis2020` names remote sensing as the overestimating mode. See D-045.
Small constructed water feature	THIS ARCHETYPE MUST NEVER BE INTERPOLATED WITH `large_water_body` BY AREA. The two are separate bodies of study, not two points on a dose–response curve: the size-threshold literature is land-surface-temperature only, and no published curve connects them on air temperature. Without this split, constructed wetlands, retention ponds, detention basins and water squares would have inherited `blue_green_corridor` at 1.0–3.0 °C on family resemblance and been overstated roughly threefold. The floor is a genuine zero because `jacobs2020` concludes the daytime effect is negligible in design practice.
Non-canopy vegetation	`armson2012` is decisive because it measures both metrics on the same plots and so isolates the surface/air conflation that inflates most published grass figures: cutting surface temperature by 24 °C while barely moving globe temperature is the whole finding. The floor is a genuine zero for dry or dormant planting, on `gill2013` and `kraemer2022`.

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Archetype	Note
Vegetated shade structure	THE keravec2026 "SHADING DEVICES 2.0 °C" FIGURE IS DELIBERATELY NOT USED HERE. That category aggregates area-scale shading, where the cooled air volume persists against advection; a pergola shades roughly 10 m^2. Applying it would rate a pergola above measured tree canopy, which ouyang2024 measures at 1.42 °C. The same figure IS used by the permeable_shaded_hardscape archetype, where the scale matches. Low confidence, with $n = 1$ for the actual intervention. The ceiling of 2.5 °C is chafer2020's measured peak; note that paper's abstract headlines "up to 5 °C", which is greenery against UNSHADED conditions rather than against the bare pergola, and is not the like-for-like contrast this envelope needs.
Productive tree canopy	Derived, not retrieved: bounded below dense_tree_canopy on the ziter2019 canopy-cover inflection and the colter2019 sparse-canopy finding, and given the same envelope as established_park with a lower evidence rating. Irrigation is NOT credited as extra degrees — the cheung2022 contrast is specific to agricultural fields, and the same paper measures −2–−1 °C in irrigated urban parks, so no general claim that models overstate irrigation cooling is made here.
Productive ground-level planting	Derived: inherits the non_canopy_vegetation envelope, because cultivated ground-level planting is ground-level vegetation whose cover is periodically broken between crops. Rated low rather than medium because the inheritance is an argument rather than a measurement of this class. The evidence gap is CITABLE (kumar2024) rather than merely unfilled, which is a stronger position than the rain-garden precedent where no source was found at all.

A.5 The 110 catalogue entries and the archetype each inherits

Entries are grouped by the source catalogue's own families, with its element/composite classification reproduced. Neither the family nor the classification enters a formula. The identifier is the entry's identifier in the source catalogue.

Table 21: The 110 curated catalogue entries, the cooling archetype each inherits, and the source catalogue's element/composite classification. An entry carries identity, family, availability, and its primary cooling mechanism; every performance value comes from the archetype named here.

ID	Entry	Archetype inherited	Kind
Tree-based (13)			
A1	Strategic individual-tree planting	Street tree canopy	element
A2	Tree cluster	Small green area with trees	element
A3	Tree grove	Dense tree canopy	element
A4	Conventional individual tree pit	Street tree canopy	element
A5	Enlarged tree pit	Street tree canopy	element
A6	Structural-soil tree pit	Street tree canopy	element
A7	Suspended-pavement or soil-cell tree pit	Street tree canopy	element
A8	Stormwater tree pit	Bioretention and drainage planting	element
A9	Bioretention tree pit	Bioretention and drainage planting	element
A11	Continuous tree trench	Street tree canopy	element
A12	Continuous open planting bed for trees	Street tree canopy	element
A13	Connected multi-layer planting bed	Street tree canopy	element
A14	Tree island	Small green area with trees	element
Street (7)			
2.2	Tree Avenue	Street tree canopy	composite
2.5	Green Street	Street tree canopy	composite
2.9	Green Median	Non-canopy vegetation	composite
2.13	Green Tram Corridor	Street tree canopy	composite
2.15	Green Pedestrian Street	Continuous shaded corridor	composite
2.16	Green Alley	Street tree canopy	composite
2.17	Green Parking Lane	Street tree canopy	composite
Park (9)			
4.1	Pocket Park	Established park	composite
4.2	Neighbourhood Park	Established park	composite
4.4	District Park	Established park	composite
4.5	Regional Park	Established park	composite
4.11	Water Park	Established park	composite
4.15	Naturalistic Park	Established park	composite
4.17	Stormwater Park	Established park	composite
4.20	Linear Park	Established park	composite

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Table 21 continued

ID	Entry	Archetype inherited	Kind
4.22	Coastal Park	Established park	composite
Public space (6)			
3.1	Tree Plaza	Small green area with trees	composite
3.2	Cooling Plaza	Permeable shaded hardscape	composite
3.3	Permeable Plaza	Permeable shaded hardscape	composite
3.4	Civic Green	Established park	composite
3.5	Courtyard Greening	Enclosed courtyard greening	composite
3.15	Green Playground	Small green area with trees	composite
Woodland and forest (6)			
5.1	Urban woodland (site)	Dense tree canopy	composite
5.3	Microforest	Dense tree canopy	composite
5.10	Degraded Woodland Restoration	Dense tree canopy	composite
5.15	Afforestation	Dense tree canopy	composite
5.16	Reforestation	Dense tree canopy	composite
5.18	Urban Woodland Buffer	Dense tree canopy	composite
Ground level (11)			
B1	Conventional planting bed	Non-canopy vegetation	element
B2	Shrub-dominated planting bed	Non-canopy vegetation	element
B3	Herbaceous perennial bed	Non-canopy vegetation	element
B4	Groundcover planting	Non-canopy vegetation	element
B6	Pocket meadow	Non-canopy vegetation	element
B7	Rain garden	Bioretention and drainage planting	element
B8	Bioswale	Bioretention and drainage planting	element
B9	Infiltration planting bed	Bioretention and drainage planting	element
B10	Depressed bioretention bed	Bioretention and drainage planting	element
B11	Vegetated planter with stormwater function	Bioretention and drainage planting	element
B12	Large-volume connected planters	Bioretention and drainage planting	element
Water landscape (12)			
6.1	Constructed Wetland	Small constructed water feature	composite

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Table 21 continued

ID	Entry	Archetype inherited	Kind
6.2	Retention Pond	Small constructed water feature	composite
6.3	Detention Basin	Small constructed water feature	composite
6.4	Water Square	Small constructed water feature	composite
6.5	Daylighted Stream	Riparian restoration	composite
6.7	River Restoration	Riparian restoration	composite
6.8	Canal Restoration	Riparian restoration	composite
6.10	Floodplain Restoration	Riparian restoration	composite
6.11	Urban Lake Restoration	Large open water body	composite
6.12	Living Shoreline	Large open water body	composite
6.13	Floating Wetland	Small constructed water feature	element
6.14	Riparian Buffer	Riparian restoration	element
Ecological network (6)			
7.1	Green Corridor	Blue-green corridor	composite
7.2	Ecological Corridor	Blue-green corridor	composite
7.4	Blue-Green Corridor	Blue-green corridor	composite
7.5	River Corridor	Riparian restoration	composite
7.6	Railway Green Corridor	Street tree canopy	composite
7.14	Coastal Ecological Corridor	Large open water body	composite
District (5)			
10.1	Green Infrastructure Network	Blue-green corridor	composite
10.2	Blue-Green Infrastructure Network	Blue-green corridor	composite
10.3	Urban Cooling Network	Dense tree canopy	composite
10.4	Green Mobility Network	Street tree canopy	composite
10.10	Forest District	Dense tree canopy	composite
Green roof (12)			
C1	Extensive green roof	Extensive green roof	element
C2	Semi-intensive green roof	Extensive green roof	element
C3	Intensive green roof	Small green area with trees	element
C5	Biodiverse green roof	Extensive green roof	element
C6	Brown roof	Extensive green roof	element
C8	Blue-green roof	Extensive green roof	element
C11	Biosolar roof	Extensive green roof	element

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Table 21 continued

ID	Entry	Archetype inherited	Kind
C13	Productive green roof	Productive ground-level planting	element
C14	Rooftop farm	Productive ground-level planting	element
C15	Roof meadow	Non-canopy vegetation	element
C16	Roof wetland	Small constructed water feature	element
C17	Tree roof or rooftop woodland	Dense tree canopy	element
Vertical greening (10)			
D1	Direct ground-rooted green façade	Green facade or living wall	element
D2	Indirect ground-rooted green façade	Green facade or living wall	element
D5	Planter-rooted green façade	Green facade or living wall	element
D6	Cascading façade planting	Green facade or living wall	element
D7	Modular living wall	Green facade or living wall	element
D8	Substrate-based living wall	Green facade or living wall	element
D9	Hydroponic living wall	Green facade or living wall	element
D11	Vegetated screen	Green facade or living wall	element
D12	Green noise barrier	Green facade or living wall	element
D13	Vegetated retaining wall	Green facade or living wall	element
Building (3)			
9.29	Green Balcony	Green facade or living wall	element
9.30	Green Terrace	Extensive green roof	element
9.33	Green Podium	Small green area with trees	composite
Vegetated shelter (3)			
E1	Green pergola	Vegetated shade structure	element
E3	Vegetated arcade	Continuous shaded corridor	element
E6	Green waiting shelter	Vegetated shade structure	element
Productive landscape (7)			
8.1	Community Garden	Productive ground-level planting	composite
8.2	Allotment Garden	Productive ground-level planting	composite
8.3	Urban Orchard	Productive tree canopy	composite
8.4	Food Forest	Productive tree canopy	composite
8.5	Urban Farm	Productive ground-level planting	composite

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Table 21 continued

ID	Entry	Archetype inherited	Kind
8.9	School Garden	Productive ground-level planting	composite
8.15	Agroforestry System	Productive tree canopy	composite

A.6 The twenty-one suitability overrides

An entry overrides an inherited suitability condition only where the source catalogue's own description of the entry makes the inherited value plainly wrong (Section 6.9). No new number enters the methodology through an override: every replacement value is one already present in the cited library. A test fails if this list grows past 25 entries.

Table 22: The 21 per-entry suitability overrides, with the reason the inherited value is wrong.

Entries	Override	Why the inherited value is wrong
Tree grove, microforest, rooftop woodland	area → 200 m^2	The dense tree canopy's 2 000 m^2 describes the woodland its *cooling evidence* comes from, not the footprint of a site element. The catalogue defines a microforest as a small dense patch on a constrained site.
Intensive green roof, productive green roof, roof meadow, roof wetland (50 m^2); rooftop farm, rooftop woodland (200 m^2)	area, and soil → none	A roof's growing medium is *built*, not ground. Requiring ground soil on a roof would flag every roof entry whose archetype happens to be a ground-level class.
Green podium, water square, floating wetland	soil → none	Built structure or open water, likewise. The water square is additionally set to 200 m^2, at plaza scale.

Entries	Override	Why the inherited value is wrong
Direct and indirect ground-rooted green façades, green noise barrier, vegetated retaining wall	soil → limited	The catalogue defines these as rooted in *natural ground*, which the containerised living-wall archetype does not require. This override makes the tool *stricter*, not more permissive.
Pocket park	area → 200 m^2	This entry *is* the tool's own cited pocket park; its minimum area is the one the previous library established.
Riparian buffer	area → 200 m^2	The catalogue distinguishes a discrete vegetated strip from the restored corridor the archetype describes.
Railway green corridor, green mobility network	area → 1 000 m^2	Corridor- and district-scale systems inheriting an element archetype.
Urban farm (500 m^2), agroforestry system (1 000 m^2)	area	The catalogue distinguishes field-scale production from garden scale.

B Complete weight tables

All weights are from `config/weights.yaml` at version 2026.08.05. Within each block the weights sum to unity. All are expert judgment in the sense of Nardo et al. (2008); where a source informs the *selection* of indicators or the *relative ordering* of weights, it is named.

B.1 Heat Exposure Score (data-rich path only)

Component	Weight
Land surface temperature anomaly	0.40
Imperviousness	0.25
Solar exposure	0.20
Vegetation deficit	0.15
Total	**1.00**

Basis. Component selection reflects established surface-energy-balance drivers of urban heat. Ziter et al. (2019) gives direct empirical support for two components: impervious cover raises air temperature approximately linearly and canopy cover reduces it nonlinearly. The relative weights are expert judgment.

B.2 Vulnerability Score

Component	Weight
Population density	0.40
Vulnerable groups	0.40
Cooling access deficit	0.20
Total	**1.00**

Basis. Indicator selection follows Reid et al. (2009), whose factor analysis of ten heat-vulnerability variables found four factors explaining over 75 % of variance, with air-conditioning prevalence emerging as an independent dimension. Density and vulnerable-group presence are weighted equally because that analysis gives no basis for ranking one above the other; cooling access carries less weight as a single, frequently unknown indicator.

B.3 Heat Priority Index and site adjustment

Heat Priority Index		**Site adjustment**	
Heat exposure	0.60	Canopy	0.40
Vulnerability	0.40	Soil–water	0.25
		Scale	0.20
		Climate	0.15
Total	**1.00**	**Total**	**1.00**

Basis for site adjustment. Canopy carries the largest weight as the condition with the strongest direct empirical support (Ziter et al. 2019). The climate component is justified by Keravec-Balbot et al. (2026): adding Köppen–Geiger subzone raises variance explained in cooling effectiveness from 12 % to 78 %.

B.4 Composite sub-scores

NbS Suitability		**Co-benefits**		**Equity**	
Space	0.30	Biodiversity	0.25	Vulnerable user benefit	0.35
Soil	0.25	Stormwater	0.25	Public accessibility	0.25
Water	0.20	Public health	0.20	Safety and comfort	0.20
Maintenance	0.15	Social inclusion	0.15	Participation relevance	0.20
Urban context	0.10	Urban quality	0.15		
Total	**1.00**	**Total**	**1.00**	**Total**	**1.00**

The rules mapping user inputs to these individual sub-indicator scores were fixed at version 2026.08.01 in `config/derived_scores.yaml` and reproduced in Table 9. The Equity Score does not enter the final aggregation (Section 7.8). The derived cost feasibility score combines the payback bracket with two delivery-risk modifiers (Equation (23)):

Cost feasibility component	**Weight**
Payback bracket score	0.50
Implementation complexity (inverted)	0.25
Maintenance intensity (inverted)	0.25
Total	**1.00**

B.5 Final aggregation and effective weights

Component	Nominal	Effective
Heat Priority Index	0.25	—
heat exposure		0.15
vulnerability		0.10
Cooling Potential	0.25	0.25
NbS Suitability	0.15	0.15
Vulnerability (direct)	0.15	0.15
Co-benefits	0.10	0.10
Cost Feasibility	0.10	0.10
Total	**1.00**	**1.00**
Restated, not added:		
Vulnerability, total		**0.25**
Heat exposure, total		**0.15**

The equity-forward double contribution of vulnerability is deliberate and is argued in Section 7.13. The sensitivity-analysis perturbation parameter is ± 25 % (Section 9).

C Input field reference

The definitive enumeration is fixed by the implemented input schema (`nature_cooling.engine.models.AssessmentInput`). Fields marked *Required* are exactly those without which a documented formula cannot be evaluated at all; every other field is *Optional*, and its absence is handled through a disclosed default or an explicit *not calculated* status, itemised in the result.

Table 23: Input fields, their status, and the normalisation applied to each.

Field	Status	Normalisation / use
Site characteristics		
`site_area_m2`	Required	Denominator of the canopy condition and of shade potential (Equation (13)).
`existing_tree_canopy_percent`	Optional	Canopy condition derivation (Table 6).
`existing_green_cover_percent`	Optional	Vegetation deficit, clamp$(100 - g)$ (Equation (5)).
`impervious_surface_percent`	Optional	Percentage used directly as a 0–100 score in Equation (4); the identity mapping is declared in configuration.
`soil_availability`	Optional	none 25 / limited 50 / moderate 75 / high 100; unknown 50. Also feeds the soil–water condition.
`irrigation_availability`	Optional	none 25 / occasional 50 / reliable 100; unknown 50. Also feeds the soil–water condition.
`current_shade_level`	Optional	very low 25 / low 50 / medium 75 / high 100; unknown 50. Counts toward cooling-block completeness; enters no formula at version 2026.08.05.

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Table 23 continued

Field	Status	Normalisation / use
`land_use`	Optional	Urban-context suitability sub-indicator: in the entry's own land-use list 100, outside it 25, unknown 50 (Table 9). That list is the availability matrix's list, so an entry is in context exactly where it is offered (Section 5.6). *campus* and *memorial* were added at version 2026.08.03.
*Availability conditions — these gate which interventions are offered and feed **no** score*		
`includes_railway`	Optional	yes / no. Gates exactly one entry, the railway green corridor. Unanswered withholds it. **Availability only**: enters no formula, no confidence block, and no aggregation (Section 5.6). New at version 2026.08.05.
`existing_woodland`	Optional	yes / no. Gates woodland *restoration* entries only; woodland *creation* entries are ungated. Unanswered withholds the restoration entries. **Availability only**. New at version 2026.08.05.
`waterfront_type`	Optional	lake / river / dry river / covered river / sea. Gates twelve water-landscape and ecological-network entries by category; the four *constructed* water features carry no waterfront condition. Unanswered withholds the gated entries. **Availability only**. New at version 2026.08.05.

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Table 23 continued

Field	Status	Normalisation / use
`productive_governance`	Optional	Multi-select over community / individual / institutional / commercial. Filters the seven productive entries by who delivers them; an *empty or unanswered* selection suppresses nothing, because absence of an answer is not evidence that no delivery model exists. Canonicalised on input so that two selections of the same models produce byte-identical results. **Availability only**. New at version 2026.08.05.
Project information		
`assessment_scale`	Required	Scale condition: city, district → excellent; neighbourhood → good; site, building → moderate.
`country`	Optional	Grid emission factor lookup in `country_defaults.yaml` (Section 7.10).
Climate and heat exposure		
`climate_zone`	Required	Climate condition lookup (Table 7).
`lst_anomaly_c`	Optional	Linear, 0 °C→0, 10 °C→100, clamped (Equation (1)). Selects the data-rich heat exposure path.
`heat_exposure_level`	Optional	Standard levels. Used directly as the Heat Exposure Score on the data-poor path.
`solar_exposure`	Optional	Standard levels.
`heat_index_concern`	Optional	Standard levels.
Vulnerability and equity		
`population_density`	Optional	Standard levels.
`vulnerable_population_presence`	Optional	Standard levels.

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Table 23 continued

Field	Status	Normalisation / use
`access_to_cooled_indoor_space`	Optional	**Inverted** levels: high 25 / medium 50 / low 100; unknown 50 (Section 5.3).
`access_to_cool_outdoor_refuge`	Optional	**Inverted** levels, as above. Second indicator of the same cooling access deficit dimension; the two are averaged over those supplied and share one confidence slot. New at version 2026.08.03.
`safety_concern`	Optional	Standard levels; equity safety-and-comfort sub-indicator (deficit reading).
`public_accessibility`	Optional	Standard levels; equity sub-indicator. New at version 2026.08.01.
`community_participation`	Optional	Standard levels; equity participation-relevance sub-indicator. New at version 2026.08.01.
NbS intervention		
`nbs_type`	Required	A *list* since version 2026.08.05: one or more catalogue entries. Selects each entry, the archetype it inherits, and that archetype's climate-family row. Duplicates are rejected; selection order is preserved in the itemised component results and changes no output. More than five components raises a warning, not an error (Section 7.7).
`new_canopy_area_at_maturity_m2`	Optional	Canopy condition and shade potential.
`expected_maturity_period_years`	Optional	Time-to-benefit class (immediate / short / medium / long term). No maturity discount is applied to cooling values.
`intervention_area_m2`	Optional	Validation warning when exceeding the site area (OQ-04); enters no score.

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Table 23 continued

Field	Status	Normalisation / use
`implementation_complexity`	Optional	Inverted levels; cost feasibility (Equation (23)) and investment readiness.
`maintenance_intensity`	Optional	Inverted levels; cost feasibility and the suitability maintenance sub-indicator.
Cost and energy (all optional)		
`nearby_building_cooling_demand_relevant`	Optional	Precondition for the energy derivation (Section 7.9).
`annual_cooling_energy_demand_kwh`	Optional	D in Equation (17). Absent: `missing_energy_demand`.
`energy_price_per_kwh`	Optional	p in Equation (20). Absent: cost savings not calculated.
`capital_cost`	Optional	C_{capital} in Equation (21). Entered **once for the whole package**: it covers every selected component and is not apportioned (Table 8). Absent: capital cost, payback, cost feasibility, and investment readiness all *not estimated*.
`currency`	Optional	ISO 4217 code, echoed with cost outputs; never used in scoring.
`grid_emission_factor_kgco2e_per_kwh`	Optional	ϕ in Equation (19); a user-supplied factor takes precedence over the country table and is itemised as an applied assumption.

D Glossary

Air temperature
The temperature of the air, here measured or estimated at pedestrian level. The quantity this methodology reports. Distinct from land surface temperature and from thermal comfort indices.

Archetype (cooling archetype)
A cited evidence class carrying every performance value — envelope, base score, evidence rating, suitability conditions, co-benefit defaults — and the citations behind them. There are 18 (Section 6.1).

Availability
Which catalogue entries the tool *offers* for a given site, decided by scale, land use, and the four gating questions of Section 5.6. Availability feeds no score.

Base cooling score
A 0–100 index expressing an archetype's relative expected cooling performance before site adjustment. Not a temperature.

Catalogue entry (typology)
One of the 110 curated entries of the source catalogue. It carries identity, family, availability, and the one archetype it inherits, and no performance value of its own.

Block confidence
A property of an *assessment*: how completely the user supplied the inputs a given output block depends on (Table 11). Distinct from evidence confidence.

Boundary-layer urban heat island
The heat island of the air mass above roof level, distinguished from the canopy-layer and surface heat islands by Oke (1976).

Canopy-layer urban heat island (CUHI)
The heat island of the air between the ground and roof level — the layer people occupy. The quantity relevant to this methodology.

Clamp
$\mathrm{clamp}(x)$ bounds a value to the 0–100 reporting scale.

Clip
$\mathrm{clip}(x,\ l,\ u)$ bounds a value to a stated interval. Used to confine reported temperature estimates to the literature envelope (Section 7.6).

Composite indicator
An index formed by normalising and aggregating several sub-indicators. Construction here follows Nardo et al. (2008).

Cooling Potential Score
The Layer 2 headline score: the typology's base cooling score scaled by the site adjustment factor (Equation (10)).

Envelope
The literature-supported interval of temperature reduction for an archetype. Its lower bound represents a poorly performing instance, its upper bound a well-executed instance under favourable conditions. Not a prediction and not a confidence interval.

Evidence confidence
A property of an *archetype*: how well grounded its adopted values are (high, medium, low). Inherited by every entry resolving to it, and propagates into an assessment by capping cooling-block confidence (Section 8.3.1).

Golden scenario
A stored input case with a hand-verified expected output, used as regression armour against unintended methodology drift and as the basis of the sensitivity analysis.

Heat Priority Index (HPI)
The Layer 1 headline score, combining heat exposure and vulnerability (Equation (8)). Independent of the proposed intervention.

Köppen–Geiger classification
A climate classification system whose subzones (Af, BWh, Csa, Cfa, Cfb, Cwa, . . .) raise the variance explained in cooling effectiveness from 12 % to 78 % (Keravec-Balbot et al. 2026).

Land surface temperature (LST)
The radiometric temperature of the ground and building surfaces, typically satellite-derived. Substantially exceeds air temperature (Venter et al. 2021); used here only as a relative screening proxy (Section 5.5).

NbS Cooling Opportunity Score
The final aggregated score (Equation (24)), on a 0–100 scale, for comparing options and sites.

Nature-based solutions (NbS)
Interventions using vegetation, water, and natural processes to deliver benefits — here, urban cooling.

Package
An assessment proposing more than one catalogue entry. Each component is scored and reported individually; the headline outputs are combined under Table 8, which caps temperature at the best-evidenced component and never sums it.

PET (physiologically equivalent temperature)
A thermal comfort index representing perceived thermal conditions. **Not** air temperature; used in this methodology only for relative ranking and caveats.

Screening level
An assessment tier producing indicative, comparable estimates for triage, positioned between unstructured judgment and detailed simulation.

Site adjustment factor
The composite multiplier $A \in [0.5, 1.2]$ derived from canopy, soil–water, scale, and climate conditions (Equation (9)).

Super-additivity

The hypothesis that combined interventions cool more than their strongest component. Not adopted, because no retrieved source quantifies it (Section 6.6.2). A package's advantage is expressed as co-benefit breadth instead.

Surface urban heat island (SUHI)

The heat island measured in surface temperatures. Reported at roughly six times the canopy-layer magnitude by Venter et al. (2021).

UTCI (universal thermal climate index)

A thermal comfort index. As with PET, not air temperature, and not reported by this tool.

E Source verification status

Every source enters the methodology's bibliography only after its bibliographic details have been checked against the publisher record, an institutional repository, or an indexing service. Table 24 reproduces the verification register from `docs/methodology/BIBLIOGRAPHY.md` at version 2026.08.05.

Three verification levels are used.

Full Bibliographic details *and* the specific quantitative finding used by the methodology were both confirmed from a retrieved source.

Metadata + finding (secondary) Bibliographic details confirmed from the publisher or repository; the quantitative finding confirmed from an authoritative secondary description — a publisher abstract page or an indexing service — rather than from the full text.

Metadata only Bibliographic details confirmed; the source is referenced qualitatively only, and no numeric value in the tool depends on it.

> This register is reproduced rather than summarised because the distinction it records is material. Several publisher sites — Elsevier, PNAS, Science — block automated retrieval, so a number of findings were confirmed from open repositories, PubMed, or publisher abstract pages. Where a value in the tool depends on a finding that is not *Full*, the configuration marks its confidence accordingly.
>
> A reader should not read *metadata + finding (secondary)* as an implied full-text reading. It means what it says: the number was confirmed from an authoritative description of the paper, not from the paper.

Table 24: Verification status of every source cited by the methodology.

Key	Verified	Basis and notes
`keravec2026`	Full	IOPscience, open access. The primary cross-typology anchor and the empirical basis for treating climate suitability as a first-order adjustment.
`ziter2019`	Full	PubMed record, PMID 30910972.

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Table 24 continued

Key	Verified	Basis and notes
`zolch2016`	Full	Lund University research portal record. *Note:* PET is a thermal-comfort index, not air temperature; used for relative ranking of intervention families and for the green-roof street-level caveat, never as an air-temperature value.
`santamouris2014`	Full	Open PDF via EEA Climate-ADAPT; abstract read directly. *Critical caveat:* the 0.3–3 K figure describes city-scale mass deployment in simulation studies, not one green roof on one building, and is therefore not used as a site-level typology value.
`reid2009`	Full	PMC open access record, PMC2801183. Grounds the vulnerability sub-indicators and the treatment of low cooling access as an independent vulnerability dimension; within that dimension, the *indoor* indicator.
`burkart2016`	Metadata + finding	PubMed record, PMID 26566198; findings from the *Environmental Health Perspectives* abstract. Grounds the *outdoor* cooling-refuge indicator added at version 2026.08.05: reachable vegetation and water quantified as modifiers of elderly heat mortality in Lisbon.
`sera2019`	Metadata + finding	PubMed record, PMID 30815699; findings from the *International Journal of Epidemiology* abstract. Corroborates `burkart2016` across 340 cities in 22 countries, and independently supports population density as a vulnerability indicator.

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Table 24 continued

Key	Verified	Basis and notes
`ember`	Full	Licence and coverage confirmed on Ember's data pages; 215 countries, CC-BY-4.0, redistribution permitted with attribution.
`beck2023`	Full	Open access at *Scientific Data*; licence, period, resolution and the 30-class legend confirmed against the published archive, whose checksum is recorded in `tools/build_datasets.py`. CC BY 4.0, redistribution permitted with attribution.
`naturalearth`	Full	Terms of use and release v5.1.2 confirmed at source; SHA-256 of both retrieved scales recorded in `tools/build_datasets.py`. Public domain; attribution recorded although not required.
`nardo2008`	Full	Full bibliographic record from the JRC repository; step sequence confirmed from OECD/JRC descriptions.
`worldbank2021`	Full	World Bank plain-text document retrieved directly. The documented justification for shipping no global default unit costs.
`jacobs2020`	Full	Open publisher PDF via the Wageningen repository, read directly. The primary source for the small constructed water archetype, and the reason constructed wetlands, ponds, basins, and water squares are not allowed to inherit the corridor envelope on family resemblance.

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Table 24 continued

Key	**Verified**	**Basis and notes**
`volker2013`	Full	Open-access publisher PDF, read directly. *Caveats recorded:* a single pooled estimate across ponds, lakes, and rivers together, so it cannot justify a size-dependent gradient; and its magnitude depends on the inclusion of remote-sensing data, which `ampatzidis2020` identifies as the overestimating mode.
`ampatzidis2020`	Full *(remote-sensing statement: metadata + finding, secondary)*	Publisher abstract read in full via the University of Bath research portal for the diurnal-variability and nocturnal findings; the compact "remote sensing overestimates" phrasing appears in the article highlights via an indexing service. Governs how the two water archetypes are read.
`armson2012`	Full	University of Manchester Research Explorer record, complete abstract. The decisive source for the non-canopy vegetation archetype, because it measures surface and globe temperature on the same plots and so isolates the surface/air conflation.
`kraemer2022`	Full	Open access, Frontiers, read directly. *Note:* an earlier draft of the review pack cited this source in connection with allotment gardens. It is a study of two Leipzig *parks* under drought and is cited as such (Section 6.7.1).

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Table 24 continued

Key	Verified	Basis and notes
`chafer2020`	Full	Publisher PDF retrieved and read; the author list was independently confirmed against the Crossref record after the review pack recorded it wrongly. *Caveat recorded:* the paper's abstract headlines "up to 5 °C", which is greenery against *unshaded* conditions; the adopted 2.5 °C is the like-for-like contrast against an identical bare pergola.
`colter2019`	Full	Arizona State University Elsevier Pure record, complete abstract. *Caveat:* PET is a comfort index, used here only for the ordering it establishes, under narrow conditions — hot summer midday, desert climate, six tree taxa.
`rost2020`	Full	Publisher-deposited abstract retrieved verbatim. *Critical caveat:* the study is **nocturnal only**, by title, abstract, and design, so it cannot support a daytime value and is not used for one (Section 6.8).
`cheung2022`	Full	Open access, IOPscience, read directly. *Caveat recorded:* the measured-against-modelled gap is specific to agricultural field studies; the same paper measures −2–−1 °C in irrigated urban parks. The methodology therefore does not claim that models overstate irrigation cooling in general.
`kumar2024`	Full	PMC open access record, PMC10909648. Two roles: a *validation* of the inherited calibration (street trees up to 2.8 °C in situ, inside the adopted envelope), and a *citable evidence gap* for productive landscapes. Its higher figures for other types are deliberately not adopted (Section 6.6.4).

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Table 24 continued

Key	Verified	Basis and notes
`bowler2010`	Metadata + finding (secondary)	Metadata from the Bangor University repository; the 0.94 °C finding from the publisher abstract description. Elsevier full text blocked.
`gunawardena2017`	Metadata + finding (secondary)	Metadata from TU Delft and Bath research portals; findings from the publisher abstract description. Elsevier full text blocked. Justifies the humidity caveat applied to blue typologies in tropical-wet climates.
`venter2021`	Metadata + finding (secondary)	Metadata from the ADS record and an author-hosted PDF; the surface-versus-canopy comparison from an indexed description. Publisher full text blocked. Basis for treating LST as a relative screening proxy only.
`akbari2001`	Metadata + finding (secondary)	Multiple indexing records and an LBNL institutional listing. Basis for *deriving* cooling-energy savings from estimated air temperature reduction, replacing unsourced per-typology energy factors.
`yao2023a`	Metadata + finding (secondary)	Arizona State University Elsevier Pure institutional record, full abstract; Elsevier full text blocked. The most direct field evidence for large water bodies, and the reason that archetype's floor was lowered from the 0.5 °C of the approved design to 0.1 °C at implementation (Section 6.7.1).
`yao2023b`	Metadata + finding (secondary)	Arizona State University Elsevier Pure record, full abstract; Elsevier full text blocked. Independent field corroboration of `jacobs2020`'s magnitudes for small water features, and one half of the nocturnal-warming caveat.

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Table 24 continued

Key	Verified	Basis and notes
`gill2013`	Metadata + finding (secondary)	Crossref record for bibliographic details; abstract via indexing services; Elsevier full text blocked. *Caveat recorded:* this is a *modelling* study, cited for the direction and mechanism of drought sensitivity and never for a numeric cooling value.
`ouyang2024`	Metadata + finding (secondary)	Crossref and NASA ADS for bibliographic details; the quantitative findings from two independent renderings of the publisher abstract; ScienceDirect blocked. Separates the comfort axis from the air-temperature axis for shade structures.
`norton2015`	Metadata + framework description (secondary)	Metadata from the Monash University research portal; framework description secondary. The closest published antecedent to this tool. No numeric value is taken from it.
`rahman2020`	Metadata only	Publisher listing. Used qualitatively — to justify that canopy condition modulates performance — never for a numeric value.
`oke1976`	Metadata only	Widely indexed. Cited for the conceptual distinction between canopy-layer, boundary-layer, and surface urban heat islands, not for a numeric value.

Which tool values depend on non-full verification

Four dependencies are worth naming explicitly.

The **energy derivation** of Section 7.9 rests entirely on `akbari2001`, verified at *metadata + finding (secondary)*. The 2–4 % per °C sensitivity is the sole input to every energy and greenhouse-gas figure the tool produces.

The **small green area and established park envelopes** depend on `bowler2010`'s 0.94 °C mean, verified at the same level, which is also the empirical anchor for the modelling-bias correction of Section 6.6.1.

The **large water body floor** of 0.1 °C rests on `yao2023a`, also at *metadata + finding (secondary)* — the publisher blocks full text, and the 0.1–0.6 °C figure was confirmed from the institutional record's full abstract. That level of verification was nonetheless sufficient to establish that the approved design had attributed to this paper a figure it does not contain, and to move the value (Section 6.7.1). The episode is the clearest illustration of what this register is for.

The **LST proxy restriction** of Section 5.5 rests on `venter2021`, again at the same level — though here the tool's use is qualitative, and the conclusion is independently supported by the conceptual distinction in `oke1976`.

In each case the tool's confidence ratings and stated limitations reflect the verification level, and in no case does a value verified only at *metadata only* carry a number.